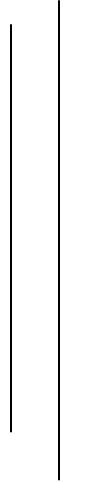


Thursday, October 27, 2016

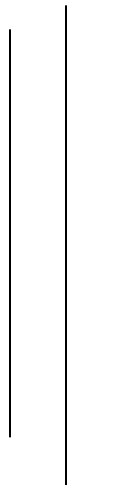
Matrices



A.N.M. Rezaul Karim

B.Sc (Honors), M.Sc in Mathematics

DCSA (BOU), PGD in ICT (BUET), Ph.D (In-Progress)



Associate Professor

Department of Computer Science & Engineering
International Islamic University Chittagong

Contents

1. Vectors
2. Determinant
3. Matrices
 - Matrix Notation
4. Vector Terminology
 - Vector Length
 - Vector Addition
 - Scalar Multiplication
 - Inner Product
 - Orthogonality
 - Normal Vector
 - Orthonormal Vectors
 - Gram-Schmidt Orthonormalization Process
5. Matrix Terminology
 - Difference between Matrix and Determinant
 - Square Matrix
 - Transpose of a Matrix
 - Diagonal Matrix
 - Identity Matrix
 - Skew-Hermitian Matrix
 - Matrix Multiplication
 - Covariance Matrix
 - Orthogonal Matrix
 - Rank of a Matrix
6. Linear System of Equations
 - Consistent
 - Inconsistent
 - Free Variable
 - Echelon Form
7. Eigenvectors and Eigenvalues
 - Geometrical Significance of Eigen Vectors and Eigen Values
 - Diagonalization
8. Different type of Inverse Method:
 - Augmented Matrix
 - Row operation Method
 - Cayley Hamilton Theorem
9. Graphical Presentation of Matrix
 - Graphical Presentation of Adjacency Matrix
 - Graphical Presentation of Transpose of a Matrix
10. Matrix Decomposition Theorem
 - 10.1 LU Decomposition
 - 10.2 QR Decomposition
 - 10.3 Eigen Decomposition Theorem:
 - 10.4 Singular Value Decomposition
11. Image Processing with SVD
12. Matrix Analysis using MATLAB

Vectors:

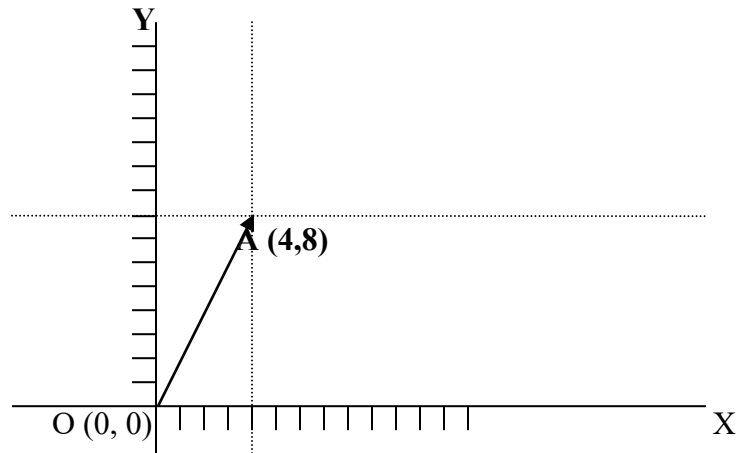


Figure no 01

The Vector $\vec{OA}$ can be written as $= [4,8]$
 $= \begin{pmatrix} 4 \\ 8 \end{pmatrix} \rightarrow$ In the form of Matrix
 $= 4\hat{i} + 8\hat{j}$

A vector can be specified by its components:

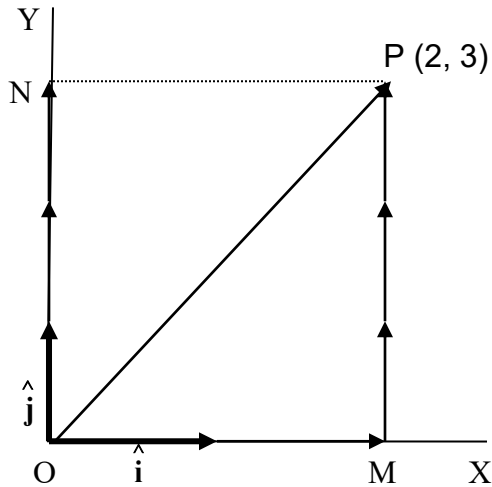


Figure no 02

A vector (Figure no 02) can be expressed as a column matrix:

$$\begin{aligned}\vec{OP} &= [2,3] \\ \Rightarrow \vec{OP} &= \begin{bmatrix} 2 \\ 3 \end{bmatrix} \\ \vec{OP} &= \vec{OM} + \vec{MP} = 2\hat{i} + 3\hat{j}\end{aligned}$$

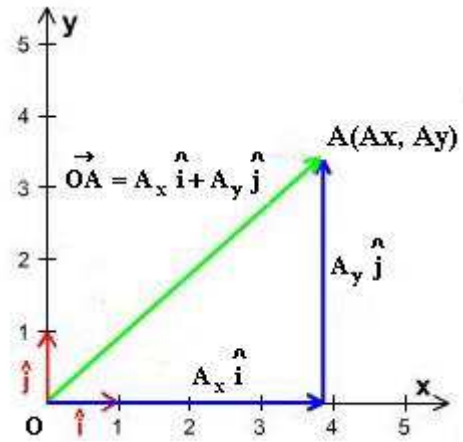


Figure 03

A vector (**Figure no 03**): 2-dimensional
(2-components)

$$\vec{OA} = [A_x, A_y]$$

$$\Rightarrow \vec{OA} = \begin{bmatrix} A_x \\ A_y \end{bmatrix}$$

$$\vec{OA} = A_x \hat{i} + A_y \hat{j}$$

A vector **Figure no 04**: 3-dimensional
(3-components)

$$\vec{a} = [a_x, a_y, a_z]$$

$$\Rightarrow \vec{a} = \begin{bmatrix} a_x \\ a_y \\ a_z \end{bmatrix}$$

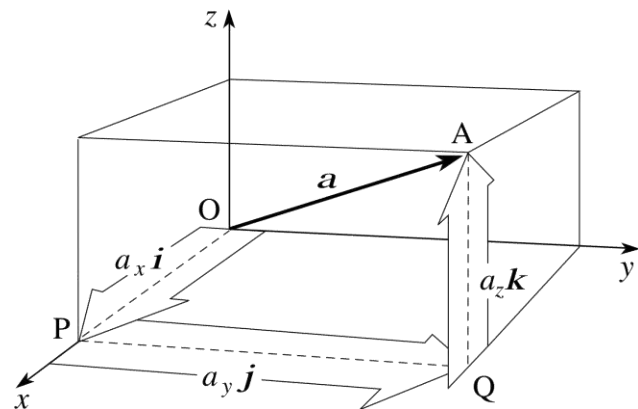


Figure no 04

The numbers comprising the vector are now called *components*, and the number of components equals the dimensionality of the vector. $\vec{x}$ is a **5-dimensional vector (5-components)**.

$$\vec{x} = \begin{pmatrix} 8 \\ 6 \\ 7 \\ 5 \\ 3 \end{pmatrix}$$

Vectors can be equivalently represented horizontally to save space, e.g. $\vec{x} = [8, 6, 7, 5, 3]$ is the same vector as above.

Determinant: A determinant is a function of a square matrix that reduces it to a single number. For example the determinant of the matrix $A = \begin{bmatrix} 4 & 1 \\ 1 & 2 \end{bmatrix}$ is

$$|A| = \begin{vmatrix} 4 & 1 \\ 1 & 2 \end{vmatrix} = 4 \times 2 - 1 \times 1 = 7$$

Matrices: A matrix is probably most familiar as a table of data, like Table 1.

Contestant	Round 1	Round 2	Round 3	Round 4	Total	Place
Carol Semple-Marzetta	17	18	5	5	45	1
Susan Curry	42	28	30	15	115	3
Monica Brant	10	10	10	21	51	2
Karen Hulse	28	5	65	39	132	5
Dale Tomita	24	26	45	21	116	4

Written as a matrix, Table 1 looks like this:

$$\begin{bmatrix} 17 & 18 & 5 & 5 & 45 & 1 \\ 42 & 28 & 30 & 15 & 115 & 3 \\ 10 & 10 & 10 & 21 & 51 & 2 \\ 28 & 5 & 65 & 39 & 132 & 5 \\ 24 & 26 & 45 & 21 & 116 & 4 \end{bmatrix}$$

The maximum number of rows is assigned to the variable m , and the number of columns is called n . Matrix *entries* (also called *elements* or *components*) are denoted by a lower-case a , and a *particular entry is referenced by its row index (labeled i) and its column index (labeled j)*.

Matrix Notation

A matrix is basically a collection of vectors. We can talk about row vectors or column vectors

A vector can be expressed as a column matrix:

$$\vec{v} = [v_x, v_y, v_z]$$

$$\Rightarrow \vec{v} = \begin{bmatrix} v_x \\ v_y \\ v_z \end{bmatrix}$$

A system of any $m \times n$ numbers arranged in a rectangular array of 'm' rows and 'n' columns is called a Matrix of order $m \times n$ or an $n \times m$ Matrix which is read as m by n or n by m Matrix.

Matrix can be denoted by [], (), || ||

Example:

$$A = (a_{ij}) = \begin{pmatrix} a_{11} & a_{12} & a_{13} & - & - & - & - & - & a_{1n} \\ a_{21} & a_{22} & a_{23} & - & - & - & - & - & a_{2n} \\ a_{31} & a_{32} & a_{33} & - & - & - & - & - & a_{3n} \\ - & - & - & - & - & - & - & - & - \\ a_{m1} & a_{m2} & a_{m3} & - & - & - & - & - & a_{mn} \end{pmatrix}$$

Where $i = 1, 2, 3, 4, \dots, m$ and $j = 1, 2, 3, 4, \dots, n$

Where $i \rightarrow$ row, $j \rightarrow$ column

[Here a_{11} means 1st row 1st column, a_{12} means 1st row 2nd column.....]

For example:

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \text{ is a } 2 \times 3 \text{ order matrix}$$

Vector Terminology: Vector Length:

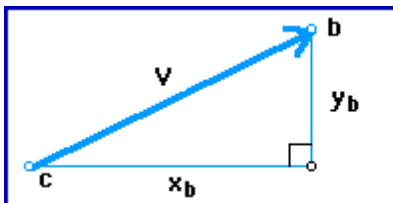


Figure no 05

Given a vector V with tail at the origin c (0,0) and head at b (x_b, y_b), what's its length?

According to Pythagoras, the length of the hypoteneuse is $\sqrt{x_b^2 + y_b^2}$

Magnitude of a vector: The magnitude of a vector $\vec{OP}$ or $\vec{V}$ is its length and is normally denoted by $|\vec{V}|$ or V . Given a vector $\vec{V}$ with tail at the origin O and head at $P(x,y)$, what's its length?

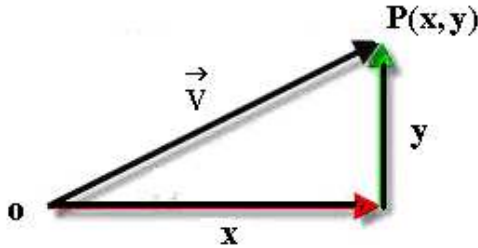


Figure: 06

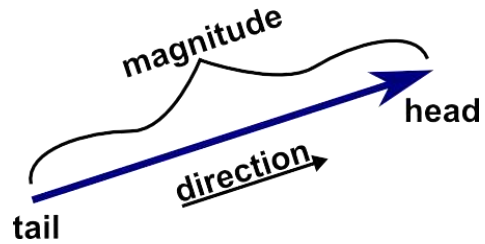


Figure: 07

According to Pythagoras, the length of the hypotenuse OP is the square root of $x^2 + y^2$.

That is, **Magnitude of a vector** $\vec{V}$ = its length = $|\vec{OP}| = |\vec{V}| = \sqrt{x^2 + y^2}$

If $\vec{u}$ is a vector, its length is denoted by $|\vec{u}|$. more concisely,

$$|\vec{u}| = \sqrt{\sum_{i=1}^n x_i^2}$$

For example, if $\vec{u} = [4, 11, 8, 10]$, then

$$|\vec{u}| = \sqrt{4^2 + 11^2 + 8^2 + 10^2} = \sqrt{301} = 17.35$$

Vector Addition:

Adding two vectors means adding each component in $\vec{u}_1$ to the component in the corresponding position in $\vec{u}_2$ to get a new vector.

For example:

$$\vec{u}_1 = [3, 2, 1, -2] \text{ and } \vec{u}_2 = [2, -1, 4, 1]$$

$$\vec{u}_1 + \vec{u}_2 = [3, 2, 1, -2] + [2, -1, 4, 1] = [(3+2), (2-1), (1+4), (-2+1)] = [5, 1, 5, -1]$$

Scalar Multiplication:

Multiplying a scalar (real number) times a vector means multiplying every component by that real number to yield a new vector.

For instance, if $\vec{u} = [3, 6, 8, 4]$, then $1.5 * \vec{u} = 1.5 * [3, 6, 8, 4] = [4.5, 9, 12, 6]$.

More generally, *scalar multiplication* means if d is a real number and $\vec{u}$ is a vector $[u_1, u_2, \dots, u_n]$, then $d * \vec{u} = [du_1, du_2, \dots, du_n]$.

Inner Product

The *inner product* of two vectors (also called the *dot product* or *scalar product*) defines multiplication of vectors. The inner product is only defined for vectors of the same

dimension. The inner product of two vectors is denoted $\left(\begin{matrix} \vec{u}_1, \vec{u}_2 \end{matrix} \right)$ or $\vec{u}_1 \cdot \vec{u}_2$ (the dot product). Thus,

$$\left(\begin{matrix} \vec{x}, \vec{y} \end{matrix} \right) = \vec{x} \cdot \vec{y} = \sum_{i=1}^n x_i y_i$$

For example, if $\vec{x} = [1, 6, 7, 4]$ and $\vec{y} = [3, 2, 8, 3]$, then

$$\vec{x} \cdot \vec{y} = 1(3) + 6(2) + 7(8) + 3(4) = 83$$

Orthogonality

Two vectors are orthonormal to each other if their inner product equals zero. In two-dimensional space this is equivalent to saying that the vectors are perpendicular, or that the only angle between them is a 90° angle. For example, the vectors $[2, 1, -2, 4]$ and $[3, -6, 4, 2]$ are orthogonal because

$$[2, 1, -2, 4] \cdot [3, -6, 4, 2] = 2(3) + 1(-6) - 2(4) + 4(2) = 0$$

Normal Vector

A *normal vector* (or *unit vector*) is a vector of length 1. Any vector with an initial length > 0 , can be normalized by dividing each component in it by the vectors length.

For example, if $\vec{v} = [2, 4, 1, 2]$, then

$$\left| \vec{v} \right| = \sqrt{2^2 + 4^2 + 1^2 + 2^2} = \sqrt{25} = 5 \text{ is not a normal vector}$$

But $\vec{u} = [2/5, 4/5, 1/5, 2/5]$ is a normal vector because

$$\left| \vec{u} \right| = \sqrt{(2/5)^2 + (4/5)^2 + (1/5)^2 + (2/5)^2} = \sqrt{25/25} = 1$$

Orthonormal Vectors

Vectors of unit length that are orthogonal to each other are said to be *orthonormal*. For example,

$$\vec{u} = [2/5, 4/5, -2/5, 4/5]$$

and $\vec{v} = [3/\sqrt{65}, -6/\sqrt{65}, 4/\sqrt{65}, 2/\sqrt{65}]$

are orthonormal because

$$\left| \vec{u} \right| = \sqrt{(2/5)^2 + (1/5)^2 + (-2/5)^2 + (4/5)^2} = 1$$

$$\left| \vec{v} \right| = \sqrt{(3/\sqrt{65})^2 + (-6/\sqrt{65})^2 + (4/\sqrt{65})^2 + (2/\sqrt{65})^2} = 1$$

$$\vec{u} \cdot \vec{v} = \frac{6}{5\sqrt{65}} - \frac{6}{5\sqrt{65}} - \frac{8}{5\sqrt{65}} + \frac{8}{5\sqrt{65}} = 0$$

What does a pair of orthonormal vectors in 2-D Euclidean space look like?

Let $\mathbf{u} = (x_1, y_1)$ and $\mathbf{v} = (x_2, y_2)$. Consider the restrictions on x_1, x_2, y_1, y_2 required to make $\mathbf{u}$ and $\mathbf{v}$ form an orthonormal pair.

- From the orthogonality restriction, $\mathbf{u} \cdot \mathbf{v} = 0$.
- From the unit length restriction on $\mathbf{u}$, $\|\mathbf{u}\| = 1$.
- From the unit length restriction on $\mathbf{v}$, $\|\mathbf{v}\| = 1$.

Expanding these terms gives 3 equations:

- $x_1x_2 + y_1y_2 = 0$
- $\sqrt{x_1^2 + y_1^2} = 1$
- $\sqrt{x_2^2 + y_2^2} = 1$

Gram-Schmidt Orthonormalization Process

The Gram-Schmidt orthonormalization process is a method for converting a set of vectors into a set of orthonormal vectors. It basically begins by normalizing the first vector under consideration and iteratively rewriting the remaining vectors in terms of themselves minus a multiplication of the already normalized vectors. For example, to convert the column vectors of

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 2 & 0 \\ 2 & 3 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

into orthonormal column vectors

$$A = \begin{bmatrix} \frac{\sqrt{6}}{6} & \frac{\sqrt{2}}{6} & \frac{2}{3} \\ 0 & \frac{2\sqrt{2}}{3} & \frac{-1}{3} \\ \frac{\sqrt{6}}{3} & 0 & 0 \\ \frac{\sqrt{6}}{6} & \frac{-\sqrt{2}}{6} & \frac{-2}{3} \end{bmatrix},$$

First normalize $\vec{v}_1 = [1, 0, 2, 1]$:

$$\vec{u}_1 = \left[\frac{1}{\sqrt{6}}, 0, \frac{2}{\sqrt{6}}, \frac{1}{\sqrt{6}} \right].$$

Next, let

$$\begin{aligned} \vec{w}_2 &= \vec{v}_2 - \vec{u}_1 \cdot \vec{v}_2 * \vec{u}_1 = [2, 2, 3, 1] - \left[\frac{1}{\sqrt{6}}, 0, \frac{2}{\sqrt{6}}, \frac{1}{\sqrt{6}} \right] \cdot [2, 2, 3, 1] * \left[\frac{1}{\sqrt{6}}, 0, \frac{2}{\sqrt{6}}, \frac{1}{\sqrt{6}} \right] \\ &= [2, 2, 3, 1] - \left(\frac{9}{\sqrt{6}} \right) * \left[\frac{1}{\sqrt{6}}, 0, \frac{2}{\sqrt{6}}, \frac{1}{\sqrt{6}} \right] \\ &= [2, 2, 3, 1] - \left[\frac{3}{2}, 0, 3, \frac{3}{2} \right] \\ &= \left[\frac{1}{2}, 2, 0, \frac{-1}{2} \right] \end{aligned}$$

Normalize $\vec{w}_2$ to get $\vec{u}_2 = \left[\frac{\sqrt{2}}{6}, \frac{2\sqrt{2}}{3}, 0, \frac{-\sqrt{2}}{6} \right]$

Now compute $\vec{u}_3$ in terms of $\vec{u}_1$ and $\vec{u}_2$ as follows. Let

$$\vec{w}_3 = \vec{v}_2 - \vec{u}_1 \cdot \vec{v}_2 * \vec{u}_1 - \vec{u}_2 \cdot \vec{v}_2 * \vec{u}_2 = \left[\frac{4}{9}, \frac{-2}{9}, 0, \frac{-4}{9} \right]$$

and normalize $\vec{w}_3$ to get $\vec{u}_3 = \left[\frac{2}{3}, \frac{-1}{3}, 0, \frac{-2}{3} \right]$

More generally, if we have an orthonormal set of vectors $\vec{u}_1, \dots, \vec{u}_{k-1}$, then $\vec{w}_k$ is

$$\text{expressed as } \vec{w}_k = \vec{v}_k - \sum_{i=1}^{k-1} \vec{u}_i \cdot \vec{v}_k * \vec{u}_i$$

We define the projection operator by

$$\text{Proj}_{\mathbf{u}} \mathbf{v} = \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\langle \mathbf{u}, \mathbf{u} \rangle} \mathbf{u} = \langle \mathbf{u}, \mathbf{v} \rangle \frac{\mathbf{u}}{\langle \mathbf{u}, \mathbf{u} \rangle},$$

Where $\langle \mathbf{u}, \mathbf{v} \rangle$ denotes the **inner product** of the vectors $\mathbf{u}$ and $\mathbf{v}$. This operator projects the vector $\mathbf{v}$ orthogonally onto the vector $\mathbf{u}$.

The Gram–Schmidt process then works as follows:

$$\mathbf{u}_1 = \mathbf{v}_1, \quad \mathbf{e}_1 = \frac{\mathbf{u}_1}{\|\mathbf{u}_1\|}$$

$$\mathbf{u}_2 = \mathbf{v}_2 - \text{proj}_{\mathbf{u}_1} \mathbf{v}_2, \quad \mathbf{e}_2 = \frac{\mathbf{u}_2}{\|\mathbf{u}_2\|}$$

$$\mathbf{u}_3 = \mathbf{v}_3 - \text{proj}_{\mathbf{u}_1} \mathbf{v}_3 - \text{proj}_{\mathbf{u}_2} \mathbf{v}_3, \quad \mathbf{e}_3 = \frac{\mathbf{u}_3}{\|\mathbf{u}_3\|}$$

$$\mathbf{u}_4 = \mathbf{v}_4 - \text{proj}_{\mathbf{u}_1} \mathbf{v}_4 - \text{proj}_{\mathbf{u}_2} \mathbf{v}_4 - \text{proj}_{\mathbf{u}_3} \mathbf{v}_4, \quad \mathbf{e}_4 = \frac{\mathbf{u}_4}{\|\mathbf{u}_4\|}$$

$$\mathbf{u}_k = \mathbf{v}_k - \sum_{j=1}^{k-1} \text{proj}_{\mathbf{u}_j} \mathbf{v}_k, \quad \mathbf{e}_k = \frac{\mathbf{u}_k}{\|\mathbf{u}_k\|}$$

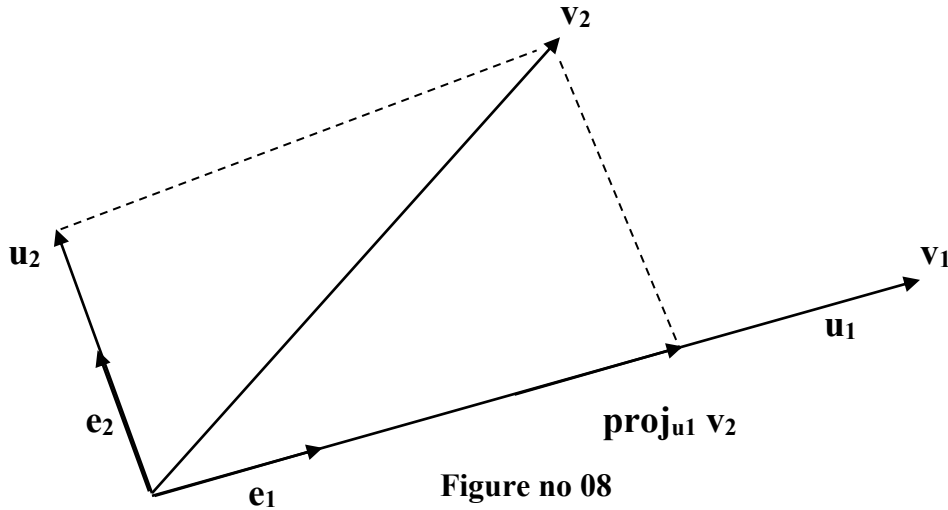


Figure no 08

The sequence $\mathbf{u}_1, \dots, \mathbf{u}_k$ is the required system of orthogonal vectors, and the normalized vectors $\mathbf{e}_1, \dots, \mathbf{e}_k$ form an *orthonormal* set.

To check that these formulas yield an orthogonal sequence, first compute $\langle \mathbf{u}_1, \mathbf{u}_2 \rangle$ by substituting the above formula for $\mathbf{u}_2$: you will get zero. Then use this to compute $\langle \mathbf{u}_1, \mathbf{u}_3 \rangle$ again by substituting the formula for $\mathbf{u}_3$: you will get zero. The general proof proceeds by mathematical induction.

Geometrically, this method proceeds as follows: to compute $\mathbf{u}_i$, it projects $\mathbf{v}_i$ orthogonally onto the subspace U generated by $\mathbf{u}_1, \dots, \mathbf{u}_{i-1}$, which is the same as the subspace generated by $\mathbf{v}_1, \dots,$

$\mathbf{v}_{i-1}$. The vector $\mathbf{u}_i$ is then defined to be the difference between $\mathbf{v}_i$ and this projection, guaranteed to be orthogonal to all of the vectors in the subspace U .

The Gram–Schmidt process also applies to a linearly independent infinite sequence $\{\mathbf{v}_i\}_i$. The result is an orthogonal (or orthonormal) sequence $\{\mathbf{u}_i\}_i$ such that for natural number n : the algebraic span of $\mathbf{v}_1, \dots, \mathbf{v}_n$ is the same as that of $\mathbf{u}_1, \dots, \mathbf{u}_n$.

If the Gram–Schmidt process is applied to a linearly dependent sequence, it outputs the $\mathbf{0}$ vector on the i th step, assuming that $\mathbf{V}_i$ is a linear combination of $\mathbf{V}_1, \mathbf{V}_2, \dots, \mathbf{V}_{i-1}$.

Example:

Consider the following set of vectors in $\mathbb{R}^2$ (with the conventional inner product)

$$S = \left\{ \mathbf{v}_1 = \begin{pmatrix} 3 \\ 1 \end{pmatrix}, \mathbf{v}_2 = \begin{pmatrix} 2 \\ 2 \end{pmatrix} \right\}.$$

Now, perform Gram-Schmidt, to obtain an orthogonal set of vectors:

$$\mathbf{u}_1 = \mathbf{v}_1 = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$$

$$\mathbf{u}_2 = \mathbf{v}_2 - \text{proj}_{\mathbf{u}_1} \mathbf{v}_2 = \begin{pmatrix} 2 \\ 2 \end{pmatrix} - \text{proj}_{\begin{pmatrix} 3 \\ 1 \end{pmatrix}} \begin{pmatrix} 2 \\ 2 \end{pmatrix} = \begin{pmatrix} -2/5 \\ 6/5 \end{pmatrix}.$$

We check that the vectors $\mathbf{u}_1$ and $\mathbf{u}_2$ are indeed orthogonal:

$$\langle \mathbf{u}_1, \mathbf{u}_2 \rangle = \left\langle \begin{pmatrix} 3 \\ 1 \end{pmatrix}, \begin{pmatrix} -2/5 \\ 6/5 \end{pmatrix} \right\rangle = -\frac{6}{5} + \frac{6}{5} = 0.$$

We can then normalize the vectors by dividing out their sizes as shown above:

$$\mathbf{e}_1 = \frac{1}{\sqrt{10}} \begin{pmatrix} 3 \\ 1 \end{pmatrix}$$

$$\mathbf{e}_2 = \frac{1}{\sqrt{40}} \begin{pmatrix} -2/5 \\ 6/5 \end{pmatrix} = \frac{1}{\sqrt{10}} \begin{pmatrix} -1 \\ 3 \end{pmatrix}.$$

Distinguish between Determinant and Matrices

Determinant	Matrices
1. Determinant can be denoted by: $\begin{vmatrix} & \\ & \end{vmatrix}$ 2. Number of rows and columns are always equal Example: $\begin{vmatrix} 2 & 9 \\ 6 & 1 \end{vmatrix}$ 3. It has a definite value at through it is an arrangement of numbers 4. If two rows (or columns) are identical, a determinant vanishes 5. Rows and columns can be interchanged 6. If a determinant is multiplied by a number m, every element of any one row (or any one column) is multiplied by m. Example: $2 \begin{vmatrix} a & b \\ c & d \end{vmatrix} = \begin{vmatrix} 2a & 2b \\ 2c & 2d \end{vmatrix}$ 7. A determinant can be resolved into two determinants by using only one row (or column) 8. The sign of a determinant changes if two rows (or columns) are interchanged 9. The product of two determinants does not change the order Example: $\begin{vmatrix} 1 & 5 \\ 3 & 1 \end{vmatrix} \times \begin{vmatrix} 3 & 1 \\ 7 & 5 \end{vmatrix} = \begin{vmatrix} 1 \times 3 + 5 \times 7 & 1 \times 1 + 5 \times 5 \\ 3 \times 3 + 1 \times 7 & 3 \times 1 + 1 \times 5 \end{vmatrix}$ $= \begin{vmatrix} 38 & 26 \\ 16 & 8 \end{vmatrix}$	1. Matrix can be denoted by: $[], (), \parallel \parallel$ 2. Number of rows and columns are not always equal Example: $\begin{pmatrix} 1 \\ 3 \end{pmatrix}$ 3. It has no definite value, it is merely an arrangement of numbers 4. Identical rows (or columns) may occur in a matrix 5. Rows and columns cannot be interchanged 6. If a matrix is multiplied by m, every element of the entire matrix is multiplied by m Example: $2 \begin{vmatrix} a & b \\ c & d \end{vmatrix} = \begin{vmatrix} 2a & 2b \\ 2c & 2d \end{vmatrix}$ 7. A matrix cannot be resolved into two matrices using only one row (or column) 8. Two rows (or columns) cannot be interchanged at all 9. The product of two matrices may change the order Example: $(2 \ 4)_{1 \times 2} \times \begin{pmatrix} 5 \\ 8 \end{pmatrix}_{2 \times 1} = (2 \times 5 + 4 \times 8) = (42)_{1 \times 1}$

Square Matrix:

If $m = n$ that is the number of rows & columns of matrix are equal then the matrix is of order $n \times n$ or $m \times m$ is called a square matrix of order n or m .

For example:

$$\begin{bmatrix} 1 & 2 & 6 \\ 2 & 4 & 7 \\ 3 & 5 & 8 \end{bmatrix} \text{ is a } 3 \times 3 \text{ order matrix}$$

Rectangular Matrix:

The number of rows & columns of a matrix need not be equal that is $m \neq n$ then the matrix is known as a rectangular matrix.

For example:

$$\begin{bmatrix} 1 & 4 & 7 & 1 \\ 2 & 5 & 8 & 2 \\ 3 & 6 & 9 & 3 \end{bmatrix} \text{ is a } 3 \times 4 \text{ order matrix}$$

Horizontal Matrix:

If in a matrix the number of columns is more than the number of rows then it is called the horizontal matrix.

For example:

$$\begin{bmatrix} 1 & 4 & 7 & 1 \\ 2 & 5 & 8 & 2 \\ 6 & 5 & 4 & 9 \end{bmatrix} \text{ is a } 3 \times 4 \text{ order matrix}$$

Vertical Matrix:

If in a matrix the number of rows is more than the number of columns then it is called the vertical matrix.

For example:

$$\begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix} \text{ is a } 3 \times 2 \text{ order matrix}$$

Row Matrix:

If in a matrix there is only one row it is called a row matrix.

For example:

$$[1 \quad 2 \quad 3 \quad 4]$$

Column Matrix:

If in a matrix there is only one column it is called a column matrix.

For example:

$$\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

Null or Zero Matrix:

If all the elements of an $m \times n$ matrix are zero then it is called a null or zero matrix. It is denoted by 0.

For example:

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \text{ or simply } 0$$

Unit Matrix/Identity Matrix:

A square matrix having unity for its elements in the leading diagonal and all other elements as zero is called a unit matrix.

For example:

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \text{ is a } 3 \times 3 \text{ order matrix}$$

Diagonal Matrix:

A square matrix in which all elements except those in the main or leading diagonal are zero is known as a diagonal matrix.

For example:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix} \text{ is a } 3 \times 3 \text{ order matrix}$$

The sum of the diagonal elements of a square matrix is called the trace of the matrix.

Sub-Matrix:

A matrix, which is obtained from a given matrix by deleting any number of rows and number of columns, is called a sub-matrix of the given matrix.

For example:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 5 & 6 & 7 \end{bmatrix} \text{ the Sub - Matrix of } A = \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix}$$

Triangular Matrix:

There are two types of triangular matrix. They are-

Upper Triangular Matrix:

A square matrix 'A' whose elements $a_{ij} = 0$ for $i > j$ is called an upper triangular matrix.

For example:

$$A = (a_{ij}) = \begin{pmatrix} a_{11} & a_{12} & a_{13} & - & - & - & - & - & - & a_{1n} \\ 0 & a_{22} & a_{23} & - & - & - & - & - & - & a_{2n} \\ 0 & 0 & a_{33} & - & - & - & - & - & - & a_{3n} \\ - & - & - & - & - & - & - & - & - & - \\ 0 & 0 & 0 & - & - & - & - & - & - & a_{mn} \end{pmatrix}$$

Where $i \rightarrow$ row, $j \rightarrow$ column

Lower Triangular Matrix:

A square matrix A whose elements $a_{ij} = 0$ for $j > i$ is called a lower triangular matrix.

For example:

$$A = (a_{ij}) = \begin{pmatrix} a_{11} & 0 & 0 & - & - & - & - & - & - & 0 \\ a_{21} & a_{22} & 0 & - & - & - & - & - & - & 0 \\ a_{31} & a_{32} & a_{33} & - & - & - & - & - & - & 0 \\ - & - & - & - & - & - & - & - & - & - \\ a_{m1} & a_{m2} & a_{m3} & - & - & - & - & - & - & a_{mn} \end{pmatrix}$$

Scalar Matrix:

If in a square matrix A all the leading diagonal elements are equal to a where $a \neq 0$ and all the remaining elements are equal to zero then it is called a scalar matrix.

For example:

$$A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

Singular Matrix: A square matrix that does not have a matrix inverse. *A matrix is singular iff its determinant is 0.* For example, there are 10 singular 2×2 (0, 1)-matrices:

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

Binary matrix: A (0, 1)-integer matrix, i.e., a matrix each of whose elements is 0 or 1, also called a binary matrix

Test the matrix $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix}$ is singular or not

Solution:

$$|A| = 1(4 \times 6 - 5 \times 5) - 2(2 \times 6 - 3 \times 5) + 3(2 \times 5 - 3 \times 4)$$

$$|A| = 1(24 - 25) - 2(12 - 15) + 3(10 - 12)$$

$$= 1(-1) - 2(-3) + 3(-2)$$

$$= -1 + 6 - 6$$

$$= -1$$

As $|A| \neq 0$, hence the matrix is not singular.

Complex Matrix: The matrix consists of complex numbers is called complex Matrix.

Example: $A = \begin{bmatrix} 1+i & 2+3i \\ 2 & 3i \end{bmatrix}$

Complex Conjugate or Conjugate Matrix:

The matrix obtained from any given matrix A of order $m \times n$ with complex elements a_{ij} by replacing its elements by the corresponding conjugate complex numbers is called the complex conjugate or conjugate of A denoted by $\bar{A}$ and read as A conjugate.

For example:

$$\text{If } A = \begin{bmatrix} 1+i & 2+3i \\ 2 & 3i \end{bmatrix}$$

$$\bar{A} = \begin{bmatrix} 1-i & 2-3i \\ 2 & -3i \end{bmatrix}$$

Real Matrix:

A matrix A is called real if it satisfies the relation $A = \bar{A}$.

For example:

$$A = \begin{bmatrix} 2 & 4 \\ 5 & 6 \end{bmatrix}$$

$$\bar{A} = \begin{bmatrix} 2 & 4 \\ 5 & 6 \end{bmatrix}$$

Imaginary Matrix:

A matrix A is called imaginary provided it satisfies the relation $A = -\bar{A}$.

For example:

$$A = \begin{bmatrix} 2i & 0 \\ 4i & 3i \end{bmatrix}$$

$$\bar{A} = \begin{bmatrix} -2i & 0 \\ -4i & -3i \end{bmatrix}$$

$$-\bar{A} = \begin{bmatrix} 2i & 0 \\ 4i & 3i \end{bmatrix} = A$$

Transpose of a Matrix: The transpose of a matrix is created by converting its rows into columns; that is, row 1 becomes column 1, row 2 becomes column 2 etc. The transpose of matrix A is A^T . For example, if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

Then its transpose is

$$A^T = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$$

Symmetric Matrix:

A square matrix $A = (a_{ij})$ is called symmetric provided $a_{ij} = a_{ji}$ for all values of i and j . For example:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix}$$

Here $a_{11} = a_{11}, a_{12} = a_{21}, a_{13} = a_{31}, a_{23} = a_{32}, a_{22} = a_{22}, a_{33} = a_{33}$

Another definition: A symmetric matrix is a square matrix that satisfies

$$A^T = A$$

Where A^T denotes the transpose, so $a_{ij} = a_{ji}$.

Skew Symmetric Matrix:

A square matrix $A = (a_{ij})$ is called skew symmetric matrix $a_{ij} = -a_{ji}$ all values of i and j .

For example:

$$A = \begin{bmatrix} 0 & 2 & 3 \\ -2 & 0 & 5 \\ -3 & -5 & 0 \end{bmatrix}$$

Here,

$$a_{ij} = -a_{ji}$$

$$a_{11} = -a_{11} [i = 1, j = 1]$$

$$\Rightarrow 2a_{11} = 0 \therefore a_{11} = 0$$

Again

$$a_{ij} = -a_{ji}$$

$$a_{22} = -a_{22} [i = 2, j = 2]$$

$$\Rightarrow 2a_{22} = 0 \therefore a_{22} = 0$$

Again

$$a_{ij} = -a_{ji}$$

$$a_{33} = -a_{33} [i = 3, j = 3]$$

$$\Rightarrow 2a_{33} = 0 \therefore a_{33} = 0$$

$$\text{and } a_{12} = -a_{21}, \quad a_{13} = -a_{31}, \quad a_{23} = -a_{32}$$

Self-Adjoint Matrix:

A matrix A for which

$$A^H \equiv \overline{A^T} = A$$

Where the conjugate transpose is denoted by A^H , A^T , is the transpose of A , and $\bar{Z}$ is the complex conjugate. If a matrix is self-adjoint, it is said to be Hermitian.

Hermitian Matrix:

A square matrix is called Hermitian if it is self-adjoint. That is it satisfies $A^H \equiv \overline{A^T} = A$.

For Example

Prove that $A = \begin{pmatrix} 3 & 2+i \\ 2-i & 1 \end{pmatrix}$ is a Hermitian matrix.

Proof:

$$A^T = \begin{pmatrix} 3 & 2-i \\ 2+i & 1 \end{pmatrix}$$

$$\therefore A^H = \overline{A^T} = \begin{pmatrix} 3 & 2+i \\ 2-i & 1 \end{pmatrix} = A$$

In linear algebra, a **rotation matrix** is a matrix that is used to perform a rotation in Euclidean space. For example the matrix

$$R = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$$

rotates points in the xy -Cartesian plane counterclockwise through an angle θ about the origin of the Cartesian coordinate system. To perform the rotation, the position of each point must be represented by a column vector $\mathbf{v}$, containing the coordinates of the point. A rotated vector is obtained by using the matrix multiplication $R\mathbf{v}$ (see below for details).

In two and three dimensions, rotation matrices are among the simplest algebraic descriptions of rotations, and are used extensively for computations in geometry, physics, and computer graphics. Though most applications involve rotations in two or three dimensions, rotation matrices can be defined for n -dimensional space.

Rotation matrices are always square, with real entries. Algebraically, a rotation matrix in n -dimensions is a $n \times n$ special orthogonal matrix, i.e. an orthogonal matrix whose determinant is 1:

$$\mathbf{R}^T = \mathbf{R}^{-1}, \det \mathbf{R} = 1$$

Unitary Matrix:

A **unitary matrix** is a square complex-valued matrix, A , whose inverse is equal to its conjugate transpose:

$$A^{-1} = \bar{A}^t.$$

A square matrix U is a unitary matrix if $U^H = U^{-1}$

Where U^H denotes the conjugate transpose and U^{-1} is the matrix inverse. For example,

$$A = \begin{pmatrix} 2^{-\frac{1}{2}} & 2^{-\frac{1}{2}} & 0 \\ -2^{-\frac{1}{2}}i & 2^{-\frac{1}{2}}i & 0 \\ 0 & 0 & i \end{pmatrix} \text{ is a unitary matrix.}$$

Skew-Hermitian Matrix:

A is said to be **skew-Hermitian** or **antihermitian** if its conjugate transpose A^* is also its negative. That is, if it satisfies the relation:

$$A^* = -A$$

or in component form, if $A = (a_{ij})$:

$$a_{i,j} = -\bar{a}_{j,i}$$

for all i and j .

Examples

For example, the following matrix is skew-Hermitian:

$$\begin{pmatrix} i & 2+i \\ -2+i & 3i \end{pmatrix}$$

Multiplication of Matrices:

If A and B be two matrices such that the number of columns in A is equal to the number of rows in B, that is if $A = (a_{ij})$, $B = (b_{jk})$ then the product of A and B denoted by AB.

Example 01:

$$A = \begin{pmatrix} 1 & 2 \\ 4 & 5 \end{pmatrix}, B = \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix}$$

$$\Rightarrow AB = \begin{pmatrix} 1 \times 5 + 2 \times 7 & 1 \times 6 + 2 \times 8 \\ 4 \times 5 + 5 \times 7 & 4 \times 6 + 5 \times 8 \end{pmatrix} \quad [\text{Row by Column}]$$

$$\Rightarrow AB = \begin{pmatrix} 5 + 14 & 6 + 16 \\ 20 + 35 & 24 + 40 \end{pmatrix}$$

$$\therefore AB = \begin{pmatrix} 19 & 22 \\ 55 & 64 \end{pmatrix}$$

$$A = \begin{pmatrix} 1 & 2 \\ 4 & 5 \end{pmatrix}, B = \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix}$$

$$\text{Example 02: } A = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \quad B = \begin{pmatrix} 2 \\ 4 \\ 6 \end{pmatrix}$$

Answer: Since the number of column in Matrix A is not equal to the number of rows in B. Hence AB does not exist.

$$\text{Example 03: } A = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \quad B = \begin{pmatrix} 1 & 2 & 3 \end{pmatrix}$$

$$\therefore AB = \begin{pmatrix} 1 \times 1 & 1 \times 2 & 1 \times 3 \\ 2 \times 1 & 2 \times 2 & 2 \times 3 \\ 3 \times 1 & 3 \times 2 & 3 \times 3 \end{pmatrix}$$

$$\therefore AB = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{pmatrix}$$

Example 04: A manufacturer produces three products: Toothpaste, Soap and Shampoo, which sell in the market. Annual sale volumes are indicated as follows:

Market	Products		
	Toothpaste	Soap	Shampoo
New Market	8000	10000	15000
Shopping Complex	10000	2000	20000

- i. If unit (mini packet) sale prices of Toothpaste, soap and shampoo are tk. 2.25, tk. 3.50 and tk. 3.00 respectively. Find the total revenue in each market with the help of matrix.
- ii. If the unit costs of the above three products are tk. 1.60, tk. 1.90 and tk. 2.40 respectively. Find the gross profit with the help of matrices.

Solution:

The product matrix gives the total revenue in each market.

Let, Unit sale price of three products be $A = \begin{pmatrix} 2.25 & 3.50 & 3.00 \end{pmatrix}$

Annual sale of three goods in two markets be $B = \begin{pmatrix} 8000 & 10000 \\ 10000 & 2000 \\ 15000 & 20000 \end{pmatrix}$

Given,

$$A = \begin{pmatrix} 2.25 & 3.50 & 3.00 \end{pmatrix} \quad B = \begin{pmatrix} 8000 & 10000 \\ 10000 & 2000 \\ 15000 & 20000 \end{pmatrix}$$

$$\therefore AB = (2.25 \times 8000 + 3.50 \times 10000 + 3 \times 15000 \quad 2.25 \times 10000 + 3.50 \times 2000 + 3 \times 20000) \\ = (98000 \quad 89500)$$

Total revenue from the new market Tk.98, 000/-

Total revenue from Shopping Complex-Tk.89, 500/-

Total revenue from two markets: Tk. 1, 87,500/-

Similarly the costs of products with manufacturers sells in the market are:

$$A = \begin{pmatrix} 1.60 & 1.90 & 2.40 \end{pmatrix} \quad B = \begin{pmatrix} 8000 & 10000 \\ 10000 & 2000 \\ 15000 & 20000 \end{pmatrix}$$

$$\therefore AB = 1.60 \times 8000 + 1.90 \times 10000 + 2.40 \times 15000 \quad 1.60 \times 10000 + 1.90 \times 2000 + 2.40 \times 20000 \\ = (67800 \quad 67800)$$

Total cost of products for new market Tk.67, 800/-

Total cost of products for Shopping Complex-Tk.67800/-

Total cost from two markets: Tk. 1, 35,600

$$\text{So, required gross profit} = \text{Total revenue} - \text{Total Cost} = (187500 - 135600) \\ = \text{Tk.51, 900/-}$$

Ans: / Tk. 51,900/-

Example 05: A trust fund has tk. 50,000 that is to be invested into two types of bonds. The first bond pays 5% interest per year and the second bond pays 6% interest per year. Using matrix multiplication determines how to divide tk. 50,000 between the two types of bond so as to obtain an annual total interest of tk. 2780.

Solution: Let tk. 50000 be divided into two parts tk. x and tk. (50,000 – x) out of which first part is invested in first type of bonds and second part is invested in second type of bonds.

The values of these bonds can be written in the form of a row matrix $A = (x \ 50000 - x)$

The amounts received as interest per Tk. annually from these two types of bonds can be

written in the form of a column matrix $B = \begin{pmatrix} \frac{5}{100} \\ \frac{6}{100} \end{pmatrix}$

Now the interest to be obtained annually is a single number

$$\begin{aligned} AB &= (x \ 50000 - x) \begin{pmatrix} \frac{5}{100} \\ \frac{6}{100} \end{pmatrix} \\ &= \left(x \times \frac{5}{100} + (50000 - x) \times \frac{6}{100} \right) \\ AB &= \left[\frac{x}{20} + \frac{3(50000 - x)}{50} \right] \\ &= \left[\frac{5x + 6(50000 - x)}{100} \right] \\ &= \left[\frac{5x + 300000 - 6x}{100} \right] \\ &= \left[\frac{300000 - x}{100} \right] \end{aligned}$$

Also we are given that the annual interest equal to tk. 2780. We must have,

$$\begin{aligned} [2780] &= \left[\frac{300000 - x}{100} \right] \\ \Rightarrow 2780 &= \frac{300000 - x}{100} \\ \Rightarrow 300000 - x &= 278000 \\ \Rightarrow x &= 300000 - 278000 \\ \Rightarrow x &= 22000 \end{aligned}$$

$\therefore$ one bond is taka 22000

and another bond is taka $(50000 - 22000) = \text{tk. } 28000$ Answer

Covariance Matrix: Let the training set of M face images be $\Gamma_1, \Gamma_2, \dots, \Gamma_M$. The average face of the set is defined by:

$$\psi = \frac{1}{M} \sum_{n=1}^M \Gamma_n \text{-----(i)}$$

Each face differs from the average face by the vector $\phi_i = \Gamma_i - \psi$ where $i = 1$ to M

We shall rearrange these vectors in a matrix $A = [\phi_1 \dots \phi_M]$ of dimension $N \times M$, which will then be subject to PCA (Principal Component Analysis). The covariance Matrix is: $C = A.A^T$

Example:

Let image-1: $\begin{pmatrix} 4 & 6 \\ 3 & 2 \end{pmatrix}$

Let image-2: $\begin{pmatrix} 1 & 5 \\ 3 & 6 \end{pmatrix}$

Let image-3: $\begin{pmatrix} 2 & 2 \\ 4 & 9 \end{pmatrix}$

Note: MATLAB Code:

1. `i=imread('mrk.jpg')` then we will get a matrix whose entries are pixel values of the image
2. `i(:);` then all pixel values will be arranged in a column.

Then three images make a single matrix named Γ

Original faces (the training set), $\Gamma_i = \begin{pmatrix} 4 & 1 & 2 \\ 3 & 3 & 4 \\ 6 & 5 & 2 \\ 2 & 6 & 9 \end{pmatrix}$ Where $i = 1, 2, 3$

i.e. $\Gamma_1 = \begin{pmatrix} 4 \\ 3 \\ 6 \\ 2 \end{pmatrix}$, $\Gamma_2 = \begin{pmatrix} 1 \\ 3 \\ 5 \\ 6 \end{pmatrix}$, $\Gamma_3 = \begin{pmatrix} 2 \\ 4 \\ 2 \\ 9 \end{pmatrix}$

The average face of the set is defined by:

$$\psi = \frac{1}{M} \sum_{n=1}^M \Gamma_n, \quad \text{Here, } M = 3$$

$$\psi = \frac{1}{3} \sum_{n=1}^3 \Gamma_n = \frac{1}{3} (\Gamma_1 + \Gamma_2 + \Gamma_3) = \frac{1}{3} \begin{pmatrix} 4 \\ 3 \\ 6 \\ 2 \end{pmatrix} + \begin{pmatrix} 1 \\ 3 \\ 5 \\ 6 \end{pmatrix} \begin{pmatrix} 2 \\ 4 \\ 2 \\ 9 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} 7 \\ 10 \\ 13 \\ 17 \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \\ 4 \\ 6 \end{pmatrix}$$

Now,

$$\phi_i = \Gamma_i - \psi$$

$$\phi_1 = \Gamma_1 - \psi = \begin{pmatrix} 4 \\ 3 \\ 6 \\ 2 \end{pmatrix} - \begin{pmatrix} 2 \\ 3 \\ 4 \\ 6 \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \\ 2 \\ -4 \end{pmatrix}$$

$$\phi_2 = \Gamma_2 - \psi = \begin{pmatrix} 1 \\ 3 \\ 5 \\ 6 \end{pmatrix} - \begin{pmatrix} 2 \\ 3 \\ 4 \\ 6 \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \\ 1 \\ 0 \end{pmatrix}$$

$$\phi_3 = \Gamma_3 - \psi \quad \Gamma_3 = \begin{pmatrix} 2 \\ 4 \\ 2 \\ 9 \end{pmatrix} - \begin{pmatrix} 2 \\ 3 \\ 4 \\ 6 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ -2 \\ 3 \end{pmatrix}$$

$$\therefore A = [\phi_1 \dots \dots \dots \phi_M]$$

$$A = [\phi_1 \quad \phi_2 \quad \phi_3]$$

$$\therefore A = \begin{pmatrix} 2 & -1 & 0 \\ 0 & 0 & 1 \\ 2 & 1 & -2 \\ -4 & 0 & 3 \end{pmatrix}$$

$$\therefore A^T = \begin{pmatrix} 2 & 0 & 2 & -4 \\ -1 & 0 & 1 & 0 \\ 0 & 1 & -2 & 3 \end{pmatrix}$$

The covariance Matrix is:

$$C = A \cdot A^T = \begin{pmatrix} 2 & -1 & 0 \\ 0 & 0 & 1 \\ 2 & 1 & -2 \\ -4 & 0 & 3 \end{pmatrix} \begin{pmatrix} 2 & 0 & 2 & -4 \\ -1 & 0 & 1 & 0 \\ 0 & 1 & -2 & 3 \end{pmatrix} = \begin{pmatrix} 5 & 0 & 3 & -8 \\ 0 & 1 & -2 & 3 \\ 3 & -2 & 9 & -14 \\ -8 & 3 & -14 & 25 \end{pmatrix}$$

Orthogonal Matrix:

A matrix A is orthogonal if $AA^T = A^T A = I$. Where I is unit matrix and A^T is the transposed matrix of A.

Example 06: Show that the following matrix $A = \begin{pmatrix} \cos\alpha & \sin\alpha \\ -\sin\alpha & \cos\alpha \end{pmatrix}$ is orthogonal

Solution:

Given that,

$$A = \begin{pmatrix} \cos\alpha & \sin\alpha \\ -\sin\alpha & \cos\alpha \end{pmatrix}$$

$$\therefore A^T = \begin{pmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{pmatrix}$$

$$\therefore AA^T = \begin{pmatrix} \cos\alpha & \sin\alpha \\ -\sin\alpha & \cos\alpha \end{pmatrix} \begin{pmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{pmatrix}$$

$$\Rightarrow AA^T = \begin{pmatrix} \cos\alpha.\cos\alpha + \sin\alpha.\sin\alpha & -\cos\alpha.\sin\alpha + \sin\alpha.\cos\alpha \\ -\sin\alpha.\cos\alpha + \cos\alpha.\sin\alpha & \sin\alpha.\sin\alpha + \cos\alpha.\cos\alpha \end{pmatrix} \quad [\text{Row by column}]$$

$$\Rightarrow AA^T = \begin{pmatrix} \cos^2\alpha + \sin^2\alpha & -\cos\alpha.\sin\alpha + \sin\alpha.\cos\alpha \\ -\sin\alpha.\cos\alpha + \cos\alpha.\sin\alpha & \sin^2\alpha + \cos^2\alpha \end{pmatrix}$$

$$\Rightarrow AA^T = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I \text{ (Proved)}$$

Example 07: Prove $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{3}{5} & -\frac{4}{5} \\ 0 & \frac{4}{5} & \frac{3}{5} \end{bmatrix}$ **is orthogonal.**

$$\therefore A^T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{3}{5} & \frac{4}{5} \\ 0 & -\frac{4}{5} & \frac{3}{5} \end{bmatrix}$$

$$AA^T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{3}{5} & -\frac{4}{5} \\ 0 & \frac{4}{5} & \frac{3}{5} \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{3}{5} & \frac{4}{5} \\ 0 & -\frac{4}{5} & \frac{3}{5} \end{bmatrix}$$

$$AA^T = \begin{bmatrix} 1 \times 1 + 0 \times 0 + 0 \times 0 & 1 \times 0 + 0 \times \frac{3}{5} + 0 \times (-\frac{4}{5}) & 1 \times 0 + 0 \times \frac{4}{5} + 0 \times \frac{3}{5} \\ 0 \times 1 + \frac{3}{5} \times 0 + (-\frac{4}{5}) \times 0 & 0 \times 0 + \frac{3}{5} \times \frac{3}{5} - \frac{4}{5} \times (-\frac{4}{5}) & 0 \times 0 + \frac{3}{5} \times \frac{4}{5} - \frac{4}{5} \times \frac{3}{5} \\ 0 \times 1 + \frac{4}{5} \times 0 + \frac{3}{5} \times 0 & 0 \times 0 + \frac{4}{5} \times \frac{3}{5} + \frac{3}{5} \times (-\frac{4}{5}) & 0 \times 0 + \frac{4}{5} \times \frac{4}{5} + \frac{3}{5} \times \frac{3}{5} \end{bmatrix}$$

$$AA^T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 + \frac{9}{25} + \frac{16}{25} & 0 + \frac{12}{25} - \frac{12}{25} \\ 0 & 0 + \frac{12}{25} - \frac{12}{25} & 0 + \frac{16}{25} + \frac{9}{25} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 + \frac{25}{25} & 0 + 0 \\ 0 & 0 + 0 & 0 + \frac{25}{25} \end{bmatrix}$$

$$\mathbf{A}\mathbf{A}^T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \mathbf{I}$$

$$\therefore \mathbf{A} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{3}{5} & -\frac{4}{5} \\ 0 & \frac{4}{5} & \frac{3}{5} \end{bmatrix} \text{ is orthogonal matrix.}$$

Adjoint of a Matrix:

$$\mathbf{A} = (\mathbf{a}_{ij}) = \begin{pmatrix} \mathbf{a}_{11} & \mathbf{a}_{12} & \mathbf{a}_{13} & - & - & - & - & - & \mathbf{a}_{1n} \\ \mathbf{a}_{21} & \mathbf{a}_{22} & \mathbf{a}_{23} & - & - & - & - & - & \mathbf{a}_{2n} \\ \mathbf{a}_{31} & \mathbf{a}_{32} & \mathbf{a}_{33} & - & - & - & - & - & \mathbf{a}_{3n} \\ - & - & - & - & - & - & - & - & - \\ \mathbf{a}_{m1} & \mathbf{a}_{m2} & \mathbf{a}_{m3} & - & - & - & - & - & \mathbf{a}_{mn} \end{pmatrix}$$

Let $\mathbf{c}_{ij}$ be the cofactor of the element $\mathbf{a}_{ij}$ in $|\mathbf{A}| = |\mathbf{a}_{ij}|$ of the $\mathbf{m} \times \mathbf{n}$ matrix $\mathbf{A} = (\mathbf{a}_{ij})$. Then

$$\mathbf{A}^c = (\mathbf{c}_{ij}) = \begin{pmatrix} \mathbf{c}_{11} & \mathbf{c}_{12} & \mathbf{c}_{13} & - & - & - & - & - & \mathbf{c}_{1n} \\ \mathbf{c}_{21} & \mathbf{c}_{22} & \mathbf{c}_{23} & - & - & - & - & - & \mathbf{c}_{2n} \\ \mathbf{c}_{31} & \mathbf{c}_{32} & \mathbf{c}_{33} & - & - & - & - & - & \mathbf{c}_{3n} \\ - & - & - & - & - & - & - & - & - \\ \mathbf{c}_{m1} & \mathbf{c}_{m2} & \mathbf{c}_{m3} & - & - & - & - & - & \mathbf{c}_{mn} \end{pmatrix}$$

[$\mathbf{c}_{11}$ =cofactor of $\mathbf{a}_{11}$, $\mathbf{c}_{12}$ =cofactor of $\mathbf{a}_{12}$].

$$\text{Adjoint of } \mathbf{A} = (\mathbf{A}^c)^t = \begin{pmatrix} \mathbf{c}_{11} & \mathbf{c}_{21} & \mathbf{c}_{31} & - & - & - & - & - & \mathbf{c}_{m1} \\ \mathbf{c}_{12} & \mathbf{c}_{22} & \mathbf{c}_{32} & - & - & - & - & - & \mathbf{c}_{m2} \\ \mathbf{c}_{13} & \mathbf{c}_{23} & \mathbf{c}_{33} & - & - & - & - & - & \mathbf{c}_{m3} \\ - & - & - & - & - & - & - & - & - \\ \mathbf{c}_{1n} & \mathbf{c}_{2n} & \mathbf{c}_{3n} & - & - & - & - & - & \mathbf{c}_{mn} \end{pmatrix}$$

Let,

$$A = (a_{ij}) = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \text{-----(i) [Where } i = 1,2,3 \text{ \& } j = 1,2,3]$$

$$A = (a_{ij}) = \begin{pmatrix} + & - & + \\ - & + & - \\ + & - & + \end{pmatrix}$$

$$\therefore \text{Co-factor of } a_{11} = c_{11} = + \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix}$$

$$\therefore \text{Co-factor of } a_{12} = c_{12} = - \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix}$$

$$\therefore \text{Co-factor of } a_{13} = c_{13} = + \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

$$\therefore \text{Co-factor of } a_{21} = c_{21} = - \begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} \end{vmatrix}$$

$$\therefore \text{Co-factor of } a_{22} = c_{22} = + \begin{vmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{vmatrix}$$

$$\therefore \text{Co-factor of } a_{23} = c_{23} = - \begin{vmatrix} a_{11} & a_{12} \\ a_{31} & a_{32} \end{vmatrix}$$

$$\therefore \text{Co-factor of } a_{31} = c_{31} = + \begin{vmatrix} a_{12} & a_{13} \\ a_{22} & a_{23} \end{vmatrix}$$

$$\therefore \text{Co-factor of } a_{32} = c_{32} = - \begin{vmatrix} a_{11} & a_{13} \\ a_{21} & a_{23} \end{vmatrix}$$

$$\therefore \text{Co-factor of } a_{33} = c_{33} = + \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}$$

$$\text{Co-factor Matrix } A^c = (c_{ij}) = \begin{pmatrix} c_{11} & c_{12} & c_{13} \\ c_{21} & c_{22} & c_{23} \\ c_{31} & c_{32} & c_{33} \end{pmatrix}$$

$$\text{Transpose of a Matrix: } (A^c)^t = (c_{ji}) = \begin{pmatrix} c_{11} & c_{21} & c_{31} \\ c_{12} & c_{22} & c_{32} \\ c_{13} & c_{23} & c_{33} \end{pmatrix}$$

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \begin{pmatrix} + & - & + \\ - & + & - \\ + & - & + \end{pmatrix}$$

Hence Adjoint Matrix of A =
$$\begin{pmatrix} c_{11} & c_{21} & c_{31} \\ c_{12} & c_{22} & c_{32} \\ c_{13} & c_{23} & c_{33} \end{pmatrix}$$

Example 08: Find the adjoint of a matrix A =
$$\begin{bmatrix} 1 & 0 & -1 \\ 3 & 4 & 5 \\ 0 & -6 & -7 \end{bmatrix}$$

We have,

$$A = (a_{ij}) = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \text{ Where } i = 1,2,3 \text{ and Where } j = 1,2,3$$

$$A = (a_{ij}) = \begin{pmatrix} + & - & + \\ - & + & - \\ + & - & + \end{pmatrix}$$

$$\therefore \text{Co-factor of } a_{11} = c_{11} = + \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix}$$

$$\therefore \text{Co-factor of } a_{12} = c_{12} = - \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix}$$

$$\therefore \text{Co-factor of } a_{13} = c_{13} = + \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

$$\therefore \text{Co-factor of } a_{21} = c_{21} = - \begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} \end{vmatrix}$$

$$\therefore \text{Co-factor of } a_{22} = c_{22} = + \begin{vmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{vmatrix}$$

$$\therefore \text{Co-factor of } a_{23} = c_{23} = - \begin{vmatrix} a_{11} & a_{12} \\ a_{31} & a_{32} \end{vmatrix}$$

$$\therefore \text{Co-factor of } a_{31} = c_{31} = + \begin{vmatrix} a_{12} & a_{13} \\ a_{22} & a_{23} \end{vmatrix}$$

$$\therefore \text{Co-factor of } a_{32} = c_{32} = - \begin{vmatrix} a_{11} & a_{13} \\ a_{21} & a_{23} \end{vmatrix}$$

$$\therefore \text{Co-factor of } a_{33} = c_{33} = + \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}$$

Here, given, $A = \begin{bmatrix} 1 & 0 & -1 \\ 3 & 4 & 5 \\ 0 & -6 & -7 \end{bmatrix}$ -----(i)

We know $A = (a_{ij}) = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$ -----(ii)

Comparing (i) and (ii)

Here,

$$a_{11} = 1, a_{12} = 0, a_{13} = -1, a_{21} = 3, a_{22} = 4, a_{23} = 5, a_{31} = 0, a_{32} = -6, a_{33} = -7$$

Now,

$$c_{11} = + \begin{vmatrix} 4 & 5 \\ -6 & -7 \end{vmatrix} = 2 \quad c_{12} = - \begin{vmatrix} 3 & 5 \\ 0 & -7 \end{vmatrix} = 21 \quad c_{13} = + \begin{vmatrix} 3 & 4 \\ 0 & -6 \end{vmatrix} = -18$$

$$c_{21} = - \begin{vmatrix} 0 & -1 \\ -6 & -7 \end{vmatrix} = 6 \quad c_{22} = + \begin{vmatrix} 1 & -1 \\ 0 & -7 \end{vmatrix} = -7 \quad c_{23} = - \begin{vmatrix} 1 & 0 \\ 0 & -6 \end{vmatrix} = 6$$

$$c_{31} = + \begin{vmatrix} 0 & -1 \\ 4 & 5 \end{vmatrix} = 4 \quad c_{32} = - \begin{vmatrix} 1 & -1 \\ 3 & 5 \end{vmatrix} = -8 \quad c_{33} = + \begin{vmatrix} 1 & 0 \\ 3 & 4 \end{vmatrix} = 4$$

$$\text{Co-factor Matrix } A^c = (c_{ij}) = \begin{pmatrix} c_{11} & c_{12} & c_{13} \\ c_{21} & c_{22} & c_{23} \\ c_{31} & c_{32} & c_{33} \end{pmatrix} = \begin{bmatrix} 2 & 21 & -18 \\ 6 & -7 & 6 \\ 4 & -8 & 4 \end{bmatrix}$$

$$\therefore \text{Adj. } A = (A^c)^t = \begin{bmatrix} 2 & 6 & 4 \\ 21 & -7 & -8 \\ -18 & 6 & 4 \end{bmatrix}$$

Inverse Matrix:

The inverse of a matrix, $A^{-1} = \frac{\text{Adjoint of } A}{|A|}$

$|A| \rightarrow$ The determinant of the matrix A.

Example 09: Find the Inverse of $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix}$ where $|A| \neq 0$

Solution:

$$|A| = 1(4 \times 6 - 5 \times 5) - 2(2 \times 6 - 3 \times 5) + 3(2 \times 5 - 3 \times 4)$$

$$|A| = 1(24 - 25) - 2(12 - 15) + 3(10 - 12)$$

$$= 1(-1) - 2(-3) + 3(-2)$$

$$= -1 + 6 - 6$$

$$= -1$$

We have

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix}$$

$$A = (a_{ij}) = \begin{pmatrix} + & - & + \\ - & + & - \\ + & - & + \end{pmatrix}$$

Now,

$$c_{11} = + \begin{vmatrix} 4 & 5 \\ 5 & 6 \end{vmatrix} = -1 \quad c_{12} = - \begin{vmatrix} 2 & 5 \\ 3 & 6 \end{vmatrix} = 3 \quad c_{13} = + \begin{vmatrix} 2 & 4 \\ 3 & 5 \end{vmatrix} = -2$$

$$c_{21} = - \begin{vmatrix} 2 & 3 \\ 5 & 6 \end{vmatrix} = 3 \quad c_{22} = + \begin{vmatrix} 1 & 3 \\ 3 & 6 \end{vmatrix} = -3 \quad c_{23} = - \begin{vmatrix} 1 & 2 \\ 3 & 5 \end{vmatrix} = 1$$

$$c_{31} = + \begin{vmatrix} 2 & 3 \\ 4 & 5 \end{vmatrix} = -2 \quad c_{32} = - \begin{vmatrix} 1 & 3 \\ 2 & 5 \end{vmatrix} = 1 \quad c_{33} = + \begin{vmatrix} 1 & 2 \\ 2 & 4 \end{vmatrix} = 0$$

$$A^c = \begin{pmatrix} -1 & 3 & -2 \\ 3 & -3 & 1 \\ -2 & 1 & 0 \end{pmatrix}$$

$$\text{Adjo int of } A = (A^c)^t = \begin{pmatrix} -1 & 3 & -2 \\ 3 & -3 & 1 \\ -2 & 1 & 0 \end{pmatrix}$$

$$A^{-1} = \frac{\text{Adjo int of } A}{|A|} = \frac{1}{-1} \begin{pmatrix} -1 & 3 & -2 \\ 3 & -3 & 1 \\ -2 & 1 & 0 \end{pmatrix} = -1 \begin{pmatrix} -1 & 3 & -2 \\ 3 & -3 & 1 \\ -2 & 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & -3 & 2 \\ -3 & 3 & -1 \\ 2 & -1 & 0 \end{pmatrix}$$

Perfection Test:

$$AA^{-1} = I$$

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix} ; A^{-1} = \begin{pmatrix} 1 & -3 & 2 \\ -3 & 3 & -1 \\ 2 & -1 & 0 \end{pmatrix}$$

$$AA^{-1} = \begin{bmatrix} 1 \times 1 + 2 \times (-3) + 3 \times 2 & 1 \times (-3) + 2 \times 3 + 3 \times (-1) & 1 \times 2 + 2 \times (-1) + 3 \times 0 \\ 2 \times 1 + 4 \times (-3) + 5 \times 2 & 2 \times (-3) + 4 \times 3 + 5 \times (-1) & 2 \times 2 + 4 \times (-1) + 5 \times 0 \\ 3 \times 1 + 5 \times (-3) + 6 \times 2 & 3 \times (-3) + 5 \times 3 + 6 \times (-1) & 3 \times 2 + 5 \times (-1) + 6 \times 0 \end{bmatrix}$$

$$AA^{-1} = \begin{bmatrix} 1-6+6 & -3+6-3 & 2-2+0 \\ 2-12+10 & -6+12-5 & 4-4+0 \\ 3-15+12 & -9+15-6 & 6-5+0 \end{bmatrix}$$

$$AA^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I$$

Example 10: Find the inverse Matrix of

$$A = \begin{pmatrix} a+ib & c+id \\ -c+id & a-ib \end{pmatrix} \text{ Where } [a^2 + b^2 + c^2 + d^2 = 1]$$

Here, $a_{11} = a+ib$, $a_{12} = c+id$, $a_{21} = -c+id$, $a_{22} = a-ib$

$$\therefore c_{11} = \text{cofactor of } a+ib = +(a-ib)$$

$$c_{12} = \text{cofactor of } c+id = -(-c+id)$$

$$c_{21} = \text{cofactor of } -c+id = -(c+id)$$

$$c_{22} = \text{cofactor of } a-ib = +(a+ib)$$

So, $c_{11} = a-ib$, $c_{12} = c-id$, $c_{21} = -c-id$, $c_{22} = a+ib$

$$A^C = \begin{pmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{pmatrix} = \begin{pmatrix} a-ib & c-id \\ -c-id & a+ib \end{pmatrix}$$

$$\text{Adj}A = (A^C)^t = \begin{pmatrix} a-ib & -c-id \\ c-id & a+ib \end{pmatrix}$$

$$\text{Given } A = \begin{pmatrix} a+ib & c+id \\ -c+id & a-ib \end{pmatrix}$$

$$\begin{aligned} \therefore |A| &= \begin{vmatrix} a+ib & c+id \\ -c+id & a-ib \end{vmatrix} = (a+ib)(a-ib) - (c+id)(-c+id) \\ &= (a^2 - iab + iab - i^2b^2) - (-c^2 + icd - icd + i^2d^2) \\ &= (a^2 - iab + iab + b^2) - (-c^2 + icd - icd - d^2) [i^2 = -1] \\ &= (a^2 + b^2 + c^2 + d^2) \\ &= 1 [a^2 + b^2 + c^2 + d^2 = 1] \end{aligned}$$

$$\therefore A^{-1} = \frac{\text{Adj}A}{|A|} = \frac{1}{1} \begin{pmatrix} a-ib & -c-id \\ c-id & a+ib \end{pmatrix} = \begin{pmatrix} a-ib & -c-id \\ c-id & a+ib \end{pmatrix}$$

$$\begin{array}{c} \begin{vmatrix} a+ib & c+id \\ -c+id & a-ib \end{vmatrix} \\ \begin{pmatrix} + & - \\ - & + \end{pmatrix} \end{array}$$

Perfection Test:

We have,

$$\begin{aligned}
 A &= \begin{pmatrix} a+ib & c+id \\ -c+id & a-ib \end{pmatrix} \text{ and } A^{-1} = \begin{pmatrix} a-ib & -c-id \\ c-id & a+ib \end{pmatrix} \\
 AA^{-1} &= \begin{pmatrix} a+ib & c+id \\ -c+id & a-ib \end{pmatrix} \begin{pmatrix} a-ib & -c-id \\ c-id & a+ib \end{pmatrix} \\
 &= \begin{pmatrix} (a+ib)(a-ib) + (c+id)(c-id) & (a+ib)(-c-id) + (c+id)(a+ib) \\ (-c+id)(a-ib) + (a-ib)(c-id) & (-c+id)(-c-id) + (a-ib)(a+ib) \end{pmatrix} \\
 &= \begin{pmatrix} a^2 - (ib)^2 + c^2 - (id)^2 & -ac - iad - ibc + bd + ac + ibc + iad - bd \\ -ac + ibc + iad + bd + ac - iad - ibc - bd & (-c)^2 - (id)^2 + (a)^2 - (ib)^2 \end{pmatrix} \\
 &= \begin{pmatrix} a^2 + b^2 + c^2 + d^2 & 0 \\ 0 & a^2 + b^2 + c^2 + d^2 \end{pmatrix} \quad [i^2 = -1] \\
 &= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I \quad [a^2 + b^2 + c^2 + d^2 = 1]
 \end{aligned}$$

Characteristics Matrix of 'A':

The Matrix $A - \lambda I$ is called the Characteristics matrix of 'A' where I is the Identity matrix

$$\text{Let, } A = \begin{pmatrix} 1 & 2 \\ 3 & 2 \end{pmatrix}$$

$$A - \lambda I = \begin{pmatrix} 1 & 2 \\ 3 & 2 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad [I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}]$$

$$A - \lambda I = \begin{pmatrix} 1 & 2 \\ 3 & 2 \end{pmatrix} - \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix}$$

$$A - \lambda I = \begin{pmatrix} 1-\lambda & 2 \\ 3 & 2-\lambda \end{pmatrix} ; \text{ Which is called Characteristics Matrix of A}$$

Characteristics Polynomial of 'A':

The determinant $|A - \lambda I|$ is called the Characteristics Polynomial of A

$$\begin{aligned}
 |A - \lambda I| &= \begin{vmatrix} 1-\lambda & 2 \\ 3 & 2-\lambda \end{vmatrix} \\
 &= (1-\lambda)(2-\lambda) - 6 = 2 - \lambda - 2\lambda + \lambda^2 - 6 \\
 &= \lambda^2 - 3\lambda - 4 ; \text{ which is polynomial of degree 2}
 \end{aligned}$$

Characteristics Equation of ‘A’:

The equation $|A - \lambda I| = 0$ is known as the Characteristics equation of ‘A’

$$\begin{aligned}
 |A - \lambda I| &= \begin{vmatrix} 1-\lambda & 2 \\ 3 & 2-\lambda \end{vmatrix} = 0 \\
 &= (1-\lambda)(2-\lambda) - 6 = 0 = 2 - \lambda - 2\lambda + \lambda^2 - 6 = 0 \\
 &= \lambda^2 - 3\lambda - 4 = 0; \text{ which is Characteristic equation of the matrix } A
 \end{aligned}$$

Characteristics root or latent roots or Eigen values:

Its roots are called Characteristics root or latent roots or Eigen values of Characteristics values.

$$\begin{aligned}
 |A - \lambda I| &= \begin{vmatrix} 1-\lambda & 2 \\ 3 & 2-\lambda \end{vmatrix} = 0 \\
 &= (1-\lambda)(2-\lambda) - 6 = 0 = 2 - \lambda - 2\lambda + \lambda^2 - 6 = 0 \\
 &= \lambda^2 - 3\lambda - 4 = 0
 \end{aligned}$$

$\lambda = 4, -1$; These are Characteristics root or latent roots or Eigen values of A.

Spectrum of A:

The set of all the Eigen values of matrix is called the spectrum of ‘A’.

$\{4, -1\}$ is the spectrum of A.

The set of eigenvalues is called the spectrum of A in analogy with the idea of the spectrum of light and the composed by various wavelengths with different intensity. For this reason, the procedure is often referred to as a spectral decomposition. If two or more eigenvalues of A are identical, the spectrum of the matrix is called degenerate.

Example 11: Find the Characteristics root or latent roots or Eigen values of a matrix

$$\begin{aligned}
 A &= \begin{pmatrix} \cos\theta & -\sin\theta \\ -\sin\theta & -\cos\theta \end{pmatrix} \\
 A - \lambda I &= \begin{pmatrix} \cos\theta & -\sin\theta \\ -\sin\theta & -\cos\theta \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \\
 A - \lambda I &= \begin{pmatrix} \cos\theta & -\sin\theta \\ -\sin\theta & -\cos\theta \end{pmatrix} - \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix} \\
 A - \lambda I &= \begin{pmatrix} \cos\theta - \lambda & -\sin\theta \\ -\sin\theta & -\cos\theta - \lambda \end{pmatrix} \\
 \therefore |A - \lambda I| &= \begin{vmatrix} \cos\theta - \lambda & -\sin\theta \\ -\sin\theta & -\cos\theta - \lambda \end{vmatrix} \\
 &= (\cos\theta - \lambda)(-\cos\theta - \lambda) - (-\sin\theta)(-\sin\theta) \\
 &= -\cos^2\theta - \lambda\cos\theta + \lambda\cos\theta + \lambda^2 - \sin^2\theta \\
 &= \lambda^2 - (\cos^2\theta + \sin^2\theta)
 \end{aligned}$$

$$= \lambda^2 - 1 [\cos^2 \theta + \sin^2 \theta = 1]$$

$$\therefore |A - \lambda I| = \lambda^2 - 1 = 0$$

$$\Rightarrow \lambda^2 - 1 = 0$$

$$\Rightarrow \lambda^2 = 1$$

$$\Rightarrow \lambda = \pm 1$$

Linear System of Equations:

Let a system of m linear equations in n unknowns $x_1, x_2, \dots, x_n$ has the general form:

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \dots + a_{1n}x_n = b_1$$

$$a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + \dots + a_{2n}x_n = b_2$$

$$a_{31}x_1 + a_{32}x_2 + a_{33}x_3 + \dots + a_{3n}x_n = b_3$$

$$\dots$$

$$\dots$$

$$\dots$$

$$a_{m1}x_1 + a_{m2}x_2 + a_{m3}x_3 + \dots + a_{mn}x_n = b_m$$

Then this system can be written in the matrix form as $AX = b$

$$\text{Where, } A = \begin{pmatrix} a_{11} & a_{12} & a_{13} & - & - & - & - & a_{1n} \\ a_{21} & a_{22} & a_{23} & - & - & - & - & a_{2n} \\ a_{31} & a_{32} & a_{33} & - & - & - & - & a_{3n} \\ - & - & - & - & - & - & - & - \\ a_{m1} & a_{m2} & a_{m3} & - & - & - & - & a_{mn} \end{pmatrix}; \quad X = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ - \\ x_n \end{pmatrix}; \quad b = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \\ - \\ b_n \end{pmatrix}$$

We have, $AX = b$

Multiplying by A^{-1} on both sides

$$AA^{-1}X = A^{-1}b$$

$$\Rightarrow IX = A^{-1}b \quad [AA^{-1} = I]$$

$$\Rightarrow X = A^{-1}b \quad [IX = X]$$

Example:

$$\text{Let, } I = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad X = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$\therefore IX = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \times x + 0 \times y + 0 \times z \\ 0 \times x + 1 \times y + 0 \times z \\ 0 \times x + 0 \times y + 1 \times z \end{pmatrix}$$

$$= \begin{pmatrix} x \\ y \\ z \end{pmatrix} \\ = \mathbf{X} \\ \therefore \mathbf{IX} = \mathbf{X}$$

Example 12: Solve by Matrix method:

$$x + 2y + 3z = 14$$

$$3x + y + 2z = 11$$

$$2x + 3y + z = 11$$

Let the system can be written $\mathbf{AX} = \mathbf{b}$ where

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \\ 2 & 3 & 1 \end{bmatrix}; \quad \mathbf{X} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}; \quad \mathbf{b} = \begin{bmatrix} 14 \\ 11 \\ 11 \end{bmatrix}$$

We have, $\mathbf{AX} = \mathbf{b}$

Multiplying by $\mathbf{A}^{-1}$ on both sides

$$\mathbf{AA}^{-1}\mathbf{X} = \mathbf{A}^{-1}\mathbf{b}$$

$$\Rightarrow \mathbf{IX} = \mathbf{A}^{-1}\mathbf{b} \quad [\mathbf{AA}^{-1} = \mathbf{I}]$$

$$\Rightarrow \mathbf{X} = \mathbf{A}^{-1}\mathbf{b} \quad [\mathbf{IX} = \mathbf{X}]$$

$$\Rightarrow \mathbf{X} = \mathbf{A}^{-1}\mathbf{b}$$

$$\Rightarrow \mathbf{X} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \mathbf{A}^{-1}\mathbf{b} \quad \text{-----(i)} \quad \left[\mathbf{X} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \right]$$

Also we have, $\mathbf{A}^{-1} = \frac{\text{Adj}\mathbf{A}}{|\mathbf{A}|}$

$$\text{We have, } \mathbf{A} = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \\ 2 & 3 & 1 \end{bmatrix}$$

$$\mathbf{A} = (\mathbf{a}_{ij}) = \begin{pmatrix} + & - & + \\ - & + & - \\ + & - & + \end{pmatrix}$$

Now,

$$\mathbf{c}_{11} = + \begin{vmatrix} 1 & 2 \\ 3 & 1 \end{vmatrix} = -5; \quad \mathbf{c}_{12} = - \begin{vmatrix} 3 & 2 \\ 2 & 1 \end{vmatrix} = 1; \quad \mathbf{c}_{13} = + \begin{vmatrix} 3 & 1 \\ 2 & 3 \end{vmatrix} = 7;$$

$$\mathbf{c}_{21} = - \begin{vmatrix} 2 & 3 \\ 3 & 1 \end{vmatrix} = 7; \quad \mathbf{c}_{22} = + \begin{vmatrix} 1 & 3 \\ 2 & 1 \end{vmatrix} = -5; \quad \mathbf{c}_{23} = - \begin{vmatrix} 1 & 2 \\ 2 & 3 \end{vmatrix} = 1;$$

$$\mathbf{c}_{31} = + \begin{vmatrix} 2 & 3 \\ 1 & 2 \end{vmatrix} = 1; \quad \mathbf{c}_{32} = - \begin{vmatrix} 1 & 3 \\ 3 & 2 \end{vmatrix} = 7; \quad \mathbf{c}_{33} = + \begin{vmatrix} 1 & 2 \\ 3 & 1 \end{vmatrix} = -5;$$

The co-factor Matrix of A: $A^C = \begin{bmatrix} -5 & 1 & 7 \\ 7 & -5 & 1 \\ 1 & 7 & -5 \end{bmatrix}$

$$\text{Adj. } A = (A^C)^T = \begin{bmatrix} -5 & 7 & 1 \\ 1 & -5 & 7 \\ 7 & 1 & -5 \end{bmatrix}$$

$$\begin{aligned} |A| &= \begin{vmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \\ 2 & 3 & 1 \end{vmatrix} = 1(1 \times 1 - 2 \times 3) - 2(3 \times 1 - 2 \times 2) + 3(3 \times 3 - 2 \times 1) \\ &= 1 \times (-5) - 2 \times (-1) + 3 \times 7 \\ &= -5 + 2 + 21 \\ &= 18 \end{aligned}$$

$$A^{-1} = \frac{\text{Adj}A}{|A|} = \frac{1}{18} \begin{bmatrix} -5 & 7 & 1 \\ 1 & -5 & 7 \\ 7 & 1 & -5 \end{bmatrix}$$

From (i),

$$\Rightarrow X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = A^{-1}b$$

$$\therefore X = A^{-1}b = \frac{1}{18} \begin{bmatrix} -5 & 7 & 1 \\ 1 & -5 & 7 \\ 7 & 1 & -5 \end{bmatrix} \begin{bmatrix} 14 \\ 11 \\ 11 \end{bmatrix}$$

$$X = \frac{1}{18} \begin{bmatrix} -5 \times 14 + 7 \times 11 + 1 \times 11 \\ 1 \times 14 - 5 \times 11 + 7 \times 11 \\ 7 \times 14 + 1 \times 11 - 5 \times 11 \end{bmatrix}$$

$$X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{18} \begin{bmatrix} -70 + 77 + 11 \\ 14 - 55 + 77 \\ 98 + 11 - 55 \end{bmatrix}$$

$$X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{18} \begin{bmatrix} 18 \\ 36 \\ 54 \end{bmatrix}$$

$$X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

$$\therefore \mathbf{X} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

$$\Rightarrow x = 1, y = 2, z = 3 \text{ Answer}$$

Row echelon form

Definition: A matrix is in row echelon form if

1. The leading term in any nonzero row is 1 and all terms below this leading 1 are zero.
2. The leading 1 in any non-zero row occurs to the right of the leading 1 in any previous row
3. The non-zero rows appear before any zero rows.
4. All rows of zeros are at the bottom of the matrix

Example:

$$\begin{bmatrix} 1 & 0 & -2 & 3 & 0 & -24 \\ 0 & 1 & -2 & 2 & 0 & -7 \\ 0 & 0 & 0 & 0 & 1 & 4 \end{bmatrix}, \begin{bmatrix} 1 & a_0 & a_1 & a_2 & a_3 \\ 0 & 0 & 2 & a_4 & a_5 \\ 0 & 0 & 0 & 1 & a_6 \end{bmatrix}, \begin{bmatrix} 1 & 1 & 3 & 3 \\ 0 & 1 & 0 & 4 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Rank of a Matrix:

The number of non zero rows of a matrix given in the echelon form is its rank.

The number of linear independent rows of the matrix is called its rank.

Example:

$$2x + 3y + z = 1$$

$$4x + 5y + 7z = 9$$

$$x + 9y + z = 0$$

$$4x + 6y + 2z = 2 \text{ -----Dependent}$$

Example 13: Find the rank of the matrix $A = \begin{bmatrix} 0 & 2 & 3 \\ 0 & 4 & 6 \\ 0 & 6 & 9 \end{bmatrix}$

$$A \sim \begin{bmatrix} 0 & 2 & 1 \\ 0 & 4 & 2 \\ 0 & 6 & 3 \end{bmatrix} \quad [C'_3 = C_3 - C_2]$$

$$A \sim \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 2 \\ 0 & 0 & 3 \end{bmatrix} \quad [C'_2 = C_2 - 2C_3]$$

$$A \sim \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad [r'_2 = r_2 - 2r_1]; [r'_3 = r_3 - 3r_1]$$

$$A \sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Then the rank of matrix is one. [1 non zero row]

Example 14: Find the rank of the matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 2 & 1 & 2 \end{bmatrix}$

$$A \sim \begin{bmatrix} 1 & 1 & 3 \\ 4 & 1 & 6 \\ 2 & 1 & 2 \end{bmatrix} \quad [C'_2 = C_3 - C_2]$$

$$A \sim \begin{bmatrix} 3 & 0 & 3 \\ 4 & 1 & 6 \\ 2 & 0 & 4 \end{bmatrix} \quad [r'_1 = r_2 - r_1]; [r'_3 = r_2 - r_3]$$

$$A \sim \begin{bmatrix} 0 & 0 & 3 \\ 2 & 1 & 6 \\ 2 & 0 & 4 \end{bmatrix} \quad [C'_1 = C_3 - C_1]$$

$$A \sim \begin{bmatrix} 0 & 0 & 3 \\ 0 & 1 & 6 \\ -2 & 0 & 4 \end{bmatrix} \quad [C'_1 = C_3 - 3C_1]$$

$$A \sim \begin{bmatrix} -2 & 0 & 4 \\ 0 & 1 & 6 \\ 0 & 0 & 3 \end{bmatrix}$$

Then the rank of matrix is 3.

Example 15: Let, $I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ Given, $E = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$

Prove that, $(cI + bE)^3 = c^3I + 3c^2bE$

L.H.S.

$$cI = c \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} c & 0 \\ 0 & c \end{pmatrix}$$

$$bE = b \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & b \\ 0 & 0 \end{pmatrix}$$

$$\therefore cI + bE = \begin{pmatrix} c & 0 \\ 0 & c \end{pmatrix} + \begin{pmatrix} 0 & b \\ 0 & 0 \end{pmatrix}$$

$$\begin{aligned}
\Rightarrow cI + bE &= \begin{pmatrix} c & b \\ 0 & c \end{pmatrix} \\
\Rightarrow (cI + bE)^2 &= \begin{pmatrix} c & b \\ 0 & c \end{pmatrix} \begin{pmatrix} c & b \\ 0 & c \end{pmatrix} \\
\Rightarrow (cI + bE)^2 &= \begin{pmatrix} c \times c + b \times 0 & c \times b + b \times c \\ 0 \times c + c \times 0 & 0 \times b + c \times c \end{pmatrix} \\
\Rightarrow (cI + bE)^2 &= \begin{pmatrix} c^2 & 2bc \\ 0 & c^2 \end{pmatrix} \\
\Rightarrow (cI + bE)^2 (cI + bE) &= \begin{pmatrix} c^2 & 2bc \\ 0 & c^2 \end{pmatrix} \begin{pmatrix} c & b \\ 0 & c \end{pmatrix} \\
\Rightarrow (cI + bE)^3 &= \begin{pmatrix} c^2 & 2bc \\ 0 & c^2 \end{pmatrix} \begin{pmatrix} c & b \\ 0 & c \end{pmatrix} \\
\Rightarrow (cI + bE)^3 &= \begin{pmatrix} c^2 \times c + 2bc \times 0 & c^2 \times b + 2bc \times c \\ 0 \times c + c^2 \times 0 & 0 \times b + c^2 \times c \end{pmatrix} \\
\Rightarrow (cI + bE)^3 &= \begin{pmatrix} c^3 & 3bc^2 \\ 0 & c^3 \end{pmatrix} \\
\text{R.H.S. } c^3I + 3c^2bE &= c^3 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + 3c^2b \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \\
&= \begin{pmatrix} c^3 & 0 \\ 0 & c^3 \end{pmatrix} + \begin{pmatrix} 0 & 3c^2b \\ 0 & 0 \end{pmatrix} \\
&= \begin{pmatrix} c^3 & 3c^2b \\ 0 & c^3 \end{pmatrix} \text{ (Proved)}
\end{aligned}$$

Linear Equation:

By a linear equation over the real field 'R' we meant an expression $a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \dots + a_{1n}x_n = b$ where $a_i b_i \in \mathbf{R}$ and x_i are in determinants (or unknown or variables).

Example: $2x_1 + 3x_2 + 4x_3 = 3$

System of linear equations:

We now consider a system of m linear equations in the n unknown $x_1, x_2, x_3, \dots, x_n$

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \dots + a_{1n}x_n = b_1$$

$$a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + \dots + a_{2n}x_n = b_2$$

$$a_{31}x_1 + a_{32}x_2 + a_{33}x_3 + \dots + a_{3n}x_n = b_3$$

$$\dots$$

$$\dots$$

$$\dots$$

$$a_{m1}x_1 + a_{m2}x_2 + a_{m3}x_3 + \dots + a_{mn}x_n = b_m$$

Where $a_i, b_i \in \mathbb{R}$.

The system is said to be homogeneous. If the constants $b_1, b_2, b_3, \dots, b_m$ are all zero. Otherwise it is called non-homogeneous system of equations

Example: $2x + 3y + z = 7$

$4x + 5y + 7z = 9 \longrightarrow$ Non-homogeneous system of Linear equations

$$x + 9y + z = 0$$

Solution of a System of Linear Equations:

Step 1: Interchange equations so that the first unknown x_1 has a nonzero coefficient in the first equation, that is, so that $a_{11} \neq 0$

Step 2: To reduce it to a simpler system for each $i > 1$ applies the operation:

$$L_i \rightarrow -a_{i1}L_1 + a_{11}L_i$$

Step 3: If an equation $0 \times x_1 + 0 \times x_2 + 0 \times x_3 + \dots + 0 \times x_n = b, b \neq 0$ occurs the System is inconsistent and has no solution.

Step 4: If an equation $0 \times x_1 + 0 \times x_2 + 0 \times x_3 + \dots + 0 \times x_n = 0$ occurs, then the Equation can be deleted without affecting the solution.

Step 5: Solution of non-homogeneous system of equation.

- i. If $r = n$, that is there are as many equations as unknowns then the system has a unique solution.
- ii. If $r < n$, that is there are fewer equations than unknowns then we can arbitrarily assign values to the $n - r$ free variables and obtain more than one solution. of the system.

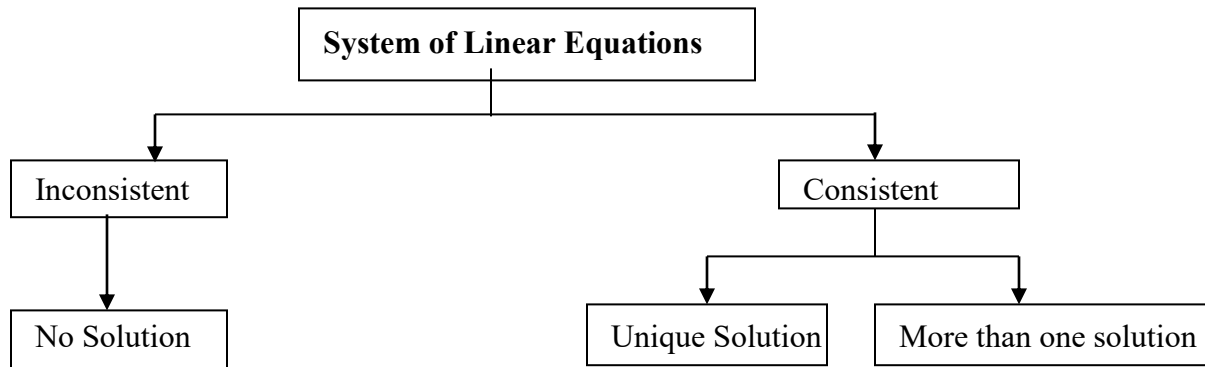
Step 6: A solution of a homogeneous system of linear equations.

- i. If $r = n$, then the system has only the zero solution.
- ii. If $r < n$, then the system has more than one non-zero solution.

$n \rightarrow$ Number of variables

$r \rightarrow$ Number of Rows or equation

Step 7: If two equations or more are identical. We can ignore them keeping one equation



Example 16: Solve the system,

$$2x + y - 2z = 10 \text{ -----}L_1$$

$$3x + 2y + 2z = 1 \text{ -----}L_2$$

$$5x + 4y + 3z = 4 \text{ -----}L_3$$

Here,

$$\begin{array}{ccc} \uparrow a_{11} & \uparrow a_{12} & \uparrow a_{13} \\ 2x + 1y - 2z = 10 \end{array}$$

$$\begin{array}{ccc} \uparrow a_{21} & \uparrow a_{22} & \uparrow a_{23} \\ 3x + 2y + 2z = 1 \end{array}$$

$$\begin{array}{ccc} \uparrow a_{31} & \uparrow a_{32} & \uparrow a_{33} \\ 5x + 4y + 3z = 4 \end{array}$$

Here, $a_{11} = 2$, $a_{12} = 1$, $a_{13} = -2$, $a_{21} = 3$, $a_{22} = 2$, $a_{23} = 2$, $a_{31} = 5$, $a_{32} = 4$, $a_{33} = 3$

We have,

$$L_i \rightarrow -a_{i1}L_1 + a_{11}L_i$$

1st time:

$$\begin{aligned} i = 2, L_2 &\rightarrow -a_{21}L_1 + a_{11}L_2 \\ &= -3(2x + y - 2z - 10) + 2(3x + 2y + 2z - 1) \\ &= -6x - 3y + 6z + 30 + 6x + 4y + 4z - 2 \\ &= y + 10z + 28 \end{aligned}$$

$$\begin{aligned} i = 3, L_3 &\rightarrow -a_{31}L_1 + a_{11}L_3 \\ &= -5(2x + y - 2z - 10) + 2(5x + 4y + 3z - 4) \\ &= -10x - 5y + 10z + 50 + 10x + 8y + 6z - 8 \\ &= 3y + 16z + 42 \end{aligned}$$

Thus we obtain the following new system

$$2x + y - 2z = 10$$

$$y + 10z = -28$$

$$3y + 16z = -42$$

2nd time:

$$2x + y - 2z = 10$$

$$y + 10z = -28 \text{-----} L_1$$

$$3y + 16z = -42 \text{-----} L_2$$

Here, $a_{11} = 1$, $a_{12} = 10$, $a_{21} = 3$, $a_{22} = 16$

We have,

$$L_i \rightarrow -a_{i1}L_1 + a_{11}L_i$$

$$i = 2, L_2 \rightarrow -a_{21}L_1 + a_{11}L_2$$

$$= -3(y + 10z + 28) + 1(3y + 16z + 42)$$

$$= -3y - 30z - 84 + 3y + 16z + 42$$

$$= -14z - 42$$

Thus we obtain the following system

$$2x + y - 2z = 10$$

$$y + 10z = -28$$

$$-14z = 42$$

Now $-14z = 42$

$$\therefore z = -3$$

Again,

$$y + 10z = -28$$

$$\Rightarrow y + 10(-3) = -28$$

$$\Rightarrow y = -28 + 30$$

$$\therefore y = 2$$

Again,

$$2x + y - 2z = 10$$

$$\Rightarrow 2x + 2 - 2(-3) = 10$$

$$\Rightarrow 2x + 2 + 6 = 10$$

$$\Rightarrow 2x = 2$$

$$\therefore x = 1$$

Ans : / (x , y , z) = (1 , 2 , - 3)

Example 17: Solve the system,

$$2x + y - 3z = 5 \text{-----} L_1$$

$$3x - 2y + 2z = 5 \text{-----} L_2$$

$$5x - 3y - z = 16 \text{-----} L_3$$

Sol:

1st time:

We have,

$$L_i \rightarrow -a_{i1}L_1 + a_{11}L_i$$

$$i = 2, L_2 \rightarrow -a_{21}L_1 + a_{11}L_2$$

$\begin{matrix} a_{11} \\ \uparrow \end{matrix}$	$\begin{matrix} a_{12} \\ \uparrow \end{matrix}$	
1	10	$y + 10z = -28$
$\begin{matrix} a_{21} \\ \uparrow \end{matrix}$	$\begin{matrix} a_{22} \\ \uparrow \end{matrix}$	
3	16	$3y + 16z = -42$

$$\begin{aligned}
&= -3(2x + y - 3z - 5) + 2(3x - 2y + 2z - 5) \\
&= -6x - 3y + 9z + 15 + 6x - 4y + 4z - 10 \\
&= -7y + 13z + 5
\end{aligned}$$

$$\begin{aligned}
i = 3, L_3 &\rightarrow -a_{31}L_1 + a_{11}L_3 \\
&= -5(2x + y - 3z - 5) + 2(5x - 3y - z - 16) \\
&= -10x - 5y + 15z + 25 + 10x - 6y - 2z - 32 \\
&= -11y + 13z - 7
\end{aligned}$$

Thus we obtain the following new system

$$\begin{aligned}
2x + y - 3z &= 5 \\
-7y + 13z &= -5 \\
-11y + 13z &= 7
\end{aligned}$$

2nd time:

$$\begin{aligned}
2x + y - 3z &= 5 \\
-7y + 13z &= -5 \text{ ----- } L_1 \\
-11y + 13z &= 7 \text{ ----- } L_2
\end{aligned}$$

Here, $a_{11} = -7$, $a_{12} = 13$, $a_{21} = -11$, $a_{22} = 13$

$$L_i \rightarrow -a_{i1}L_1 + a_{11}L_i$$

$$\begin{aligned}
i = 2, L_2 &\rightarrow -a_{21}L_1 + a_{11}L_2 \\
&= -(-11)(-7y + 13z + 5) + (-7)(-11y + 13z - 7) \\
&= -77y + 143z + 55 + 77y - 91z + 49 \\
&= 52z + 104
\end{aligned}$$

Thus we obtain the following system

$$\begin{aligned}
2x + y - 3z &= 5 \\
-7y + 13z &= -5 \\
52z &= -104
\end{aligned}$$

Now,

$$\begin{aligned}
52z &= -104 \\
\Rightarrow z &= -\frac{104}{52} \\
\therefore z &= -2
\end{aligned}$$

And

$$\begin{aligned}
-7y - 13 \times 2 &= -5 \\
\Rightarrow -7y &= -5 + 26 \\
\Rightarrow y &= \frac{21}{-7} \\
\therefore y &= -3
\end{aligned}$$

And

$$\begin{aligned}
2x - 3 + 6 &= 5 \\
\Rightarrow 2x &= 5 - 3
\end{aligned}$$

$ \begin{array}{cc} \begin{array}{c} \uparrow a_{11} \\ -7 \end{array} & \begin{array}{c} \uparrow a_{12} \\ 13 \end{array} \\ -7y + 13z &= -5 \\ \begin{array}{c} \uparrow a_{21} \\ -11 \end{array} & \begin{array}{c} \uparrow a_{22} \\ 13 \end{array} \\ -11y + 13z &= 7 \end{array} $

$$\Rightarrow 2x = 2$$

$$\therefore x = 1$$

$$\text{Ans : / (x, y, z) = (1, -3, -2)}$$

Example 18: Solve the system:

$$2x + 3y - 2z = 5 \text{-----} L_1$$

$$x - 2y + 3z = 2 \text{-----} L_2$$

$$4x - y + 4z = 1 \text{-----} L_3$$

Sol: 1st time:

$$L_i \rightarrow -a_{i1}L_1 + a_{i1}L_i$$

$$\begin{aligned} i = 2, L_2 &\rightarrow -a_{21}L_1 + a_{11}L_2 \\ &= -1(2x + 3y - 2z - 5) + 2(x - 2y + 3z - 2) \\ &= -2x - 3y + 2z + 5 + 2x - 4y + 6z - 4 \\ &= -7y + 8z + 1 \end{aligned}$$

$$\begin{aligned} i = 3, L_3 &\rightarrow -a_{31}L_1 + a_{11}L_3 \\ &= -4(2x + 3y - 2z - 5) + 2(4x - y + 4z - 1) \\ &= -8x - 12y + 8z + 20 + 8x - 2y + 8z - 2 \\ &= -14y + 16z + 18 \end{aligned}$$

Thus we obtain the following system

$$\begin{aligned} 2x + 3y - 2z &= 5 \\ -7y + 8z &= -1 \\ -14y + 16z &= -18 \end{aligned}$$

Again, 2nd time:

$$\begin{aligned} 2x + 3y - 2z &= 5 \\ -7y + 8z &= -1 \text{-----} L_1 \\ -14y + 16z &= -18 \text{-----} L_2 \end{aligned}$$

$$L_i \rightarrow -a_{i1}L_1 + a_{11}L_i$$

$$\begin{aligned} i = 2, L_2 &\rightarrow -a_{21}L_1 + a_{11}L_2 \\ &= -(-14)(-7y + 8z + 1) + (-7)(-14y + 16z + 18) \\ &= -98y + 112z + 14 + 98y - 112z - 126 \\ &= -112 \end{aligned}$$

Thus we obtain the following system

$$\begin{aligned} 2x + 3y - 2z &= 5 \\ -7y + 8z &= -1 \\ 0 &= 112 \end{aligned}$$

Here,

$$0 \times x + 0 \times y + 0 \times z = 112$$

$$\Rightarrow 0 = 112$$

$\Rightarrow$ but $112 \neq 0$; It is impossible (inconsistent)

$\begin{array}{cc} a_{11} & a_{12} \\ \uparrow & \uparrow \\ -7y + 8z = -1 \\ \\ a_{21} & a_{22} \\ \uparrow & \uparrow \\ -14y + 16z = -18 \end{array}$
--

So, the system is inconsistent and has no solution (*see step-03*)

Ans: The system has no solution

How to block diagonal of a matrix

Let a matrix

$$\begin{bmatrix} 3 & 7 & 4 \\ 2 & 5 & 1 \\ 8 & 6 & 9 \end{bmatrix}$$

Multiplying this matrix by a column matrix:

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$\begin{bmatrix} 3 & 7 & 4 \\ 2 & 5 & 1 \\ 8 & 6 & 9 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3x+7y+4z \\ 2x+5y+z \\ 8x+6y+9z \end{bmatrix}$$

Example 19: Solve the system,

$$x + 2y - 3z = 6$$

$$2x - y + 4z = 2$$

$$4x + 3y - 2z = 14$$

Sol: We have,

1st time:

$$L_i \rightarrow -a_{i1}L_1 + a_{11}L_i$$

$$i = 2, L_2 \rightarrow -a_{21}L_1 + a_{11}L_2$$

$$= -2(x + 2y - 3z - 6) + 1(2x - y + 4z - 2)$$

$$= -2x - 4y + 6z + 12 + 2x - y + 4z - 2$$

$$= -5y + 10z + 10$$

$$i = 3, L_3 \rightarrow -a_{31}L_1 + a_{11}L_3$$

$$= -4(x + 2y - 3z - 6) + 1(4x + 3y - 2z - 14)$$

$$= -4x - 8y + 12z + 24 + 4x + 3y - 2z - 14$$

$$= -5y + 10z + 10$$

Thus we obtain the following system

$$\begin{aligned}x + 2y - 3z &= 6 \\-5y + 10z &= -10 \\-5y + 10z &= -10\end{aligned}$$

$\begin{array}{cc} \begin{matrix} a_{11} \\ \uparrow \end{matrix} & \begin{matrix} a_{12} \\ \uparrow \end{matrix} \\ -5y + 10z = -10 \\ \begin{matrix} a_{21} \\ \uparrow \end{matrix} & \begin{matrix} a_{22} \\ \uparrow \end{matrix} \\ -5y + 10z = -10 \end{array}$
--

Again, 2nd time:

$$L_i \rightarrow -a_{i1}L_1 + a_{11}L_i$$

$$i = 2, L_2 \rightarrow -a_{21}L_1 + a_{11}L_2$$

$$\begin{aligned}&= -(-5)(-5y + 10z + 10) + (-5)(-5y + 10z + 10) \\&= -25y + 50z + 50 + 25y - 50z - 50 \\&= 0\end{aligned}$$

Thus we obtain the following system

$$\begin{aligned}x + 2y - 3z &= 6 \\-5y + 10z &= -10 \\0 &= 0\end{aligned}$$

(see step-04)

It occurs that,

$$0 \times x + 0 \times y + 0 \times z = 0$$

Hence we can delete the equation without affecting the solution

Thus we obtain the following new system

$$\begin{aligned}x + 2y - 3z &= 6 \\-5y + 10z &= -10\end{aligned}$$

In echelon form, there are only two equations in three unknowns, then the system has a non-zero solution and in particular $(3-2)=1$ free variable, which is z .

[Note Here total variables are three, i.e x, y, z and blocked variable is x and y , hence free variable is z]

Let,

$$\begin{aligned}z &= 1 \\-5y + 10 &= -10 \\\Rightarrow -5y &= -20 \\\therefore y &= 4\end{aligned}$$

Again,

$$\begin{aligned}x + 2 \times 4 - 3 \times 1 &= 6 \\\Rightarrow x + 8 - 3 &= 6 \\\Rightarrow x + 5 &= 6 \\\Rightarrow x &= 6 - 5 \\\therefore x &= 1\end{aligned}$$

$$\text{Ans: } (x, y, z) = (1, 4, 1)$$

Example 20: Solve the system

$$x + 2y - z = 0$$

$$2x + 5y + 2z = 0$$

$$x + 4y + 7z = 0$$

$$x + 3y + 3z = 0$$

Sol: We have,

1st time:

$$L_i \rightarrow -a_{i1}L_1 + a_{11}L_i$$

$$\begin{aligned} i = 2, L_2 &\rightarrow -a_{21}L_1 + a_{11}L_2 \\ &= -2(x + 2y - z) + 1(2x + 5y + 2z) \\ &= -2x - 2y + 2z + 2x + 5y + 2z \\ &= y + 4z \end{aligned}$$

$$\begin{aligned} i = 3, L_3 &\rightarrow -a_{31}L_1 + a_{11}L_3 \\ &= -1(x + 2y - z) + 1(x + 4y + 7z) \\ &= -x - 2y + z + x + 4y + 7z \\ &= 2y + 8z \end{aligned}$$

$$\begin{aligned} i = 4, L_4 &\rightarrow -a_{41}L_1 + a_{11}L_4 \\ &= -1(x + 2y - z) + 1(x + 3y + 3z) \\ &= -x - 2y + z + x + 3y + 3z \\ &= y + 4z \end{aligned}$$

Then we obtain the system of equations

$$x + 2y - z = 0$$

$$y + 4z = 0$$

$$2y + 8z = 0$$

$$y + 4z = 0$$

Then we obtain the system of equations

$$x + 2y - z = 0$$

$$y + 4z = 0$$

$$y + 4z = 0 \quad [\text{Dividing by 2}]$$

$$y + 4z = 0$$

Again, **2nd time:**

Since the 2nd, 3rd & 4th equations are identical. We can disregard one of them

$$\begin{aligned} x + 2y - z &= 0 \\ y + 4z &= 0 \end{aligned}$$

In echelon form, there are only two equations in three unknowns, then the system has a non-zero solution and in particular $(3-2)=1$ free variable, which is z .

[Note Here total variables are three, i.e x, y, z and blocked variable is x and y , hence free variable is z]

Let,

$$\begin{aligned}
 z &= 1 \\
 y &= -4 \\
 x &= -2y + z \\
 \Rightarrow x &= -2 \times (-4) + 1 \\
 \Rightarrow x &= 8 + 1 \\
 \therefore x &= 9
 \end{aligned}$$

Ans: $(x, y, z) = (9, -4, 1)$

Example 21: Solve the system:

$$\begin{aligned}
 x + y - z &= 0 \\
 2x + 4y - z &= 0 \\
 3x + 2y + 2z &= 0
 \end{aligned}$$

Sol:

We have,

$$L_i \rightarrow -a_{i1}L_1 + a_{i1}L_i$$

1st time:

$$i > 1$$

$$\begin{aligned}
 i = 2, L_2 &\rightarrow -a_{21}L_1 + a_{11}L_2 \\
 &= -2(x + y - z) + 1(2x + 4y - z) \\
 &= -2x - 2y + 2z + 2x + 4y - z \\
 &= 2y + z
 \end{aligned}$$

$$\begin{aligned}
 i = 3, L_3 &\rightarrow -a_{31}L_1 + a_{11}L_3 \\
 &= -3(x + y - z) + 1(3x + 2y + 2z) \\
 &= -3x - 3y + 3z + 3x + 2y + 2z \\
 &= -y + 5z
 \end{aligned}$$

Then we obtain the system of equations

$$\begin{aligned}
 x + y - z &= 0 \\
 2y + z &= 0 \\
 -y + 5z &= 0
 \end{aligned}$$

2nd time:

$$L_i \rightarrow -a_{i1}L_1 + a_{i1}L_i$$

$$\begin{aligned}
 i = 2, L_2 &\rightarrow -a_{21}L_1 + a_{11}L_2 \\
 &= -(-1)(2y + z) + 2(-y + 5z) \\
 &= 2y + z - 2y + 10z \\
 &= 11z
 \end{aligned}$$

Then we obtain the system of equations

$$\begin{aligned}
 x + y - z &= 0 \\
 2y + z &= 0 \\
 11z &= 0
 \end{aligned}$$

$ \begin{array}{cc} \overset{a_{11}}{\uparrow} & \overset{a_{12}}{\uparrow} \\ 2y + 1z = 0 \\ \overset{a_{21}}{\uparrow} & \overset{a_{22}}{\uparrow} \\ -1y + 5z = 0 \end{array} $
--

In echelon form there are three equations in three unknowns. Hence the system has a unique solution.

$$z = 0$$

$$y = 0$$

$$x = 0$$

$$\text{Ans : / (x, y, z) = (0, 0, 0)}$$

Example 22: Solve the system,

$$2x + y - 2z + 3w = 1$$

$$3x + 2y - z + 2w = 4$$

$$3x + 3y + 3z - 3w = 5$$

Sol:

We have,

$$L_i \rightarrow -a_{i1}L_1 + a_{i1}L_i$$

1st step:

$$i > 1$$

$$i = 2, L_2 \rightarrow -a_{21}L_1 + a_{11}L_2$$

$$= -3(2x + y - 2z + 3w - 1) + 2(3x + 2y - z + 2w - 4)$$

$$= -6x - 3y + 6z - 9w + 3 + 6x + 4y - 2z + 4w - 8$$

$$= y + 4z - 5w - 5$$

$$i = 3, L_3 \rightarrow -a_{31}L_1 + a_{11}L_3$$

$$= -3(2x + y - 2z + 3w - 1) + 2(3x + 3y + 3z - 3w - 5)$$

$$= -6x - 3y + 6z - 9w + 3 + 6x + 6y + 6z - 6w - 10$$

$$= 3y + 12z - 15w - 7$$

Then we obtain the system of equations

$$2x + y - 2z + 3w = 1$$

$$y + 4z - 5w = 5$$

$$3y + 12z - 15w = 7$$

2nd time:

$$2x + y - 2z + 3w = 1$$

$$y + 4z - 5w = 5 \text{ ----- } L_1$$

$$3y + 12z - 15w = 7 \text{ ----- } L_2$$

$$L_i \rightarrow -a_{i1}L_1 + a_{i1}L_i$$

$$i = 2, L_2 \rightarrow -a_{21}L_1 + a_{11}L_2$$

$$= -3(y + 4z - 5w - 5) + 1(3y + 12z - 15w - 7)$$

$$= -3y - 12z + 15w + 15 + 3y + 12z - 15w - 7$$

$$= 8$$

$\uparrow$	$\uparrow$	$\uparrow$	
a_{11}	a_{12}	a_{13}	
1	y	4	$z - 5w = 5$
$\uparrow$	$\uparrow$	$\uparrow$	
a_{21}	a_{22}	a_{23}	
3	y	12	$z - 15w = 7$

Then we obtain the new system of equations

$$2x + y - 2z + 3w = 1$$

$$y + 4z - 5w = 5$$

$$0 = -8$$

$$\text{Here, } 0 \times x + 0 \times y + 0 \times z + 0 \times w = -8$$

$$\Rightarrow 0 = -8$$

$$\text{But, } -8 \neq 0 \quad [\text{See Step-03}]$$

So, the system is inconsistent and has no solution.

Ans: The system has no solution.

Example 23: Solve the system,

$$x + y + z = 0$$

$$x + 3y + z = 0$$

$$x + y + z = 0$$

We have,

$$L_i \rightarrow -a_{i1}L_1 + a_{i1}L_i$$

$$i = 2, L_2 \rightarrow -a_{21}L_1 + a_{11}L_2$$

$$L_2 \rightarrow -1(x + y + z) + 1(x + 3y + z) \\ = 2y$$

$$i = 3, L_3 \rightarrow -a_{31}L_1 + a_{11}L_3$$

$$L_3 \rightarrow -1(x + y + z) + 1(x + y + z) \\ = 0$$

Thus we obtain the following system

$$x + y + z = 0$$

$$2y = 0$$

$$0 = 0$$

$$\Rightarrow x + y + z = 0$$

$$y = 0$$

We get, $y = 0$ then

$$\Rightarrow x + 0 + z = 0$$

Thus we obtain the following system

$$\Rightarrow x + z = 0$$

In echelon form, there are only one equation in two unknowns, then the system has a non-zero solution and in particular $(2-1)=1$ free variable, which is z .

Let,

$$z = 1$$

$$\therefore x = -1$$

$$\text{Ans: } (x, y, z) = (-1, 0, 1)$$

Home Task

1. A manufacturer receives daily shipments of 70,000 springs and 45,000 pounds of stuffing for producing regular and support mattresses. Regular mattresses r requires 50 springs and 30 pounds of stuffing; support mattresses s requires 60 springs and 40 pounds of stuffing. How many mattresses of each type should be produced daily to use all available inventories?
2. A manufacturer produces desks and bookcases. Desks d require 5 hr of cutting time and 10 hr of assembling time. Bookcases b requires 15 min of cutting time and 1 hr of assembling time. Each day the manufacturer has available 200 personnel hours for cutting and 500 personnel hours for assembling. How many desks and bookcases should be scheduled for completion each day to use all available workpower?

Polynomials of Matrices and Linear Operators

Consider a polynomial $f(t)$ over a field k

$$f(t) = a_n t^n + a_{n-1} t^{n-1} + a_{n-2} t^{n-2} + \dots + a_1 t + a_0$$

If A is a square matrix over k

Then we define

$$f(A) = a_n A^n + a_{n-1} A^{n-1} + a_{n-2} A^{n-2} + \dots + a_1 A + a_0 I$$

Example: Let $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $f(t) = 2t^2 - 3t + 7$; $g(t) = t^2 - 5t - 2$ then $f(A)$ and $g(A)$.

Solution:

$$\text{Given, } f(t) = 2t^2 - 3t + 7$$

$$\therefore f(A) = 2A^2 - 3A + 7I$$

$$\Rightarrow f(A) = 2 \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} - 3 \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} + 7 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\Rightarrow f(A) = 2 \begin{bmatrix} 1 \times 1 + 2 \times 3 & 1 \times 2 + 2 \times 4 \\ 3 \times 1 + 4 \times 3 & 3 \times 2 + 4 \times 4 \end{bmatrix} - \begin{bmatrix} 3 & 6 \\ 9 & 12 \end{bmatrix} + \begin{bmatrix} 7 & 0 \\ 0 & 7 \end{bmatrix}$$

$$\Rightarrow f(A) = 2 \begin{bmatrix} 7 & 10 \\ 15 & 22 \end{bmatrix} - \begin{bmatrix} 3 & 6 \\ 9 & 12 \end{bmatrix} + \begin{bmatrix} 7 & 0 \\ 0 & 7 \end{bmatrix}$$

$$\Rightarrow f(A) = \begin{bmatrix} 14 & 20 \\ 30 & 44 \end{bmatrix} - \begin{bmatrix} 3 & 6 \\ 9 & 12 \end{bmatrix} + \begin{bmatrix} 7 & 0 \\ 0 & 7 \end{bmatrix}$$

$$\Rightarrow f(A) = \begin{bmatrix} 18 & 14 \\ 21 & 39 \end{bmatrix}$$

$$\text{Given, } g(t) = t^2 - 5t - 2$$

$$\therefore g(A) = A^2 - 5A - 2I$$

$$\Rightarrow g(A) = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} - 5 \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} - 2 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\Rightarrow g(A) = \begin{bmatrix} 1 \times 1 + 2 \times 3 & 1 \times 2 + 2 \times 4 \\ 3 \times 1 + 4 \times 3 & 3 \times 2 + 4 \times 4 \end{bmatrix} - \begin{bmatrix} 5 & 10 \\ 15 & 20 \end{bmatrix} - \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

$$\Rightarrow g(A) = \begin{bmatrix} 7 & 10 \\ 15 & 22 \end{bmatrix} - \begin{bmatrix} 5 & 10 \\ 15 & 20 \end{bmatrix} - \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

$$\Rightarrow g(A) = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Geometrical Significance of Eigen values and Eigen Vectors (Eigenfaces)

Suppose we have a matrix A: $A = \begin{bmatrix} 4 & 1 \\ 1 & 4 \end{bmatrix}$

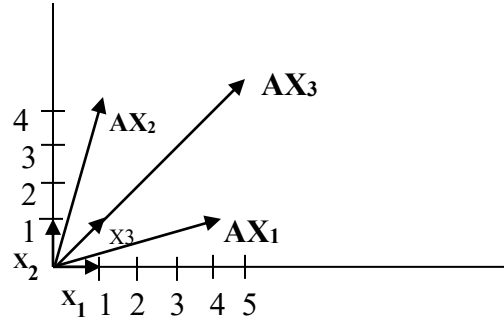


Figure 09: Matrix Vector multiplication for various vectors

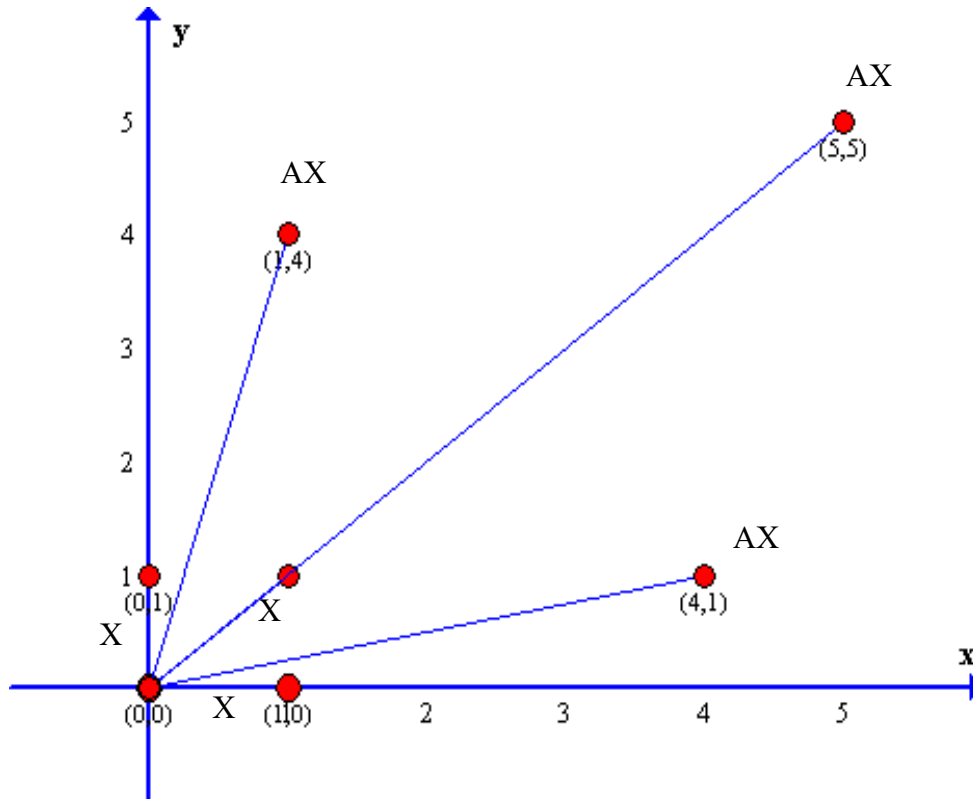


Figure no 10

Suppose we take the product AX_1 , where $X_1 = [1,0] = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, as shown in Fig 10

$$\text{Then, } \mathbf{AX}_1 = \begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 4 \times 1 + 1 \times 0 \\ 1 \times 1 + 4 \times 0 \end{pmatrix} = \begin{pmatrix} 4 \\ 1 \end{pmatrix} \text{-----(i)}$$

By comparing the vectors $\mathbf{X}_1$ and $\mathbf{AX}_1$ we see that the product vector is scaled and rotated counter-clockwise with respect to $\mathbf{X}_1$

Now consider the case where $\mathbf{X}_2 = [0, 1] = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$,

$$\text{Then } \mathbf{AX}_2 = \begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 4 \times 0 + 1 \times 1 \\ 1 \times 0 + 4 \times 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 4 \end{pmatrix} = [1, 4] \text{-----(ii)}$$

Here, we note a clockwise rotation of $\mathbf{AX}_2$ with respect to $\mathbf{X}_2$

Now let's consider a more interesting case. Suppose $\mathbf{X}_3 = [1, 1] = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

$$\text{Then } \mathbf{AX}_3 = \begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 4 \times 1 + 1 \times 1 \\ 1 \times 1 + 4 \times 1 \end{pmatrix} = \begin{pmatrix} 5 \\ 5 \end{pmatrix} = [5, 5] \text{-----(iii)}$$

Now the product vector points in the same direction as $\mathbf{X}_3$. The vector $\mathbf{AX}_3$ is scaled version of the vector $\mathbf{X}_3$. Because of this property, $\mathbf{X}_3 = [1, 1]$ is an Eigen vector of A. The scale factor (which in this case is 5) is given the symbol λ and is referred to as an Eigen value.

$$\text{Thus } \mathbf{AX}_3 = \begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 4 \times 1 + 1 \times 1 \\ 1 \times 1 + 4 \times 1 \end{pmatrix} = \begin{pmatrix} 5 \\ 5 \end{pmatrix} = 5 \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 5 \\ 5 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 5 \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\Rightarrow \mathbf{AX}_3 = 5 \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\Rightarrow \mathbf{AX}_3 = 5\mathbf{X}_3 \text{ [} \mathbf{X}_3 = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \text{]} \text{-----(iv)}$$

Here $\mathbf{X}_3$ is an Eigen vector 5 is an Eigen value

In general we can say the relation of Eigen value and Eigen vector is $\mathbf{AX} = \lambda\mathbf{X}$

Summary of Eigen vectors (Eigenfaces) and Eigen values:

An Eigen vector is a nonzero vector that satisfies the equation $A\vec{V} = \lambda\vec{V}$, where A is a square matrix, λ is a scalar, and $\vec{v}$ is the eigenvector. λ is called an eigenvalue.

Or An Eigen vector is a nonzero vector that satisfies the equation $A\vec{X} = \lambda\vec{X}$, where A is a square matrix, λ is a scalar value, and $\vec{X}$ is the eigenvector. λ is called an eigenvalue

That is, the relation between Eigen-vectors and Eigen values of a matrix $A\vec{X} = \lambda\vec{X}$

Where A is given matrix, λ is Eigen values of the given matrix, $\vec{X}$ is corresponding Eigen vectors of the matrix

Example 23: Determine which of the following vectors are Eigen vectors for

$A = \begin{pmatrix} 1 & 2 \\ -4 & 7 \end{pmatrix}$ showing your analysis procedure graphically using graph paper.

(a) $\begin{pmatrix} 1 \\ -1 \end{pmatrix}$ (b) $\begin{pmatrix} 2 \\ 3 \end{pmatrix}$ (c) $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$ (d) $\begin{pmatrix} 2 \\ 2 \end{pmatrix}$ (e) $\begin{pmatrix} 2 \\ 4 \end{pmatrix}$ (f) $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$

Answer:

We have the relation between Eigen-vectors and Eigen values of a matrix $A\vec{X} = \lambda\vec{X}$

Where A is given matrix, λ is Eigen values of the given matrix, $\vec{X}$ is corresponding Eigen vectors of the matrix.

Test (a):

Given, $A = \begin{pmatrix} 1 & 2 \\ -4 & 7 \end{pmatrix}$, $X = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$

$$\therefore AX = \begin{pmatrix} 1 & 2 \\ -4 & 7 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 \times 1 - 2 \times 1 \\ -4 \times 1 - 7 \times 1 \end{pmatrix} = \begin{pmatrix} -1 \\ -11 \end{pmatrix} = -1 \begin{pmatrix} 1 \\ 11 \end{pmatrix}$$

$A\vec{X} = \lambda\vec{X}$ is not satisfied since $\vec{X}$ is not returned

Test (b):

$A = \begin{pmatrix} 1 & 2 \\ -4 & 7 \end{pmatrix}$, $X = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$

$$\therefore AX = \begin{pmatrix} 1 & 2 \\ -4 & 7 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 1 \times 2 + 2 \times 3 \\ -4 \times 2 + 7 \times 3 \end{pmatrix} = \begin{pmatrix} 8 \\ 13 \end{pmatrix}$$

$AX = \lambda X$ is not satisfied, since $\vec{X}$ is not returned

Test (c):

$A = \begin{pmatrix} 1 & 2 \\ -4 & 7 \end{pmatrix}$, $X = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$

$$\therefore AX = \begin{pmatrix} 1 & 2 \\ -4 & 7 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 \times 1 + 2 \times 2 \\ -4 \times 1 + 7 \times 2 \end{pmatrix} = \begin{pmatrix} 5 \\ 10 \end{pmatrix} = 5 \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

$$\therefore \mathbf{AX} = \begin{pmatrix} 1 & 2 \\ -4 & 7 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \end{pmatrix} = 5 \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

$$\begin{array}{cccc} \downarrow & \downarrow & \downarrow & \downarrow \\ \mathbf{A} & \mathbf{X} & \lambda & \mathbf{X} \end{array}$$

$\therefore \mathbf{AX} = \lambda \mathbf{X}$ is satisfied; since $\vec{X}$ is returned

Test (d):

$$\mathbf{A} = \begin{pmatrix} 1 & 2 \\ -4 & 7 \end{pmatrix}, \mathbf{X} = \begin{pmatrix} 2 \\ 2 \end{pmatrix}$$

$$\therefore \mathbf{AX} = \begin{pmatrix} 1 & 2 \\ -4 & 7 \end{pmatrix} \begin{pmatrix} 2 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 \times 2 + 2 \times 2 \\ -4 \times 2 + 7 \times 2 \end{pmatrix} = \begin{pmatrix} 6 \\ 6 \end{pmatrix} = 3 \begin{pmatrix} 2 \\ 2 \end{pmatrix}$$

$$\therefore \mathbf{AX} = \begin{pmatrix} 1 & 2 \\ -4 & 7 \end{pmatrix} \begin{pmatrix} 2 \\ 2 \end{pmatrix} = 3 \begin{pmatrix} 2 \\ 2 \end{pmatrix}$$

$$\begin{array}{cccc} \downarrow & \downarrow & \downarrow & \downarrow \\ \mathbf{A} & \mathbf{X} & \lambda & \mathbf{X} \end{array}$$

$\therefore \mathbf{AX} = \lambda \mathbf{X}$ is satisfied, since $\vec{X}$ is returned

Test (e):

$$\mathbf{A} = \begin{pmatrix} 1 & 2 \\ -4 & 7 \end{pmatrix}, \mathbf{X} = \begin{pmatrix} 2 \\ 4 \end{pmatrix}$$

$$\therefore \mathbf{AX} = \begin{pmatrix} 1 & 2 \\ -4 & 7 \end{pmatrix} \begin{pmatrix} 2 \\ 4 \end{pmatrix} = \begin{pmatrix} 1 \times 2 + 2 \times 4 \\ -4 \times 2 + 7 \times 4 \end{pmatrix} = \begin{pmatrix} 10 \\ 20 \end{pmatrix} = 5 \begin{pmatrix} 2 \\ 4 \end{pmatrix}$$

$$\therefore \mathbf{AX} = \begin{pmatrix} 1 & 2 \\ -4 & 7 \end{pmatrix} \begin{pmatrix} 2 \\ 4 \end{pmatrix} = 5 \begin{pmatrix} 2 \\ 4 \end{pmatrix}$$

$$\begin{array}{cccc} \downarrow & \downarrow & \downarrow & \downarrow \\ \mathbf{A} & \mathbf{X} & \lambda & \mathbf{X} \end{array}$$

$\therefore \mathbf{AX} = \lambda \mathbf{X}$ is satisfied; since $\vec{X}$ is returned

Test (f):

$$\mathbf{A} = \begin{pmatrix} 1 & 2 \\ -4 & 7 \end{pmatrix}, \mathbf{X} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\therefore \mathbf{AX} = \begin{pmatrix} 1 & 2 \\ -4 & 7 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \times 1 + 2 \times 1 \\ -4 \times 1 + 7 \times 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 3 \end{pmatrix} = 3 \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\therefore \mathbf{AX} = \begin{pmatrix} 1 & 2 \\ -4 & 7 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 5 \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\begin{array}{cccc} \downarrow & \downarrow & \downarrow & \downarrow \\ \mathbf{A} & \mathbf{X} & \lambda & \mathbf{X} \end{array}$$

$\therefore \mathbf{AX} = \lambda \mathbf{X}$ is satisfied; since $\vec{X}$ is returned

Graphical Analysis:

Test (a):

Given, $A = \begin{pmatrix} 1 & 2 \\ -4 & 7 \end{pmatrix}$, $X = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$

$$\therefore AX = \begin{pmatrix} 1 & 2 \\ -4 & 7 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 \times 1 - 2 \times 1 \\ -4 \times 1 - 7 \times 1 \end{pmatrix} = \begin{pmatrix} -1 \\ -11 \end{pmatrix}$$

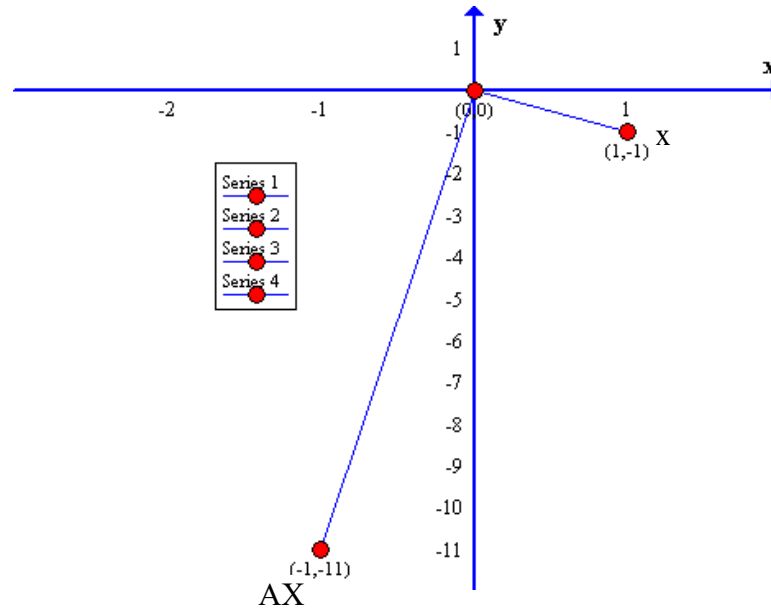


Figure no 11

The product vector AX is not in the same direction as X, So X is not Eigen vector of A

Test (b):

$A = \begin{pmatrix} 1 & 2 \\ -4 & 7 \end{pmatrix}$, $X = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$

$$\therefore AX = \begin{pmatrix} 1 & 2 \\ -4 & 7 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 1 \times 2 + 2 \times 3 \\ -4 \times 2 + 7 \times 3 \end{pmatrix} = \begin{pmatrix} 8 \\ 13 \end{pmatrix}$$

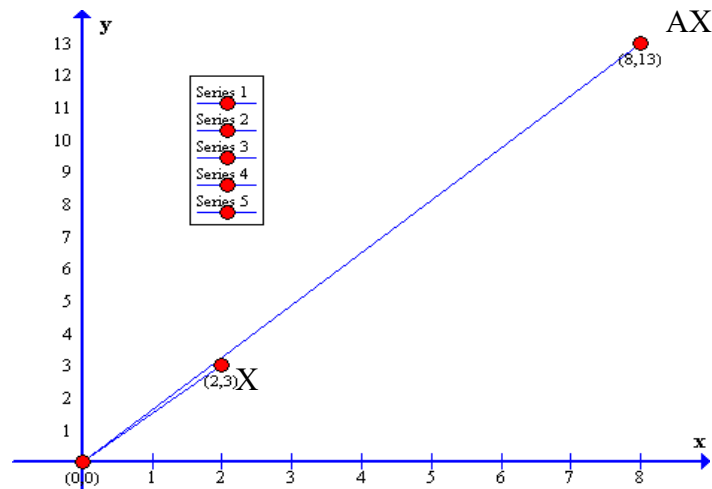


Figure no 12

The product vector AX is not in the same direction as X , So X is not eigen vector of A

Test (c):

$$A = \begin{pmatrix} 1 & 2 \\ -4 & 7 \end{pmatrix}, X = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

$$\therefore AX = \begin{pmatrix} 1 & 2 \\ -4 & 7 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 \times 1 + 2 \times 2 \\ -4 \times 1 + 7 \times 2 \end{pmatrix} = \begin{pmatrix} 5 \\ 10 \end{pmatrix}$$

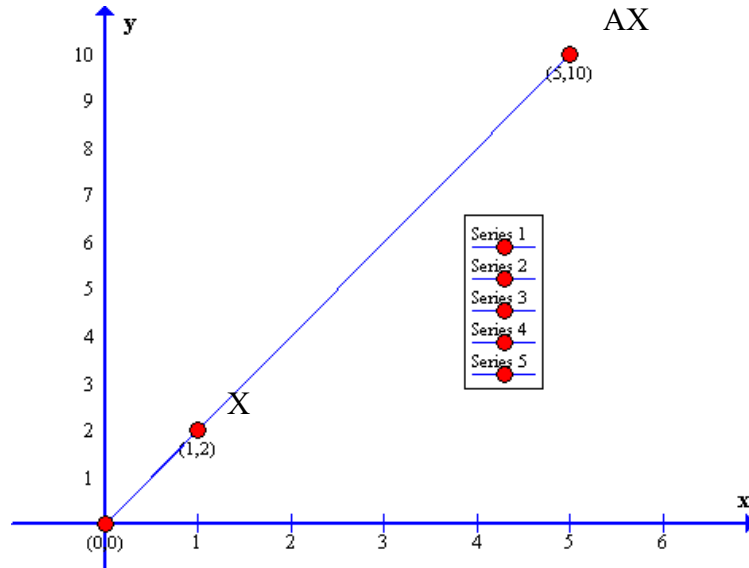


Figure no 13

The product vector AX is in the same direction as X , So X is an Eigen vector of A .

Test (d):

$$A = \begin{pmatrix} 1 & 2 \\ -4 & 7 \end{pmatrix}, X = \begin{pmatrix} 2 \\ 2 \end{pmatrix}$$

$$\therefore AX = \begin{pmatrix} 1 & 2 \\ -4 & 7 \end{pmatrix} \begin{pmatrix} 2 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 \times 2 + 2 \times 2 \\ -4 \times 2 + 7 \times 2 \end{pmatrix} = \begin{pmatrix} 6 \\ 6 \end{pmatrix}$$

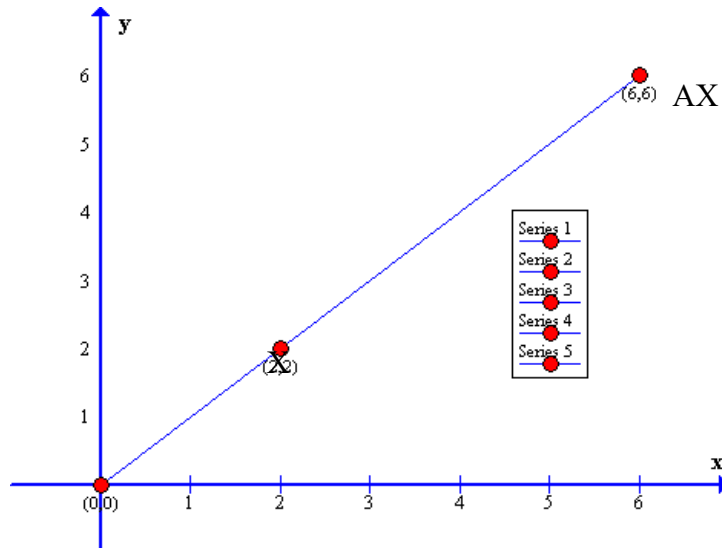


Figure no 14

The product vector AX is in the same direction as X , So X is an Eigen vector of A .

Test (e):

$$A = \begin{pmatrix} 1 & 2 \\ -4 & 7 \end{pmatrix}, X = \begin{pmatrix} 2 \\ 4 \end{pmatrix}$$

$$\therefore AX = \begin{pmatrix} 1 & 2 \\ -4 & 7 \end{pmatrix} \begin{pmatrix} 2 \\ 4 \end{pmatrix} = \begin{pmatrix} 1 \times 2 + 2 \times 4 \\ -4 \times 2 + 7 \times 4 \end{pmatrix} = \begin{pmatrix} 10 \\ 20 \end{pmatrix}$$

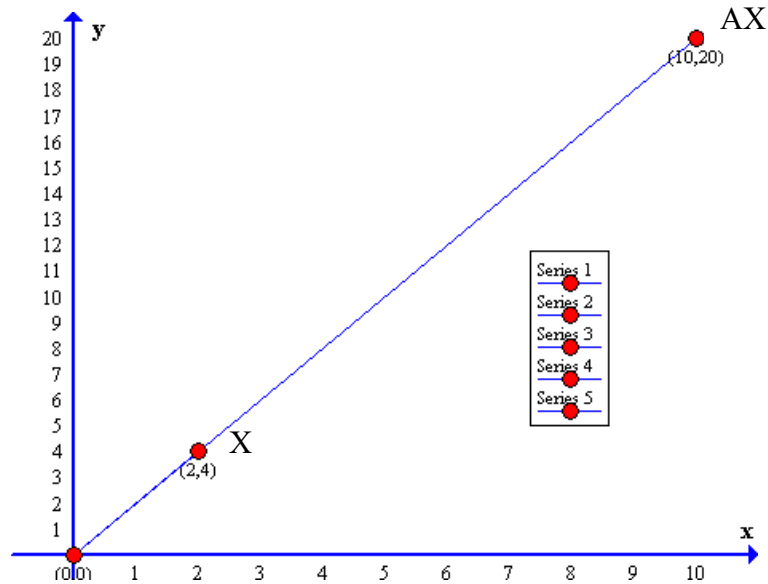


Figure no 15

The product vector AX is in the same direction as X , So X is an Eigen vector of A .

Test (f):

$$A = \begin{pmatrix} 1 & 2 \\ -4 & 7 \end{pmatrix}, X = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\therefore AX = \begin{pmatrix} 1 & 2 \\ -4 & 7 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \times 1 + 2 \times 1 \\ -4 \times 1 + 7 \times 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 3 \end{pmatrix}$$

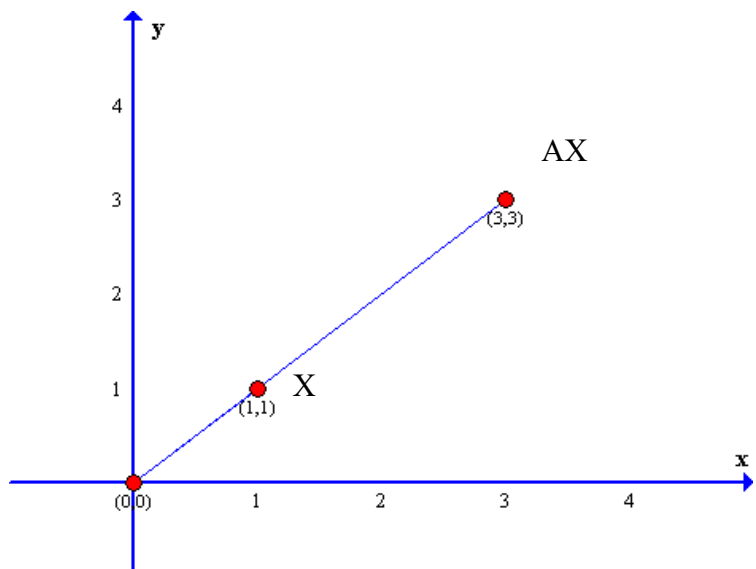


Figure no 16

The product vector AX is in the same direction as X , So X is an Eigen vector of A .

Three Dimensional Vector Representations

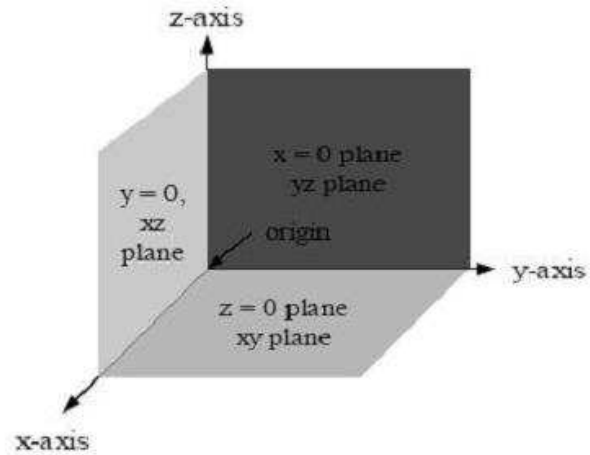


Figure 17

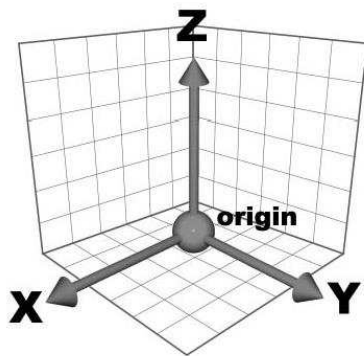


Figure no 18

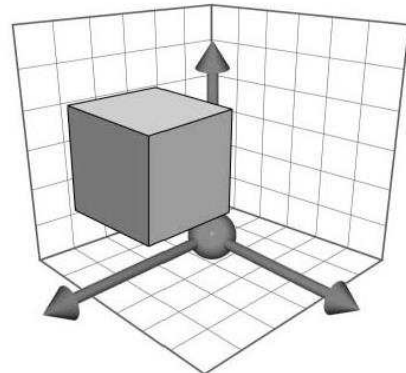


Figure no 19

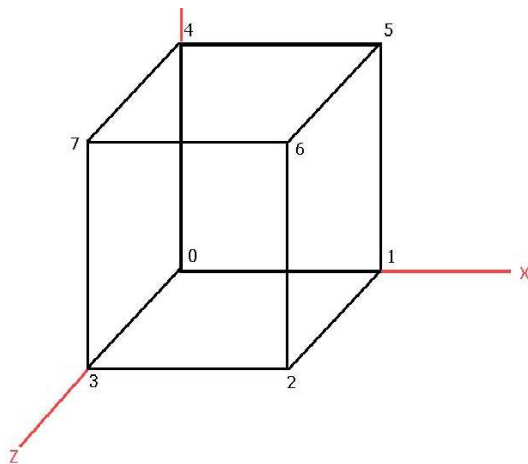


Figure no 20

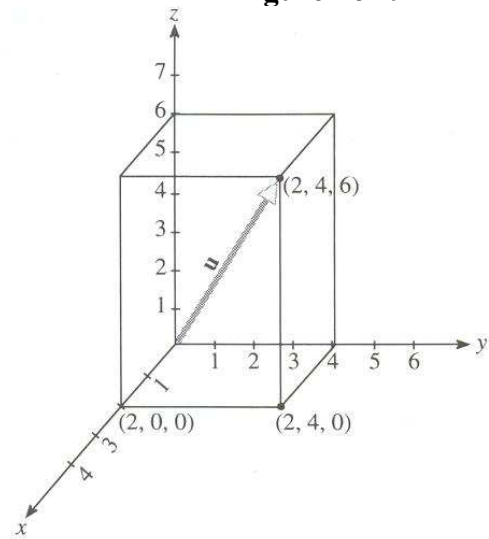


Figure no 21

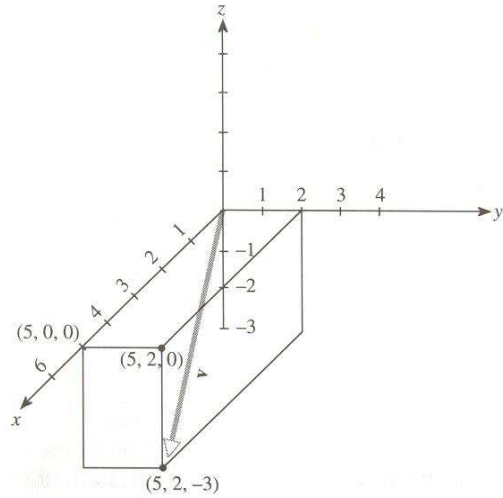


Figure 22

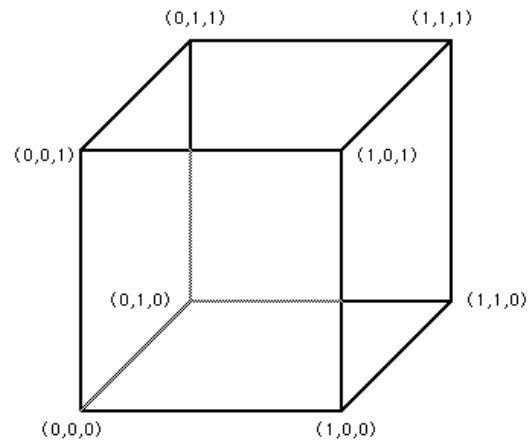


Figure 23

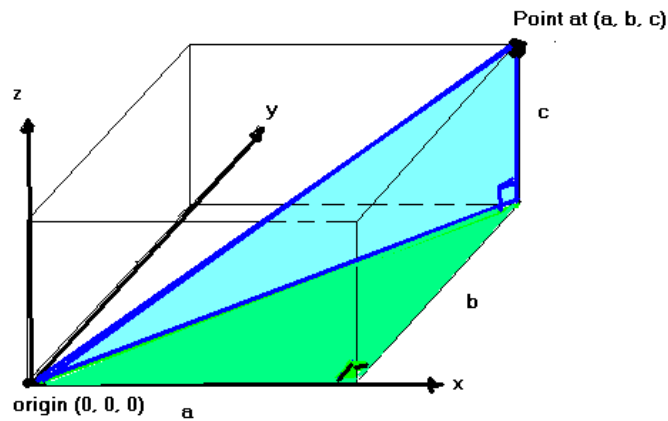


Figure no 24

Each axis of the cube represents values of red, green, or blue in the range [0,255].

- $(0, 0, 0)$ is black
- $(255, 255, 255)$ is white
- $(255, 0, 0)$ is red
- $(0, 255, 0)$ is green
- $(0, 0, 255)$ is blue
- $(255, 255, 0)$ is yellow
- $(0, 255, 255)$ is cyan
- $(255, 0, 255)$ is magenta

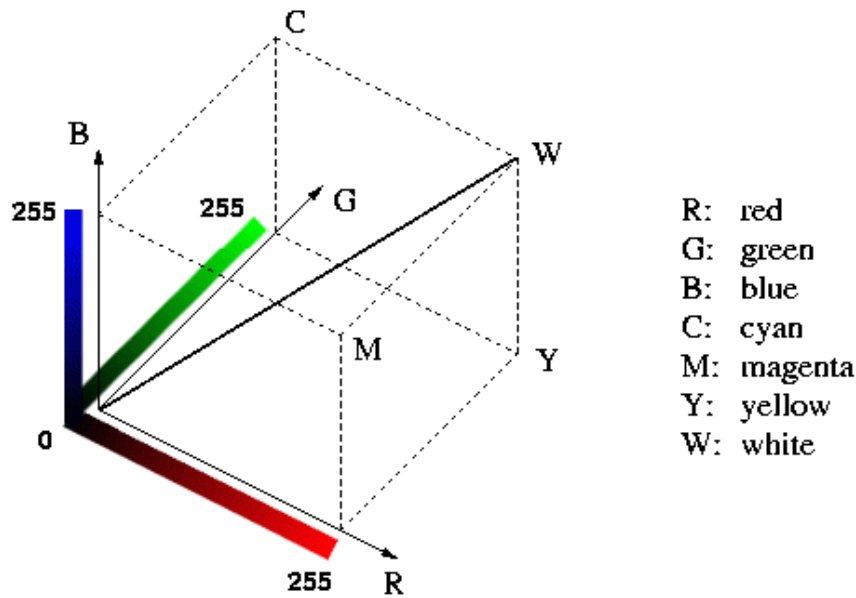


Figure 25: The RGB Color Cube

Note:

1. Each 8 bits represents the level of a primary colour - Red / Green / Blue (RGB)
2. Each pixel can be stored as a 24-bit number.
3. In hex it becomes obvious: - 2 digits / colour
4. $00FF00_{16}$ is the brightest **pure green**
5. Single hex digit = four bits
6. 2 hex digits per **byte**

Example 24: Find the Eigen values and Eigen vectors of the matrix $A = \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix}$

Solution:

$$A - \lambda I = \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 1-\lambda & 2 \\ 3 & 2-\lambda \end{bmatrix}$$

$$\therefore |A - \lambda I| = \begin{vmatrix} 1-\lambda & 2 \\ 3 & 2-\lambda \end{vmatrix}$$

$$\begin{aligned}
&= (1-\lambda)(2-\lambda) - 6 \\
&= 2 - \lambda - 2\lambda + \lambda^2 - 6 \\
&= \lambda^2 - 3\lambda - 4 \\
&= (\lambda^2 - 4\lambda + \lambda - 4)
\end{aligned}$$

$$|\mathbf{A} - \lambda \mathbf{I}| = 0$$

$$\begin{aligned}
&\Rightarrow (\lambda^2 - 4\lambda + \lambda - 4) = 0 \\
&\Rightarrow (\lambda - 4)(\lambda + 1) = 0 \\
&\Rightarrow \lambda = 4 \text{ and } \lambda = -1
\end{aligned}$$

Let a non-zero vector $\mathbf{v} = \begin{bmatrix} x \\ y \end{bmatrix}$

We have,

$$\begin{aligned}
&\mathbf{A}\mathbf{v} = \lambda\mathbf{v} \\
&\begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \lambda \begin{bmatrix} x \\ y \end{bmatrix} \\
&\Rightarrow \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = -1 \begin{bmatrix} x \\ y \end{bmatrix} \quad [\text{When } \lambda = -1] \\
&\Rightarrow \begin{bmatrix} x + 2y \\ 3x + 2y \end{bmatrix} = -1 \begin{bmatrix} x \\ y \end{bmatrix}
\end{aligned}$$

Then we obtain the system of equations

$$x + 2y = -x$$

$$3x + 2y = -y$$

$$\begin{aligned}
&\Rightarrow 2x + 2y = 0 \\
&3x + 3y = 0
\end{aligned}$$

$$\Rightarrow x + y = 0 \quad [\text{Dividing by 2 on both sides}]$$

$$x + y = 0 \quad [\text{Dividing by 3 on both sides}]$$

Since the 1st and 2nd equations are identical. We can disregard one of them

$$x + y = 0$$

In echelon form, there are only one equation in two unknowns, then the system has a non-zero solution and in particular $(2-1) = 1$ free variable, which is y .

[Note Here total variables are two, i.e x, y and blocked variable is x , hence free variable is y]

Let, $y = 2 \therefore x = -2$

$$\mathbf{v} = \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -2 \\ 2 \end{bmatrix} \text{ is a non-zero Eigen vector belonging to the Eigen value } \lambda = -1$$

Again, for $\lambda = 4$

$$A\mathbf{v} = \lambda\mathbf{v}$$

$$\begin{aligned} & \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \lambda \begin{bmatrix} x \\ y \end{bmatrix} \\ \Rightarrow & \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 4 \begin{bmatrix} x \\ y \end{bmatrix} \\ \Rightarrow & \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 4x \\ 4y \end{bmatrix} \\ \Rightarrow & \begin{bmatrix} x + 2y \\ 3x + 2y \end{bmatrix} = \begin{bmatrix} 4x \\ 4y \end{bmatrix} \end{aligned}$$

Then we obtain the system of equations

$$x + 2y = 4x$$

$$3x + 2y = 4y$$

$$\Rightarrow -3x + 2y = 0$$

$$3x - 2y = 0$$

$$\Rightarrow 3x - 2y = 0 \quad [\text{Multiplying by -1 on both sides}]$$

$$3x - 2y = 0$$

Since the 1st and 2nd equations are identical. We can disregard one of them

So, $3x - 2y = 0$

In echelon form, there are only one equation in two unknowns, then the system has a non-zero solution and in particular $(2-1)=1$ free variable, which is y .

[Note Here total variables are two, i.e x, y and blocked variable is x , hence free variable is y]

Let, $y = 3$

$$3x = 6 \therefore x = 2$$

$$\mathbf{v} = \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \end{bmatrix} \text{ is a non-zero eigen vector belonging to the eigen value } \lambda = 4.$$

$\therefore -1, 4$ are the eigen values of the matrix A and their respective eigen vectors are

$$\begin{bmatrix} -2 \\ 2 \end{bmatrix} \text{ and } \begin{bmatrix} 2 \\ 3 \end{bmatrix}.$$

Example 25: Find the Eigen vectos of the matrix $\begin{bmatrix} 3 & 1 & 1 \\ 1 & 5 & 1 \\ 1 & 1 & 3 \end{bmatrix}$

$$\text{Let, } A = \begin{bmatrix} 3 & 1 & 1 \\ 1 & 5 & 1 \\ 1 & 1 & 3 \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 3 & 1 & 1 \\ 1 & 5 & 1 \\ 1 & 1 & 3 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 3 & 1 & 1 \\ 1 & 5 & 1 \\ 1 & 1 & 3 \end{bmatrix} - \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 3-\lambda & 1 & 1 \\ 1 & 5-\lambda & 1 \\ 1 & 1 & 3-\lambda \end{bmatrix}$$

$$\therefore |A - \lambda I| = \begin{vmatrix} 3-\lambda & 1 & 1 \\ 1 & 5-\lambda & 1 \\ 1 & 1 & 3-\lambda \end{vmatrix}$$

$$= (3-\lambda)[(5-\lambda)(3-\lambda)-1] - 1[1(3-\lambda)-1] + 1(1-(5-\lambda))$$

$$= (3-\lambda)(\lambda^2 - 8\lambda + 15 - 1) - 1(3-\lambda-1) + 1(1-5+\lambda)$$

$$= (3-\lambda)(\lambda^2 - 8\lambda + 14) - (2-\lambda) + (\lambda-4)$$

$$= 3\lambda^2 - 24\lambda + 42 - \lambda^3 + 8\lambda^2 - 14\lambda - 2 + \lambda + \lambda - 4$$

$$= -\lambda^3 + 11\lambda^2 - 36\lambda + 36$$

$$|A - \lambda I| = 0$$

$$-\lambda^3 + 11\lambda^2 - 36\lambda + 36 = 0$$

$$\Rightarrow \lambda^3 - 11\lambda^2 + 36\lambda - 36 = 0 \text{ -----(i)}$$

If $\lambda = 3$ then the equation (i) is satisfied.

So $\lambda - 3$ is a factor of $\lambda^3 - 11\lambda^2 + 36\lambda - 36 = 0$

Now, Follow the Remainder System

$$\begin{array}{r|l} \lambda - 3 & \lambda^3 - 11\lambda^2 + 36\lambda - 36 \\ & \lambda^3 - 3\lambda^2 \\ & - \quad + \\ & \hline & -8\lambda^2 + 36\lambda - 36 \\ & -8\lambda^2 + 24\lambda \\ & + \quad - \\ & \hline & 12\lambda - 36 \\ & 12\lambda - 36 \\ & - \quad + \\ & \hline & 0 \end{array}$$

$$\Rightarrow \lambda^3 - 11\lambda^2 + 36\lambda - 36 = 0$$

$$\Rightarrow (\lambda - 3)(\lambda^2 - 8\lambda + 12) = 0 \quad \dots\dots\dots(ii)$$

$$\Rightarrow (\lambda - 3)(\lambda^2 - 6\lambda - 2\lambda + 12) = 0$$

$$\Rightarrow (\lambda - 3)(\lambda - 6)(\lambda - 2) = 0$$

$$\therefore \lambda = 3, 6, 2$$

Let a non-zero vector, $v = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$

We have,

$$Av = \lambda v$$

$$\therefore Av = \lambda v$$

$$\begin{bmatrix} 3 & 1 & 1 \\ 1 & 5 & 1 \\ 1 & 1 & 3 \end{bmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \lambda \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$\begin{bmatrix} 3 & 1 & 1 \\ 1 & 5 & 1 \\ 1 & 1 & 3 \end{bmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 3 \begin{pmatrix} x \\ y \\ z \end{pmatrix} \quad [\text{For } \lambda = 3]$$

$$\Rightarrow \begin{pmatrix} 3x + y + z \\ x + 5y + z \\ x + y + 3z \end{pmatrix} = \begin{pmatrix} 3x \\ 3y \\ 3z \end{pmatrix}$$

Thus we obtain following system:

$$3x + y + z = 3x$$

$$x + 5y + z = 3y$$

$$x + y + 3z = 3z$$

$$\Rightarrow y + z = 0$$

$$x + 2y + z = 0$$

$$x + y = 0$$

Thus we obtain following system:

$$x + 2y + z = 0$$

$$\begin{aligned}x + y + 0.z &= 0 \\ y + z &= 0\end{aligned}$$

We have,

$$L_i \rightarrow -a_{i1}L_1 + a_{i1}L_i$$

$$i = 2, L_2 \rightarrow -a_{21}L_1 + a_{11}L_2$$

$$\begin{aligned}&= -1(x + 2y + z) + 1(x + y + 0.z) \\ &= -y - z\end{aligned}$$

Thus we obtain following system:

$$\therefore x + 2y + z = 0$$

$$-y - z = 0$$

$$y + z = 0$$

Thus we obtain following system:

$$\therefore x + 2y + z = 0$$

$$y + z = 0$$

[Multiplying by -1 on both sides]

$$y + z = 0$$

Since the 2nd and 3rd equations are identical. We can disregard one of them

Thus we obtain following system:

$$x + 2y + z = 0$$

$$y + z = 0$$

In echelon form, there are only two equations in three unknowns, then the system has a non-zero solution and in particular $(3-2)=1$ free variable, which is z .

[Note Here total variables are three, i.e x, y, z and blocked variable is x and y , hence free variable is z]

Let $z = 1$

$$\therefore y = -1$$

$$x = 1$$

$$v = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \text{ is a non-zero eigen vector belonging to the eigen value } \lambda = 3$$

Again

$$\text{Let a non-zero vector, } v = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

We have,

$$Av = \lambda v$$

$$\begin{bmatrix} 3 & 1 & 1 \\ 1 & 5 & 1 \\ 1 & 1 & 3 \end{bmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \lambda \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$\begin{bmatrix} 3 & 1 & 1 \\ 1 & 5 & 1 \\ 1 & 1 & 3 \end{bmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 6 \begin{pmatrix} x \\ y \\ z \end{pmatrix} \text{ [For } \lambda = 6]$$

$$\Rightarrow \begin{pmatrix} 3x + y + z \\ x + 5y + z \\ x + y + 3z \end{pmatrix} = \begin{pmatrix} 6x \\ 6y \\ 6z \end{pmatrix}$$

$$\Rightarrow 3x + y + z = 6x$$

$$x + 5y + z = 6y$$

$$x + y + 3z = 6z$$

$$\Rightarrow -3x + y + z = 0$$

$$x - y + z = 0$$

$$x + y - 3z = 0$$

$$\Rightarrow 3x - y - z = 0$$

$$x - y + z = 0$$

$$x + y - 3z = 0$$

Thus we obtain the following system

$$3x - y - z = 0$$

$$x - y + z = 0$$

$$x + y - 3z = 0$$

We have,

$$L_i \rightarrow -a_{i1}L_1 + a_{i1}L_i$$

$$i = 2, L_2 \rightarrow -a_{21}L_1 + a_{11}L_2$$

$$L_2 \rightarrow -1(3x - y - z) + 3(x - y + z)$$

$$= -3x + y + z + 3x - 3y + 3z$$

$$= -2y + 4z$$

$$i = 3, L_3 \rightarrow -a_{31}L_1 + a_{11}L_3$$

$$L_3 \rightarrow -1(3x - y - z) + 3(x + y - 3z)$$

$$= -3x + y + z + 3x + 3y - 9z = 0$$

$$\Rightarrow 4y - 8z = 0$$

Thus we obtain following system:

$$\begin{aligned}
3x - y - z &= 0 \\
-2y + 4z &= 0 \\
4y - 8z &= 0 \\
\Rightarrow 3x - y - z &= 0 \\
y - 2z &= 0 \\
y - 2z &= 0
\end{aligned}$$

Since the 2nd and 3rd equations are identical. We can disregard one of them

$$\begin{aligned}
\cancel{3x - y - z} &= 0 \\
y - 2z &= 0
\end{aligned}$$

In echelon form, there are only two equations in three unknowns, then the system has a non-zero solution and in particular (3-2)=1 free variable, which is z.

[Note Here total variables are three, i.e x, y, z and blocked variable is x and y, hence free variable is z]

Let,

$$z = 1$$

$$\therefore y = 2$$

$$x = 1$$

$$v = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} \text{ is a non-zero eigen vector belonging to the eigen value } \lambda = 6$$

Again

$$\text{Let a non-zero vector, } v = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

We have,

$$Av = \lambda v$$

$$\therefore Av = \lambda v$$

$$\begin{aligned}
\begin{bmatrix} 3 & 1 & 1 \\ 1 & 5 & 1 \\ 1 & 1 & 3 \end{bmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} &= \lambda \begin{pmatrix} x \\ y \\ z \end{pmatrix} \\
\begin{bmatrix} 3 & 1 & 1 \\ 1 & 5 & 1 \\ 1 & 1 & 3 \end{bmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} &= 2 \begin{pmatrix} x \\ y \\ z \end{pmatrix} \text{ [For } \lambda = 2]
\end{aligned}$$

$$\Rightarrow \begin{pmatrix} 3x + y + z \\ x + 5y + z \\ x + y + 3z \end{pmatrix} = \begin{pmatrix} 2x \\ 2y \\ 2z \end{pmatrix}$$

$$\Rightarrow 3x + y + z = 2x$$

$$x + 5y + z = 2y$$

$$x + y + 3z = 2z$$

$$\Rightarrow x + y + z = 0$$

$$x + 3y + z = 0$$

$$x + y + z = 0$$

We have,

$$L_i \rightarrow -a_{i1}L_1 + a_{i1}L_i$$

$$i = 2, L_2 \rightarrow -a_{21}L_1 + a_{11}L_2$$

$$L_2 \rightarrow -1(x + y + z) + 1(x + 3y + z) \\ = 2y$$

$$i = 3, L_3 \rightarrow -a_{31}L_1 + a_{11}L_3$$

$$L_3 \rightarrow -1(x + y + z) + 1(x + y + z) \\ = 0$$

Thus we obtain the following system

$$x + y + z = 0$$

$$2y = 0$$

$$0 = 0$$

$$\Rightarrow \begin{matrix} x + y + z = 0 \\ y = 0 \end{matrix}$$

We get, $y = 0$ then

$$x + 0 + z = 0$$

Thus we obtain the following system

$$\Rightarrow \textcircled{x + z} = 0$$

In echelon form, there is only one equation in two unknowns, then the system has a non-zero solution and in particular $(2-1)=1$ free variable, which is z .

Let,

$$\begin{aligned} \mathbf{z} &= \mathbf{1} \\ \therefore \mathbf{x} &= -\mathbf{1} \\ \mathbf{v} &= \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} \text{ is a non-zero eigen vector belonging to the eigen value } \lambda = 2 \end{aligned}$$

Let P be the matrix whose columns are the Eigen vectors of the given matrix

$$\mathbf{P} = \begin{bmatrix} 1 & -1 & 1 \\ -1 & 0 & 2 \\ 1 & 1 & 1 \end{bmatrix}$$

Diagonalization: Let $\mathbf{T}: \mathbf{v} \rightarrow \mathbf{v}$ be a linear operator on a vector space $\mathbf{v}$ with finite dimension n . Note that $\mathbf{T}$ can be represented by a diagonal matrix.

$$\begin{bmatrix} k_1 & 0 & 0 & 0 & - & - & 0 \\ 0 & k_2 & 0 & 0 & - & - & 0 \\ 0 & 0 & k_3 & 0 & - & - & 0 \\ 0 & 0 & 0 & k_4 & - & - & 0 \\ - & - & - & - & - & - & - \\ - & - & - & - & - & - & - \\ 0 & 0 & 0 & 0 & - & - & k_n \end{bmatrix}$$

If and only if there exists a basis $(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \dots, \mathbf{v}_n)$ of $\mathbf{V}$ for which

$$\mathbf{T}(\mathbf{v}_1) = k_1 \mathbf{v}_1$$

$$\mathbf{T}(\mathbf{v}_2) = k_2 \mathbf{v}_2$$

$$\mathbf{T}(\mathbf{v}_3) = k_3 \mathbf{v}_3$$

.....

.....

$$\mathbf{T}(\mathbf{v}_n) = k_n \mathbf{v}_n$$

That is, such that the vectors $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \dots, \mathbf{v}_n$ are eigen vectors of $\mathbf{T}$ belonging respectively to eigen values $k_1, k_2, k_3, \dots, k_n$.

Example 26: If $\mathbf{D}$ be the different operator on the vector space $\mathbf{V}$ of differentiable functions. Then $\mathbf{D}(\mathbf{e}^{a_k t}) = a_k \mathbf{e}^{a_k t}$ accordingly $\mathbf{e}^{a_1 t}, \mathbf{e}^{a_2 t}, \dots, \mathbf{e}^{a_n t}$ are Eigen vectors of $\mathbf{D}$ belonging to the distinct Eigen values $a_1, a_2, a_3, \dots, a_n$

Diagonalization of a Matrix:

$$\mathbf{D} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \begin{bmatrix} \lambda_1 & 0 & 0 & - & 0 \\ 0 & \lambda_2 & 0 & - & 0 \\ 0 & 0 & \lambda_3 & - & 0 \\ - & - & - & - & - \\ 0 & 0 & 0 & 0 & \lambda_n \end{bmatrix}$$

Where,

D is a diagonalized matrix consisting of Eigen values.

P is a matrix consisting of Eigen vectors

$\mathbf{P}^{-1}$ is a inverse matrix of P

A is given Matrix

$\lambda_1, \lambda_2, \lambda_3, \dots, \lambda_n$ are eigen values of A.

Example 27: Diagonalize the matrix $\begin{bmatrix} 1 & -3 & 3 \\ 3 & -5 & 3 \\ 6 & -6 & 4 \end{bmatrix}$

Solution:

Let, $\mathbf{A} = \begin{bmatrix} 1 & -3 & 3 \\ 3 & -5 & 3 \\ 6 & -6 & 4 \end{bmatrix}$

$$\Rightarrow \mathbf{A} - \lambda\mathbf{I} = \begin{bmatrix} 1 & -3 & 3 \\ 3 & -5 & 3 \\ 6 & -6 & 4 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\Rightarrow \mathbf{A} - \lambda\mathbf{I} = \begin{bmatrix} 1 & -3 & 3 \\ 3 & -5 & 3 \\ 6 & -6 & 4 \end{bmatrix} - \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix}$$

$$\Rightarrow \mathbf{A} - \lambda\mathbf{I} = \begin{bmatrix} 1-\lambda & -3 & 3 \\ 3 & -5-\lambda & 3 \\ 6 & -6 & 4-\lambda \end{bmatrix}$$

$$|\mathbf{A} - \lambda\mathbf{I}| = \begin{vmatrix} 1-\lambda & -3 & 3 \\ 3 & -5-\lambda & 3 \\ 6 & -6 & 4-\lambda \end{vmatrix}$$

$$= (1-\lambda)[(-5-\lambda)(4-\lambda) - (-6)(3)] - (-3)[3(4-\lambda) - 3 \times 6] + 3[3(-6) - 6(-5-\lambda)]$$

$$= (1-\lambda)[(-20+5\lambda-4\lambda+\lambda^2)+18] + 3[12-3\lambda-18] + 3[-18+30+6\lambda]$$

$$= (1-\lambda)[\lambda+\lambda^2-2] + 3[-3\lambda-6] + 3[12+6\lambda]$$

$$= \lambda + \lambda^2 - 2 - \lambda^2 - \lambda^3 + 2\lambda - 9\lambda - 18 + 36 + 18\lambda$$

$$-\lambda^3 + 12\lambda + 16$$

Now,

$$|\mathbf{A} - \lambda \mathbf{I}| = 0$$

$$\Rightarrow -\lambda^3 + 12\lambda + 16 = 0$$

$$\Rightarrow \lambda^3 - 12\lambda - 16 = 0$$

$$\Rightarrow \lambda^3 + 2\lambda^2 - 2\lambda^2 - 4\lambda - 8\lambda - 16 = 0$$

$$\Rightarrow \lambda^2(\lambda + 2) - 2\lambda(\lambda + 2) - 8(\lambda + 2) = 0$$

$$\Rightarrow (\lambda + 2)(\lambda^2 - 2\lambda - 8) = 0$$

$$\Rightarrow (\lambda + 2)(\lambda^2 - 4\lambda + 2\lambda - 8) = 0$$

$$\Rightarrow (\lambda + 2)(\lambda - 4)(\lambda + 2) = 0$$

$$\Rightarrow \lambda = -2, 4, -2$$

$$\text{Let, } \mathbf{v}_1 = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

We have, $\mathbf{A}\mathbf{v}_1 = \lambda\mathbf{v}_1$

$$\begin{bmatrix} 1 & -3 & 3 \\ 3 & -5 & 3 \\ 6 & -6 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \lambda \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1 & -3 & 3 \\ 3 & -5 & 3 \\ 6 & -6 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 4 \begin{bmatrix} x \\ y \\ z \end{bmatrix} \quad [\lambda = 4]$$

$$\Rightarrow \begin{bmatrix} 1 & -3 & 3 \\ 3 & -5 & 3 \\ 6 & -6 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 4x \\ 4y \\ 4z \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x - 3y + 3z \\ 3x - 5y + 3z \\ 6x - 6y + 4z \end{bmatrix} = \begin{bmatrix} 4x \\ 4y \\ 4z \end{bmatrix}$$

$$\Rightarrow x - 3y + 3z = 4x$$

$$3x - 5y + 3z = 4y$$

$$6x - 6y + 4z = 4z$$

$$\Rightarrow -3x - 3y + 3z = 0$$

$$3x - 9y + 3z = 0$$

$$6x - 6y = 0$$

$$\Rightarrow x + y - z = 0$$

$$\begin{aligned}
 & x - 3y + z = 0 \\
 & x - y = 0 \\
 \Rightarrow & \begin{aligned} x + y - z &= 0 \\ x - 3y + z &= 0 \\ x - y + 0.z &= 0 \end{aligned}
 \end{aligned}$$

We have,

$$L_i \rightarrow -a_{i1}L_1 + a_{i1}L_i$$

$$i = 2, L_2 \rightarrow -a_{21}L_1 + a_{11}L_2$$

$$\begin{aligned}
 &= -1(x + y - z) + 1(x - 3y + z) \\
 &= -4y + 2z
 \end{aligned}$$

$$i = 3, L_3 \rightarrow -a_{31}L_1 + a_{11}L_3$$

$$\begin{aligned}
 L_3 &\rightarrow -1(x + y - z) + 1(x - y + 0.z) \\
 &= -2y + z
 \end{aligned}$$

Thus we obtain following system

$$\begin{aligned}
 & x + y - z = 0 \\
 & -4y + 2z = 0 \\
 & -2y + z = 0 \\
 \Rightarrow & x + y - z = 0 \\
 & 2y - z = 0 \quad \text{[Dividing by -2 on both sides]} \\
 & 2y - z = 0 \quad \text{[Multiplying by -1 on both sides]}
 \end{aligned}$$

Since the 2nd and 3rd equations are identical. We can disregard one of them

$$\begin{aligned}
 \Rightarrow & x + y - z = 0 \\
 & 2y - z = 0
 \end{aligned}$$

In echelon form, there are only two equations in three unknowns, then the system has a non-zero solution and in particular (3-2)=1 free variable, which is z.

[Note Here total variables are three, i.e x, y, z and blocked variable is x and y, hence free variable is z]

Let, $z = 2$

$$\therefore y = 1 \text{ and } x = 1$$

$$\therefore v_1 = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$$

$$\text{Let, } \mathbf{v}_2 = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

We have,

$$\mathbf{A}\mathbf{v}_2 = \lambda\mathbf{v}_2$$

$$\begin{bmatrix} 1 & -3 & 3 \\ 3 & -5 & 3 \\ 6 & -6 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \lambda \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1 & -3 & 3 \\ 3 & -5 & 3 \\ 6 & -6 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = -2 \begin{bmatrix} x \\ y \\ z \end{bmatrix} \quad [\lambda = -2]$$

$$\Rightarrow \begin{bmatrix} 1 & -3 & 3 \\ 3 & -5 & 3 \\ 6 & -6 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -2x \\ -2y \\ -2z \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x - 3y + 3z \\ 3x - 5y + 3z \\ 6x - 6y + 4z \end{bmatrix} = \begin{bmatrix} -2x \\ -2y \\ -2z \end{bmatrix}$$

$$\Rightarrow \begin{aligned} x - 3y + 3z &= -2x \\ 3x - 5y + 3z &= -2y \\ 6x - 6y + 4z &= -2z \end{aligned}$$

$$\Rightarrow \begin{aligned} 3x - 3y + 3z &= 0 \\ 3x - 3y + 3z &= 0 \\ 6x - 6y + 6z &= 0 \end{aligned}$$

$$\Rightarrow \begin{aligned} x - y + z &= 0 \\ x - y + z &= 0 \\ x - y + z &= 0 \end{aligned}$$

Since the 1st, 2nd and 3rd equations are identical. We can disregard two of them

$$\Rightarrow \textcircled{x} - y + z = 0$$

In echelon form, there are only one equation in three unknowns, then the system has a non-zero solution and in particular (3-1)=2 free variable, which is y and z.

Let, y = 1 and z = 0

$$\therefore x = 1$$

$$\therefore \mathbf{v}_2 = \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$$

$$\text{Let, } \mathbf{v}_3 = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

We have,

$$\mathbf{A}\mathbf{v}_3 = \lambda\mathbf{v}_3$$

$$\begin{bmatrix} 1 & -3 & 3 \\ 3 & -5 & 3 \\ 6 & -6 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \lambda \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1 & -3 & 3 \\ 3 & -5 & 3 \\ 6 & -6 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = -2 \begin{bmatrix} x \\ y \\ z \end{bmatrix} \quad [\lambda = -2]$$

$$\Rightarrow \begin{bmatrix} 1 & -3 & 3 \\ 3 & -5 & 3 \\ 6 & -6 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -2x \\ -2y \\ -2z \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x - 3y + 3z \\ 3x - 5y + 3z \\ 6x - 6y + 4z \end{bmatrix} = \begin{bmatrix} -2x \\ -2y \\ -2z \end{bmatrix}$$

$$\Rightarrow \begin{aligned} x - 3y + 3z &= -2x \\ 3x - 5y + 3z &= -2y \\ 6x - 6y + 4z &= -2z \end{aligned}$$

$$\Rightarrow \begin{aligned} 3x - 3y + 3z &= 0 \\ 3x - 3y + 3z &= 0 \\ 6x - 6y + 6z &= 0 \end{aligned}$$

$$\Rightarrow \begin{aligned} x - y + z &= 0 && [\text{Dividing by 3 on both sides}] \\ x - y + z &= 0 && [\text{Dividing by 3 on both sides}] \\ x - y + z &= 0 && [\text{Dividing by 6 on both sides}] \end{aligned}$$

Since the 1st, 2nd and 3rd equations are identical. We can disregard two of them

$$\Rightarrow \textcircled{x} - y + z = 0$$

In echelon form, there are only one equation in three unknowns, then the system has a non-zero solution and in particular (3-1)=2 free variable, which is y and z.

Let, $y = 0$ and $z = 1$

$$\therefore x = -1$$

$$\therefore \mathbf{v}_3 = \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$$

$$\mathbf{P} = [\mathbf{v}_1 \quad \mathbf{v}_2 \quad \mathbf{v}_3]$$

$$\mathbf{P} = \begin{bmatrix} 1 & 1 & -1 \\ 1 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix}$$

We know that, $\mathbf{D} = \mathbf{P}^{-1} \mathbf{A} \mathbf{P}$

Now,

$$\mathbf{P} = \begin{bmatrix} 1 & 1 & -1 \\ 1 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix}$$

$$\therefore \text{Co-factor of } \mathbf{a}_{11}(\mathbf{1}) = \mathbf{c}_{11} = + \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 1$$

$$\therefore \text{Co-factor of } \mathbf{a}_{12}(\mathbf{1}) = \mathbf{c}_{12} = - \begin{vmatrix} 1 & 0 \\ 2 & 1 \end{vmatrix} = -1$$

$$\therefore \text{Co-factor of } \mathbf{a}_{13}(-1) = \mathbf{c}_{13} = + \begin{vmatrix} 1 & 1 \\ 2 & 0 \end{vmatrix} = -2$$

$$\therefore \text{Co-factor of } \mathbf{a}_{21}(\mathbf{1}) = \mathbf{c}_{21} = - \begin{vmatrix} 1 & -1 \\ 0 & 1 \end{vmatrix} = -1$$

$$\therefore \text{Co-factor of } \mathbf{a}_{22}(\mathbf{1}) = \mathbf{c}_{22} = + \begin{vmatrix} 1 & -1 \\ 2 & 1 \end{vmatrix} = 3$$

$$\therefore \text{Co-factor of } \mathbf{a}_{23}(\mathbf{0}) = \mathbf{c}_{23} = - \begin{vmatrix} 1 & 1 \\ 2 & 0 \end{vmatrix} = 2$$

$$\therefore \text{Co-factor of } \mathbf{a}_{31}(\mathbf{2}) = \mathbf{c}_{31} = + \begin{vmatrix} 1 & -1 \\ 1 & 0 \end{vmatrix} = 1$$

$$\therefore \text{Co-factor of } \mathbf{a}_{32}(\mathbf{0}) = \mathbf{c}_{32} = - \begin{vmatrix} 1 & -1 \\ 1 & 0 \end{vmatrix} = -1$$

$$\therefore \text{Co-factor of } \mathbf{a}_{33}(\mathbf{1}) = \mathbf{c}_{33} = + \begin{vmatrix} 1 & 1 \\ 1 & 1 \end{vmatrix} = 0$$

$$\text{Co-factor Matrix: } \mathbf{P}^C = \begin{bmatrix} 1 & -1 & -2 \\ -1 & 3 & 2 \\ 1 & -1 & 0 \end{bmatrix}$$

$$\text{Adjoint of } \mathbf{P} = (\mathbf{P}^C)^t = \begin{bmatrix} 1 & -1 & 1 \\ -1 & 3 & -1 \\ -2 & 2 & 0 \end{bmatrix}$$

$$\text{Given, } \mathbf{P} = \begin{bmatrix} 1 & 1 & -1 \\ 1 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix}$$

$$|\mathbf{P}| = \begin{vmatrix} 1 & 1 & -1 \\ 1 & 1 & 0 \\ 2 & 0 & 1 \end{vmatrix} = 1(1 \times 1 - 0) - 1(1 - 0) + (-1)(0 - 2) = 1 - 1 + 2 = 2$$

$$\therefore \mathbf{P}^{-1} = \frac{\text{Adjoint of } \mathbf{P}}{|\mathbf{P}|} = \frac{1}{2} \begin{bmatrix} 1 & -1 & 1 \\ -1 & 3 & -1 \\ -2 & 2 & 0 \end{bmatrix}$$

$$\mathbf{D} = \mathbf{P}^{-1} \mathbf{A} \mathbf{P} = \frac{1}{2} \begin{bmatrix} 1 & -1 & 1 \\ -1 & 3 & -1 \\ -2 & 2 & 0 \end{bmatrix} \begin{bmatrix} 1 & -3 & 3 \\ 3 & -5 & 3 \\ 6 & -6 & 4 \end{bmatrix} \begin{bmatrix} 1 & 1 & -1 \\ 1 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix}$$

$$= \frac{1}{2} \begin{bmatrix} 1 & -1 & 1 \\ -1 & 3 & -1 \\ -2 & 2 & 0 \end{bmatrix} \begin{bmatrix} 1 \times 1 - 3 \times 1 + 3 \times 2 & 1 \times 1 - 3 \times 1 + 3 \times 0 & 1(-1) + (-3)(0) + 3 \times 1 \\ 3 \times 1 - 5 \times 1 + 3 \times 2 & 3 \times 1 - 5 \times 1 + 3 \times 0 & 3(-1) - 5(0) + 3 \times 1 \\ 6 \times 1 - 6 \times 1 + 4 \times 2 & 6 \times 1 - 6 \times 1 + 4 \times 0 & 6(-1) - 6 \times 0 + 4 \times 1 \end{bmatrix}$$

$$= \frac{1}{2} \begin{bmatrix} 1 & -1 & 1 \\ -1 & 3 & -1 \\ -2 & 2 & 0 \end{bmatrix} \begin{bmatrix} 1-3+6 & 1-3+0 & -1+0+3 \\ 3-5+6 & 3-5+0 & -3-0+3 \\ 6-6+8 & 6-6+0 & -6-0+4 \end{bmatrix}$$

$$= \frac{1}{2} \begin{bmatrix} 1 & -1 & 1 \\ -1 & 3 & -1 \\ -2 & 2 & 0 \end{bmatrix} \begin{bmatrix} 4 & -2 & 2 \\ 4 & -2 & 0 \\ 8 & 0 & -2 \end{bmatrix}$$

$$= \frac{1}{2} \begin{bmatrix} 1 \times 4 - 1 \times 4 + 1 \times 8 & 1(-2) + (-1)(-2) + 1 \times 0 & 1 \times 2 - 1 \times 0 + 1(-2) \\ -1 \times 4 + 3 \times 4 + (-1) \times 8 & (-1)(-2) + 3(-2) + (-1) \times 0 & (-1) \times 2 + 3 \times 0 + (-1)(-2) \\ -2 \times 4 + 2 \times 4 + 0 \times 8 & (-2)(-2) + 2(-2) + 0 \times 0 & -2 \times 2 + 2 \times 0 + 0(-2) \end{bmatrix}$$

$$= \frac{1}{2} \begin{bmatrix} 4 - 4 + 8 & -2 + 2 & 2 - 2 \\ -4 + 12 - 8 & 2 - 6 & -2 + 2 \\ -8 + 8 + 0 & 4 - 4 & -4 \end{bmatrix}$$

$$= \frac{1}{2} \begin{bmatrix} 8 & 0 & 0 \\ 0 & -4 & 0 \\ 0 & 0 & -4 \end{bmatrix}$$

$$D = \begin{bmatrix} 4 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{bmatrix}$$

Example 28: Diagonalize the matrix: $\begin{bmatrix} 4 & 1 \\ 1 & 4 \end{bmatrix}$

Solution:

$$\text{Let, } A = \begin{bmatrix} 4 & 1 \\ 1 & 4 \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 4 & 1 \\ 1 & 4 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 4 & 1 \\ 1 & 4 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 4 - \lambda & 1 \\ 1 & 4 - \lambda \end{bmatrix}$$

$$|A - \lambda I| = \begin{vmatrix} 4 - \lambda & 1 \\ 1 & 4 - \lambda \end{vmatrix}$$

$$|A - \lambda I| = (4 - \lambda)(4 - \lambda) - 1$$

$$|A - \lambda I| = 16 - 4\lambda - 4\lambda + \lambda^2 - 1$$

$$|A - \lambda I| = \lambda^2 - 8\lambda + 15$$

Now,

$$|A - \lambda I| = 0$$

$$\Rightarrow |A - \lambda I| = \lambda^2 - 8\lambda + 15 = 0$$

$$\Rightarrow |A - \lambda I| = \lambda^2 - 3\lambda - 5\lambda + 15 = 0$$

$$\Rightarrow |\mathbf{A} - \lambda \mathbf{I}| = \lambda^2 - 3\lambda - 5\lambda + 15 = 0$$

$$\Rightarrow |\mathbf{A} - \lambda \mathbf{I}| = (\lambda - 3)(\lambda - 5) = 0$$

$$\Rightarrow (\lambda - 3)(\lambda - 5) = 0$$

$$\Rightarrow \lambda = 3, \lambda = 5$$

We have, $\mathbf{A}\mathbf{v}_1 = \lambda\mathbf{v}_1$

$$\text{Let, } \mathbf{v}_1 = \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\begin{bmatrix} 4 & 1 \\ 1 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \lambda \begin{bmatrix} x \\ y \end{bmatrix}$$

$$\begin{bmatrix} 4 & 1 \\ 1 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 3 \begin{bmatrix} x \\ y \end{bmatrix} \quad [\text{When, } \lambda = 3]$$

$$\begin{bmatrix} 4 & 1 \\ 1 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 3x \\ 3y \end{bmatrix}$$

$$\begin{bmatrix} 4x + 1y \\ 1x + 4y \end{bmatrix} = \begin{bmatrix} 3x \\ 3y \end{bmatrix}$$

$$\begin{bmatrix} 4x + y \\ x + 4y \end{bmatrix} = \begin{bmatrix} 3x \\ 3y \end{bmatrix}$$

$$\Rightarrow \begin{aligned} 4x + y &= 3x \\ x + 4y &= 3y \end{aligned}$$

$$\Rightarrow \begin{aligned} x + y &= 0 \\ x + y &= 0 \end{aligned}$$

Since the 1st and 2nd equations are identical. We can disregard one of them

$$\Rightarrow \textcircled{x} + y = 0$$

In echelon form, there are only one equation in two unknowns, then the system has a non-zero solution and in particular $(2-1)=1$ free variable, which is y .

Let $y = 1, x = -1$

$$\therefore \mathbf{v}_1 = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

We have, $A\mathbf{v}_2 = \lambda\mathbf{v}_2$

$$\text{Let, } \mathbf{v}_2 = \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\begin{bmatrix} 4 & 1 \\ 1 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \lambda \begin{bmatrix} x \\ y \end{bmatrix}$$

$$\begin{bmatrix} 4 & 1 \\ 1 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 5 \begin{bmatrix} x \\ y \end{bmatrix} \quad [\text{When, } \lambda = 5]$$

$$\begin{bmatrix} 4 & 1 \\ 1 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 5x \\ 5y \end{bmatrix}$$

$$\begin{bmatrix} 4 \times x + 1 \times y \\ 1 \times x + 4 \times y \end{bmatrix} = \begin{bmatrix} 5x \\ 5y \end{bmatrix}$$

$$\begin{bmatrix} 4x + y \\ x + 4y \end{bmatrix} = \begin{bmatrix} 5x \\ 5y \end{bmatrix}$$

$$\Rightarrow \quad \begin{aligned} 4x + y &= 5x \\ x + 4y &= 5y \end{aligned}$$

$$\Rightarrow \quad \begin{aligned} -x + y &= 0 \\ x - y &= 0 \end{aligned}$$

$$\Rightarrow \quad \begin{aligned} x - y &= 0 \\ x - y &= 0 \end{aligned} \quad [\text{Multiplying by -1 on both sides}]$$

Since the 1st and 2nd equations are identical. We can disregard one of them

$$\Rightarrow \quad \cancel{x} - y = 0$$

In echelon form, there are only one equation in two unknowns, then the system has a non-zero solution and in particular $(2-1) = 1$ free variable, which is y .

$$\text{Let, } y = 1 \quad \therefore x = 1$$

$$\mathbf{v}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\mathbf{P} = [\mathbf{v}_1 \quad \mathbf{v}_2]$$

$$\therefore \mathbf{P} = \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix}$$

$$|\mathbf{P}| = (-1) \times 1 - 1 = -2$$

We know that, $\mathbf{D} = \mathbf{P}^{-1} \mathbf{A} \mathbf{P}$

Now,

$$\therefore \mathbf{P} = \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix}$$

$$\therefore \text{Co-factor of } \mathbf{a}_{11}(-1) = \mathbf{c}_{11} = +1$$

$$\therefore \text{Co-factor of } \mathbf{a}_{12}(1) = \mathbf{c}_{12} = -1$$

$$\therefore \text{Co-factor of } \mathbf{a}_{21}(1) = \mathbf{c}_{21} = -1$$

$$\therefore \text{Co-factor of } \mathbf{a}_{22}(1) = \mathbf{c}_{22} = +(-1) = -1$$

$$\text{Co-factor Matrix: } \mathbf{P}^C = \begin{bmatrix} 1 & -1 \\ -1 & -1 \end{bmatrix}$$

$$\text{Adjoint of } \mathbf{P} = (\mathbf{P}^C)^t = \begin{bmatrix} 1 & -1 \\ -1 & -1 \end{bmatrix}$$

$$\therefore \mathbf{P}^{-1} = \frac{\text{Adjoint of } \mathbf{P}}{|\mathbf{P}|} = \frac{1}{-2} \begin{bmatrix} 1 & -1 \\ -1 & -1 \end{bmatrix}$$

$$\mathbf{D} = \mathbf{P}^{-1} \mathbf{A} \mathbf{P} = \frac{1}{-2} \begin{bmatrix} 1 & -1 \\ -1 & -1 \end{bmatrix} \begin{bmatrix} 4 & 1 \\ 1 & 4 \end{bmatrix} \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix}$$

$$\mathbf{D} = \mathbf{P}^{-1} \mathbf{A} \mathbf{P} = \frac{1}{-2} \begin{bmatrix} 1 & -1 \\ -1 & -1 \end{bmatrix} \begin{bmatrix} 4(-1) + 1 \times 1 & 4 \times 1 + 1 \times 1 \\ 1(-1) + 4 \times 1 & 1 \times 1 + 4 \times 1 \end{bmatrix}$$

$$\mathbf{D} = \mathbf{P}^{-1} \mathbf{A} \mathbf{P} = \frac{1}{-2} \begin{bmatrix} 1 & -1 \\ -1 & -1 \end{bmatrix} \begin{bmatrix} -4 + 1 & 4 + 1 \\ -1 + 4 & 1 + 4 \end{bmatrix}$$

$$\mathbf{D} = \mathbf{P}^{-1} \mathbf{A} \mathbf{P} = \frac{1}{-2} \begin{bmatrix} 1 & -1 \\ -1 & -1 \end{bmatrix} \begin{bmatrix} -3 & 5 \\ 3 & 5 \end{bmatrix}$$

$$\mathbf{D} = \mathbf{P}^{-1} \mathbf{A} \mathbf{P} = \frac{1}{-2} \begin{bmatrix} 1(-3) + (-1) \times 3 & 1 \times 5 + (-1) \times 5 \\ (-1)(-3) + (-1) \times 3 & (-1) \times 5 + (-1) \times 5 \end{bmatrix}$$

$$\mathbf{D} = \mathbf{P}^{-1} \mathbf{A} \mathbf{P} = \frac{1}{-2} \begin{bmatrix} -3 - 3 & 5 - 5 \\ 3 - 3 & -5 - 5 \end{bmatrix}$$

$$\mathbf{D} = \mathbf{P}^{-1} \mathbf{A} \mathbf{P} = \frac{1}{-2} \begin{bmatrix} -6 & 0 \\ 0 & -10 \end{bmatrix}$$

$$\mathbf{D} = \mathbf{P}^{-1} \mathbf{A} \mathbf{P} = \begin{bmatrix} 3 & 0 \\ 0 & 5 \end{bmatrix}$$

$$\therefore \mathbf{D} = \mathbf{P}^{-1} \mathbf{A} \mathbf{P} = \begin{bmatrix} 3 & 0 \\ 0 & 5 \end{bmatrix} \text{ Answer}$$

Example 29: Diagonalize the matrix: $\begin{bmatrix} 5 & 1 & 2 \\ 0 & 3 & 0 \\ 2 & 1 & 5 \end{bmatrix}$

Solution:

$$\text{Given, } A = \begin{bmatrix} 5 & 1 & 2 \\ 0 & 3 & 0 \\ 2 & 1 & 5 \end{bmatrix}$$

For Eigen values,

$$\Rightarrow A - \lambda I = \begin{bmatrix} 5 & 1 & 2 \\ 0 & 3 & 0 \\ 2 & 1 & 5 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\Rightarrow A - \lambda I = \begin{bmatrix} 5 & 1 & 2 \\ 0 & 3 & 0 \\ 2 & 1 & 5 \end{bmatrix} - \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix}$$

$$\Rightarrow A - \lambda I = \begin{bmatrix} 5-\lambda & 1 & 2 \\ 0 & 3-\lambda & 0 \\ 2 & 1 & 5-\lambda \end{bmatrix}$$

$$\Rightarrow |A - \lambda I| = \begin{vmatrix} 5-\lambda & 1 & 2 \\ 0 & 3-\lambda & 0 \\ 2 & 1 & 5-\lambda \end{vmatrix}$$

$$= (5-\lambda)[(3-\lambda)(5-\lambda) - 0] - 1[0(5-\lambda) - 2 \times 0] + 2[0 \times 1 - 2(3-\lambda)]$$

$$= (5-\lambda)[15 - 3\lambda - 5\lambda + \lambda^2] - 1[0 - 0] + 2[0 - 6 + 2\lambda]$$

$$= (5-\lambda)[15 - 8\lambda + \lambda^2] + 2[-6 + 2\lambda]$$

$$= (5-\lambda)[15 - 8\lambda + \lambda^2] - 12 + 4\lambda$$

$$= 75 - 40\lambda + 5\lambda^2 - 15\lambda + 8\lambda^2 - \lambda^3 - 12 + 4\lambda$$

$$= 63 - 51\lambda + 13\lambda^2 - \lambda^3$$

$$= -\lambda^3 + 13\lambda^2 - 51\lambda + 63$$

$$\therefore |A - \lambda I| = 0$$

$$\Rightarrow -\lambda^3 + 13\lambda^2 - 51\lambda + 63 = 0$$

$$\Rightarrow \lambda^3 - 13\lambda^2 + 51\lambda - 63 = 0$$

$$\Rightarrow \lambda^3 - 3\lambda^2 - 10\lambda^2 + 30\lambda + 21\lambda - 63 = 0$$

$$\Rightarrow \lambda^2(\lambda - 3) - 10\lambda(\lambda - 3) + 21(\lambda - 3) = 0$$

$$\Rightarrow (\lambda - 3)(\lambda^2 - 10\lambda + 21) = 0$$

$$\Rightarrow (\lambda - 3)(\lambda^2 - 7\lambda - 3\lambda + 21) = 0$$

$$\Rightarrow (\lambda - 3)[\lambda(\lambda - 7) - 3(\lambda - 7)] = 0$$

$$\Rightarrow (\lambda - 3)(\lambda - 7)(\lambda - 3) = 0$$

$$\Rightarrow \lambda = 3, \lambda = 7, \lambda = 3$$

$$\Rightarrow \lambda = 3, \lambda = 3, \lambda = 7$$

Let a nonzero vector $\mathbf{v}_1 = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$

We have, $A\mathbf{v}_1 = \lambda\mathbf{v}_1$

$$\Rightarrow \begin{bmatrix} 5 & 1 & 2 \\ 0 & 3 & 0 \\ 2 & 1 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \lambda \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 5 & 1 & 2 \\ 0 & 3 & 0 \\ 2 & 1 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 3 \begin{bmatrix} x \\ y \\ z \end{bmatrix} \quad [\text{When, } \lambda = 3]$$

$$\Rightarrow \begin{bmatrix} 5 & 1 & 2 \\ 0 & 3 & 0 \\ 2 & 1 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3x \\ 3y \\ 3z \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 5x + y + 2z \\ 0 \times x + 3y + 0 \times z \\ 2x + y + 5z \end{bmatrix} = \begin{bmatrix} 3x \\ 3y \\ 3z \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 5x + y + 2z \\ 3y \\ 2x + y + 5z \end{bmatrix} = \begin{bmatrix} 3x \\ 3y \\ 3z \end{bmatrix}$$

$$\Rightarrow 5x + y + 2z = 3x$$

$$3y = 3y$$

$$2x + y + 5z = 3z$$

$$\Rightarrow 2x + y + 2z = 0$$

$$3y - 3y = 0$$

$$2x + y + 2z = 0$$

$$\Rightarrow 2x + y + 2z = 0$$

$$0 = 0$$

$$2x + y + 2z = 0$$

$$\Rightarrow 2x + y + 2z = 0$$

$$2x + y + 2z = 0$$

Since the 1st and 2nd equations are identical. We can disregard one of them

$$\Rightarrow 2x + y + 2z = 0$$

In echelon form, there are only one equation in three unknowns, then the system has a non-zero solution and in particular $(3-1)=2$ free variable, which is y and z.

Let,

$$z = 1$$

$$y = 2$$

$$\therefore 2x + 2 + 2 = 0$$

$$\Rightarrow x = -2$$

$$\therefore \mathbf{v}_1 = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -2 \\ 2 \\ 1 \end{bmatrix} \text{ For } \lambda = 3$$

$$\text{Let, } \mathbf{v}_2 = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

We have, $\mathbf{A}\mathbf{v}_2 = \lambda\mathbf{v}_2$

$$\Rightarrow \begin{bmatrix} 5 & 1 & 2 \\ 0 & 3 & 0 \\ 2 & 1 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \lambda \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 5 & 1 & 2 \\ 0 & 3 & 0 \\ 2 & 1 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 3 \begin{bmatrix} x \\ y \\ z \end{bmatrix} \text{ [When, } \lambda = 3 \text{]}$$

$$\Rightarrow \begin{bmatrix} 5 & 1 & 2 \\ 0 & 3 & 0 \\ 2 & 1 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3x \\ 3y \\ 3z \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 5x + y + 2z \\ 0 \times x + 3y + 0 \times z \\ 2x + y + 5z \end{bmatrix} = \begin{bmatrix} 3x \\ 3y \\ 3z \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 5x + y + 2z \\ 3y \\ 2x + y + 5z \end{bmatrix} = \begin{bmatrix} 3x \\ 3y \\ 3z \end{bmatrix}$$

$$\Rightarrow 5x + y + 2z = 3x$$

$$3y = 3y$$

$$2x + y + 5z = 3z$$

$$\Rightarrow 2x + y + 2z = 0$$

$$3y - 3y = 0$$

$$2x + y + 2z = 0$$

$$\Rightarrow 2x + y + 2z = 0$$

$$0 = 0$$

$$2x + y + 2z = 0$$

$$\Rightarrow 2x + y + 2z = 0$$

$$2x + y + 2z = 0$$

Since the 1st and 2nd equations are identical. We can disregard one of them

$$\Rightarrow 2x + y + 2z = 0$$

In echelon form, there are only one equation in three unknowns, then the system has a non-zero solution and in particular $(3-1)=2$ free variable, which is y and z.

Let,

$$z = 1$$

$$y = 0$$

$$\therefore 2x + 0 + 2 = 0$$

$$\Rightarrow x = -1$$

$$\therefore v_2 = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

$$\text{Let, } v_3 = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

We have, $Av_3 = \lambda v_3$

$$\Rightarrow \begin{bmatrix} 5 & 1 & 2 \\ 0 & 3 & 0 \\ 2 & 1 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \lambda \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 5 & 1 & 2 \\ 0 & 3 & 0 \\ 2 & 1 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 7 \begin{bmatrix} x \\ y \\ z \end{bmatrix} \quad [\text{When, } \lambda = 7]$$

$$\Rightarrow \begin{bmatrix} 5 & 1 & 2 \\ 0 & 3 & 0 \\ 2 & 1 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 7x \\ 7y \\ 7z \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 5x + y + 2z \\ 0 \times x + 3y + 0 \times z \\ 2x + y + 5z \end{bmatrix} = \begin{bmatrix} 7x \\ 7y \\ 7z \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 5x + y + 2z \\ 3y \\ 2x + y + 5z \end{bmatrix} = \begin{bmatrix} 7x \\ 7y \\ 7z \end{bmatrix}$$

$$\Rightarrow 5x + y + 2z = 7x$$

$$3y = 7y$$

$$2x + y + 5z = 7z$$

$$\Rightarrow -2x + y + 2z = 0$$

$$-4y = 0$$

$$2x + y - 2z = 0$$

$$\Rightarrow -2x + y + 2z = 0$$

$$4y = 0$$

$$2x + y - 2z = 0$$

$$\Rightarrow 2x - y - 2z = 0$$

$$y = 0$$

$$2x + y - 2z = 0$$

[Multiplying by -1 on both sides]

We get, $y = 0$

$$\Rightarrow 2x - y - 2z = 0$$

$$2x + y - 2z = 0$$

We have,

$$L_i \rightarrow -a_{i1}L_1 + a_{i1}L_i$$

$$i = 2, L_2 \rightarrow -a_{21}L_1 + a_{11}L_2$$

$$= -2(2x - y - 2z) + 2(2x + y - 2z)$$

$$= -4x + 2y + 4z + 4x + 2y - 4z$$

$$= 4y$$

Thus we obtain following system

$$\Rightarrow 2x - y - 2z = 0$$

$$4y = 0$$

$$\Rightarrow 2x - y - 2z = 0$$

$$y = 0$$

$$\Rightarrow 2x - 0 - 2z = 0$$

$$\Rightarrow \textcircled{2x} - 2z = 0$$

In echelon form, there are only one equation in two unknowns, then the system has a non-zero solution and in particular $(2-1)=1$ free variable, which is z .

Let,

$$z = 1$$

$$\therefore y = 0 \text{ [obtained]}$$

$$\therefore 2x - 2 = 0$$

$$\Rightarrow x = 1$$

$$\therefore \mathbf{v}_3 = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \text{ [For } \lambda = 7 \text{]}$$

Let P be a matrix whose columns are the above Eigen vectors.

$$P = [\mathbf{v}_1 \quad \mathbf{v}_2 \quad \mathbf{v}_3]$$

$$P = \begin{bmatrix} -2 & -1 & 1 \\ 2 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

$$|P| = (-2)(0-0) - (-1)(2-0) + 1(2-0)$$

$$|P| = 2 + 2 = 4$$

Then, $D = P^{-1}AP$ is the diagonalized matrix whose leading diagonal are the respective Eigen values.

Now

$$P = \begin{bmatrix} -2 & -1 & 1 \\ 2 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

$$\therefore \text{Co-factor of } a_{11}(-2) = c_{11} = + \begin{vmatrix} 0 & 0 \\ 1 & 1 \end{vmatrix} = 0$$

$$\therefore \text{Co-factor of } a_{12}(-1) = c_{12} = - \begin{vmatrix} 2 & 0 \\ 1 & 1 \end{vmatrix} = -2$$

$$\therefore \text{Co-factor of } a_{13}(1) = c_{13} = + \begin{vmatrix} 2 & 0 \\ 1 & 1 \end{vmatrix} = 2$$

$$\therefore \text{Co-factor of } a_{21}(2) = c_{21} = - \begin{vmatrix} -1 & 1 \\ 1 & 1 \end{vmatrix} = 2$$

$$\therefore \text{Co-factor of } a_{22}(0) = c_{22} = + \begin{vmatrix} -2 & 1 \\ 1 & 1 \end{vmatrix} = -3$$

$$\therefore \text{Co-factor of } \mathbf{a}_{23}(\mathbf{0}) = \mathbf{c}_{23} = - \begin{vmatrix} -2 & -1 \\ 1 & 1 \end{vmatrix} = 1$$

$$\therefore \text{Co-factor of } \mathbf{a}_{31}(\mathbf{1}) = \mathbf{c}_{31} = + \begin{vmatrix} -1 & 1 \\ 0 & 0 \end{vmatrix} = 0$$

$$\therefore \text{Co-factor of } \mathbf{a}_{32}(\mathbf{1}) = \mathbf{c}_{32} = - \begin{vmatrix} -2 & 1 \\ 2 & 0 \end{vmatrix} = 2$$

$$\therefore \text{Co-factor of } \mathbf{a}_{33}(\mathbf{1}) = \mathbf{c}_{33} = + \begin{vmatrix} -2 & -1 \\ 2 & 0 \end{vmatrix} = 2$$

$$\text{Co-factor Matrix: } \mathbf{P}^C = \begin{bmatrix} 0 & -2 & 2 \\ 2 & -3 & 1 \\ 0 & 2 & 2 \end{bmatrix}$$

$$\text{Adjoint of } \mathbf{P} = (\mathbf{P}^C)^t = \begin{bmatrix} 0 & 2 & 0 \\ -2 & -3 & 2 \\ 2 & 1 & 2 \end{bmatrix}$$

$$\therefore \mathbf{P}^{-1} = \frac{\text{Adjoint of } \mathbf{P}}{|\mathbf{P}|} = \frac{1}{4} \begin{bmatrix} 0 & 2 & 0 \\ -2 & -3 & 2 \\ 2 & 1 & 2 \end{bmatrix}$$

$$\mathbf{D} = \mathbf{P}^{-1} \mathbf{A} \mathbf{P} = \frac{1}{4} \begin{bmatrix} 0 & 2 & 0 \\ -2 & -3 & 2 \\ 2 & 1 & 2 \end{bmatrix} \begin{bmatrix} 5 & 1 & 2 \\ 0 & 3 & 0 \\ 2 & 1 & 5 \end{bmatrix} \begin{bmatrix} -2 & -1 & 1 \\ 2 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

$$= \frac{1}{4} \begin{bmatrix} 0 & 2 & 0 \\ -2 & -3 & 2 \\ 2 & 1 & 2 \end{bmatrix} \begin{bmatrix} 5(-2)+1 \times 2+2 \times 1 & 5(-1)+1 \times 0+2 \times 1 & 5 \times 1+1 \times 0+2 \times 1 \\ 0(-2)+3 \times 2+0 \times 1 & 0(-1)+3 \times 0+0 \times 1 & 0 \times 1+3 \times 0+0 \times 1 \\ 2(-2)+1 \times 2+5 \times 1 & 2(-1)+1 \times 0+5 \times 1 & 2 \times 1+1 \times 0+5 \times 1 \end{bmatrix}$$

$$= \frac{1}{4} \begin{bmatrix} 0 & 2 & 0 \\ -2 & -3 & 2 \\ 2 & 1 & 2 \end{bmatrix} \begin{bmatrix} -10+2+2 & -5+0+2 & 5+0+2 \\ 0+6+0 & 0+0+0 & 0+0+0 \\ -4+2+5 & -2+0+5 & 2+0+5 \end{bmatrix}$$

$$= \frac{1}{4} \begin{bmatrix} 0 & 2 & 0 \\ -2 & -3 & 2 \\ 2 & 1 & 2 \end{bmatrix} \begin{bmatrix} -6 & -3 & 7 \\ 6 & 0 & 0 \\ 3 & 3 & 7 \end{bmatrix}$$

$$= \frac{1}{4} \begin{bmatrix} 0(-6)+2 \times 6+0 \times 3 & 0(-3)+2 \times 0+0 \times 3 & 0 \times 7+2 \times 0+0 \times 7 \\ (-2)(-6)+(-3)(6)+2 \times 3 & (-2)(-3)+(-3) \times 0+2 \times 3 & (-2) \times 7+(-3) \times 0+2 \times 7 \\ 2(-6)+1 \times 6+2 \times 3 & 2(-3)+1 \times 0+2 \times 3 & 2 \times 7+1 \times 0+2 \times 7 \end{bmatrix}$$

$$= \frac{1}{4} \begin{bmatrix} 0+12+0 & 0+0+0 & 0+0+0 \\ 12-18+6 & 6+6 & -14-0+14 \\ -12+6+6 & -6+0+6 & 14+0+14 \end{bmatrix}$$

$$= \frac{1}{4} \begin{bmatrix} 12 & 0 & 0 \\ 0 & 12 & 0 \\ 0 & 0 & 28 \end{bmatrix}$$

$$D = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 7 \end{bmatrix}$$

$$\therefore D = P^{-1}AP = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 7 \end{bmatrix} \text{ Answer}$$

Home Task:

Diagonalize the matrix

$$i. \begin{bmatrix} 2 & -1 \\ -3 & 4 \end{bmatrix} \quad ii. \begin{bmatrix} 1 & -1 & 2 \\ 0 & 2 & -1 \\ 0 & 0 & 3 \end{bmatrix} \quad iii. \begin{bmatrix} 5 & 0 & 0 \\ 1 & 0 & 3 \\ 0 & 0 & -2 \end{bmatrix} \quad iv. \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix} \quad v. \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad vi. \begin{bmatrix} 3 & 4 \\ 4 & -3 \end{bmatrix}$$

Solve the diagonalization problem for $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$

Hints:

$$\lambda = 1, 1$$

We cannot obtain two linearly independent eigenvectors for the given matrix. Thus the matrix is *not* diagonalizable.

Use Elementary row operations to transform the row-reduced matrix to the identity matrix:

$$\begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (\text{By adding to the second row } -3 \text{ times the third row})$$

$$\rightarrow \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (\text{By adding to the first row } -1 \text{ times the third row})$$

$$\rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (\text{By adding to the first row } -2 \text{ times the second row})$$

Thus, a square matrix A has an inverse if and only if A can be transformed into an identity matrix with elementary row operations.

Finding Inverse using Linear Equations:

Find A^{-1} where $A = \begin{bmatrix} 1 & 2 \\ 3 & 1 \end{bmatrix}$

Answer:

Given, $A = \begin{bmatrix} 1 & 2 \\ 3 & 1 \end{bmatrix}$, Let $X = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ and $B = \begin{bmatrix} d_1 \\ d_2 \end{bmatrix}$

$$\therefore AX = B$$

$$\begin{bmatrix} 1 \times x_1 + 2 \times x_2 \\ 3 \times x_1 + 1 \times x_2 \end{bmatrix} = \begin{bmatrix} d_1 \\ d_2 \end{bmatrix}$$

The matrix can be written as

$$x_1 + 2x_2 = d_1 \text{ (say) } \text{-----(i)}$$

$$3x_1 + x_2 = d_2 \text{ (say) } \text{-----(ii)}$$

$$(ii) \times 2 - (i) :$$

$$5x_1 = 2d_2 - d_1$$

$$x_1 = -\frac{1}{5}d_1 + \frac{2}{5}d_2 \text{ -----(iii)}$$

$$(i) \times 3 - (ii) :$$

$$5x_2 = 3d_1 - d_2$$

$$x_2 = \frac{3}{5}d_1 - \frac{1}{5}d_2 \text{ -----(iv)}$$

FROM (iii) and (iv) we get,

$$\therefore A^{-1} = \begin{bmatrix} -\frac{1}{5} & \frac{2}{5} \\ \frac{3}{5} & -\frac{1}{5} \end{bmatrix}$$

Question: Define Augmented Matrix with example:

Answer: The augmented matrix for $AX=b$ is the partitioned matrix $[A|b]$.

All the scalars contained in the system of equations $AX=b$ appear in the coefficient matrix A and the column matrix b . These scalars can be combined into the single partitioned matrix $[A|b]$, known as the augmented matrix for the system of equations.

Example 30: $x_1 + x_2 - 2x_3 = -3$

$$2x_1 + 5x_2 + 3x_3 = 11$$

$$-x_1 + 3x_2 + x_3 = 5$$

can be written as the matrix equation

$$\begin{bmatrix} 1 & 1 & -2 \\ 2 & 5 & 3 \\ -1 & 3 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -3 \\ 11 \\ 5 \end{bmatrix}$$

Which has as its augmented matrix?

$$[A|b] = \left[\begin{array}{ccc|c} 1 & 1 & -2 & -3 \\ 2 & 5 & 3 & 11 \\ -1 & 3 & 1 & 5 \end{array} \right]$$

Elementary Row and Column Operations

The matrix operations of

1. Interchanging two rows or columns,
2. Adding a multiple of one row or column to another,
3. Multiplying any row or column by a nonzero element.

Calculating Inverses by the elementary row operation method

Step 1: Create an augment matrix $[A | I]$, where A is the $n \times n$ matrix to be inverted and I is the $n \times n$ identity matrix.

Step 2: Use elementary row operations on $[A | I]$ to transform the left partition A to row-reduced form, applying each operation to the full augmented matrix.

Step 3: If the left partition of the row-reduced matrix has zero elements on its main diagonal, stop-A does not have an inverse-otherwise, continue.

Step 4: Use elementary row operations on the row-reduced augmented matrix to transform the left partition to the $n \times n$ identity matrix, applying each operation to the full augmented matrix.

Step 5: The right partition of the final augmented matrix is the inverse of A.

[Note: It is easy to show that if any diagonal element in a diagonal matrix is (zero) 0, then that matrix is singular.]

Example 31: Find the inverse of $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$

$$\text{Solution: } [A|I] = \left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 3 & 4 & 0 & 1 \end{array} \right]$$

$$r_2' = r_2 + (-3)r_1$$

$$\approx \left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 3 + (-3).1 & 4 + (-3).2 & 0 + (-3).1 & 1 + (-3).0 \end{array} \right]$$

$$\approx \left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 3-3 & 4-6 & -3 & 1+0 \end{array} \right]$$

$$\approx \left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & -2 & -3 & 1 \end{array} \right]$$

$$r_2' = r_2 \times \left(-\frac{1}{2}\right)$$

$$\approx \left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 \times \left(-\frac{1}{2}\right) & -2 \times \left(-\frac{1}{2}\right) & -3 \times \left(-\frac{1}{2}\right) & 1 \times \left(-\frac{1}{2}\right) \end{array} \right]$$

$$\approx \left[\begin{array}{cc|cc} 1 & 2 & \frac{1}{2} & 0 \\ 0 & 1 & \frac{3}{2} & -\frac{1}{2} \end{array} \right]$$

$$r_1' = r_1 + (-2)r_2$$

$$\approx \left[\begin{array}{cc|cc} 1+(-2).0 & 2+(-2).1 & 1+(-2).\frac{3}{2} & 0+(-2).\left(-\frac{1}{2}\right) \\ 0 & 1 & \frac{3}{2} & -\frac{1}{2} \end{array} \right]$$

$$\approx \left[\begin{array}{cc|cc} 1+0 & 2-2 & 1-\frac{3}{2} & 0+\frac{1}{2} \\ 0 & 1 & \frac{3}{2} & -\frac{1}{2} \end{array} \right]$$

$$\approx \left[\begin{array}{cc|cc} 1 & 0 & -\frac{2}{3} & \frac{1}{2} \\ 0 & 1 & \frac{3}{2} & -\frac{1}{2} \end{array} \right]$$

$$\text{Thus } A^{-1} = \begin{bmatrix} -\frac{2}{3} & \frac{1}{2} \\ \frac{3}{2} & -\frac{1}{2} \end{bmatrix}$$

$$\text{Example 32: Find the inverse of } A = \begin{bmatrix} 5 & 8 & 1 \\ 0 & 2 & 1 \\ 4 & 3 & -1 \end{bmatrix}$$

$$\text{Solution: } \left[\begin{array}{ccc|ccc} 5 & 8 & 1 & 1 & 0 & 0 \\ 0 & 2 & 1 & 0 & 1 & 0 \\ 4 & 3 & -1 & 0 & 0 & 1 \end{array} \right]$$

$$\rightarrow \left[\begin{array}{ccc|ccc} 1 & 1.6 & 0.2 & 0.2 & 0 & 0 \\ 0 & 2 & 1 & 0 & 1 & 0 \\ 4 & 3 & -1 & 0 & 0 & 1 \end{array} \right] \text{ (By multiplying the first row by 0.2)}$$

$$\rightarrow \left[\begin{array}{ccc|ccc} 1 & 1.6 & 0.2 & 0.2 & 0 & 0 \\ 0 & 2 & 1 & 0 & 1 & 0 \\ 0 & -3.4 & -1.8 & -0.8 & 0 & 1 \end{array} \right] \text{ (By adding to the third row } -4 \text{ times the first row)}$$

$$\rightarrow \left[\begin{array}{ccc|ccc} 1 & 1.6 & 0.2 & 0.2 & 0 & 0 \\ 0 & 1 & 0.5 & 0 & 0.5 & 0 \\ 0 & -3.4 & -1.8 & -0.8 & 0 & 1 \end{array} \right] \text{ (By multiplying the second row by } \frac{1}{2} \text{)}$$

$$\rightarrow \left[\begin{array}{ccc|ccc} 1 & 1.6 & 0.2 & 0.2 & 0 & 0 \\ 0 & 1 & 0.5 & 0 & 0.5 & 0 \\ 0 & 0 & -0.1 & -0.8 & 1.7 & 1 \end{array} \right] \text{ (By adding to the third row } 3.4 \text{ times the second row)}$$

$$\rightarrow \left[\begin{array}{ccc|ccc} 1 & 1.6 & 0.2 & 0.2 & 0 & 0 \\ 0 & 1 & 0.5 & 0 & 0.5 & 0 \\ 0 & 0 & 1 & 8 & -17 & -10 \end{array} \right] \text{ (by multiplying the third row by } -10 \text{)}$$

A has been transformed into row-reduced form with a main diagonal of only ones; A has an inverse. Continuing with the transformation process, we get:

$$\rightarrow \left[\begin{array}{ccc|ccc} 1 & 1.6 & 0.2 & 0.2 & 0 & 0 \\ 0 & 1 & 0 & -4 & 9 & 5 \\ 0 & 0 & 1 & 8 & -17 & -10 \end{array} \right] \text{ (By adding to the second row } -0.5 \text{ times the third row)}$$

$$\rightarrow \left[\begin{array}{ccc|ccc} 1 & 1.6 & 0 & -1.4 & 3.4 & 2 \\ 0 & 1 & 0 & -4 & 9 & 5 \\ 0 & 0 & 1 & 8 & -17 & -10 \end{array} \right] \text{ (by adding to the first row } -0.2 \text{ times the third row)}$$

$$\rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 5 & -11 & -6 \\ 0 & 1 & 0 & -4 & 9 & 5 \\ 0 & 0 & 1 & 8 & -17 & -10 \end{array} \right] \text{ (by adding to the first row } -1.6 \text{ times the second row)}$$

second row)

$$\text{Thus, } \mathbf{A}^{-1} = \begin{bmatrix} 5 & -11 & -6 \\ -4 & 9 & 5 \\ 8 & -17 & -10 \end{bmatrix}$$

Question: State Cayley Hamilton Theorem with example:

Answer:

Statement: Every Square matrix satisfies its characteristic equation or if $|A - \lambda I| = (-1)^n [\lambda^n + a_1 \lambda^{n-1} + a_2 \lambda^{n-2} + \dots - a_n]$ be the characteristic polynomial of $n \times n$ matrix $A = [a_{ij}]$, then the matrix equation $X^n + a_1 X^{n-1} + \dots - a_n I = 0$ is satisfied by $X=A$
i.e $A^n + a_1 A^{n-1} + \dots - a_n I = 0$

Example 33: Show that the matrix $A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$ satisfies Cayley-Hamilton Theorem

and hence compute A^{-1}

Answer: Hence we have,

$$A - \lambda I = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 1-\lambda & 4-0 \\ 2-0 & 3-\lambda \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 1-\lambda & 4 \\ 2 & 3-\lambda \end{bmatrix}$$

$$|A - \lambda I| = \begin{vmatrix} 1-\lambda & 4 \\ 2 & 3-\lambda \end{vmatrix}$$

$$|A - \lambda I| = (1-\lambda)(3-\lambda) - 8$$

$$|A - \lambda I| = (3-\lambda-3\lambda+\lambda^2) - 8$$

$$|A - \lambda I| = 3-4\lambda+\lambda^2-8$$

$$|A - \lambda I| = \lambda^2 - 4\lambda - 5$$

$\therefore$ Characteristic equation of A is $|A - \lambda I| = 0$

$$\lambda^2 - 4\lambda - 5 = 0 \text{ -----(i)}$$

Replace λ by A in L.H.S of (i)

Now $A^2 - 4A - 5I$

$$\begin{aligned} &= \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix} - 4 \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix} - 5 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 1+8 & 4+12 \\ 2+6 & 8+9 \end{bmatrix} - \begin{bmatrix} 4 & 16 \\ 8 & 12 \end{bmatrix} - \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix} \\ &= \begin{bmatrix} 9 & 16 \\ 8 & 17 \end{bmatrix} - \begin{bmatrix} 4 & 16 \\ 8 & 12 \end{bmatrix} - \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix} \end{aligned}$$

$$= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = \text{NULL MATRIX}$$

That is $A^2 - 4A - 5I = 0$

------(ii)

Hence the given matrix A satisfies its characteristic equation (i) i.e satisfy Caley Hamilton Theorem.

Again, from (ii)

$A^2 - 4A - 5I = 0$ gives

$$A^2 - 4A = 5I$$

------(iii)

Multiplying (iii) by A^{-1} on both sides:

$$A^2 \cdot A^{-1} - 4AA^{-1} = 5A^{-1}I$$

$$A - 4I = 5A^{-1}$$

$$[AA^{-1} = I \text{ \& } A^{-1}I = A^{-1}]$$

$$\therefore A^{-1} = \frac{1}{5}(A - 4I)$$

------(iv)

Putting the value of A and I in (iv), we get

$$\therefore A^{-1} = \frac{1}{5}(A - 4I)$$

$$\therefore A^{-1} = \frac{1}{5} \left\{ \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix} - 4 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right\}$$

$$\therefore A^{-1} = \frac{1}{5} \left\{ \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix} - \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix} \right\}$$

$$\therefore A^{-1} = \frac{1}{5} \left\{ \begin{bmatrix} 1-4 & 4-0 \\ 2-0 & 3-4 \end{bmatrix} \right\}$$

$$= \frac{1}{5} \begin{bmatrix} -3 & 4 \\ 2 & -1 \end{bmatrix}$$

Perfection Test:

$$AA^{-1} = \frac{1}{5} \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} -3 & 4 \\ 2 & -1 \end{bmatrix}$$

$$AA^{-1} = \frac{1}{5} \begin{bmatrix} 1 \times (-3) + 4 \times 2 & 1 \times 4 + 4 \times (-1) \\ 2 \times (-3) + 3 \times 2 & 2 \times 4 + 3 \times (-1) \end{bmatrix}$$

$$AA^{-1} = \frac{1}{5} \begin{bmatrix} -3+8 & 4-4 \\ -6+6 & 8-3 \end{bmatrix}$$

$$AA^{-1} = \frac{1}{5} \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix}$$

$$AA^{-1} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$AA^{-1} = I$$

Geometrical Applications of Matrices

A matrix is a rectangular array of numbers, which can be used to give information. The coordinates in two dimensions of a point can be represented by a matrix, so that the coordinates (2,3) of a point can be represented by a matrix $\begin{pmatrix} 2 \\ 3 \end{pmatrix}$.

The components of a vector can be represented by a matrix

Example:

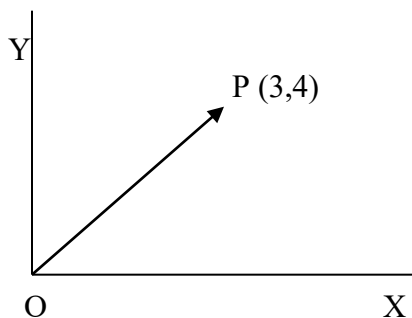


Figure no 26

$$\vec{OP} = \begin{pmatrix} 3 \\ 4 \end{pmatrix} = (3\hat{i} + 4\hat{j})$$

We can even write the coordinates of several points in a single matrix, so that the points coordinates O(0,0), P(1,0), Q(1,1) and R(0,1), the vertices of the unit square, can be displayed in the matrix $\begin{pmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix}$ (Fig:1) and the points O(0,0), P(2,0), Q(2,1) and

R(0,1) can be displayed in the matrix $\begin{pmatrix} 0 & 2 & 2 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix}$ (Fig:2). Since these points are the vertices of a rectangle, call this matrix R.

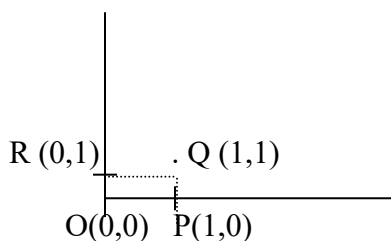


Figure no 27

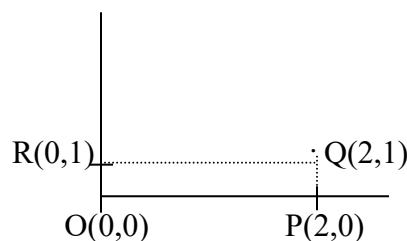


Figure no 28

When the matrix R is multiplied by 2×2 -unit matrixes, it is unaltered.

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 2 & 2 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 2 & 2 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix}$$

Reflection:

When R is multiplied by $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$,

$$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 0 & 2 & 2 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 2 & 2 & 0 \\ 0 & 0 & -1 & -1 \end{pmatrix}.$$

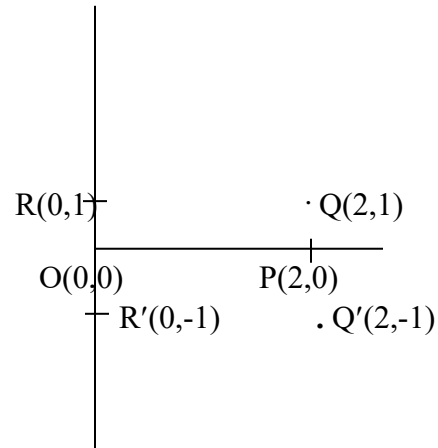
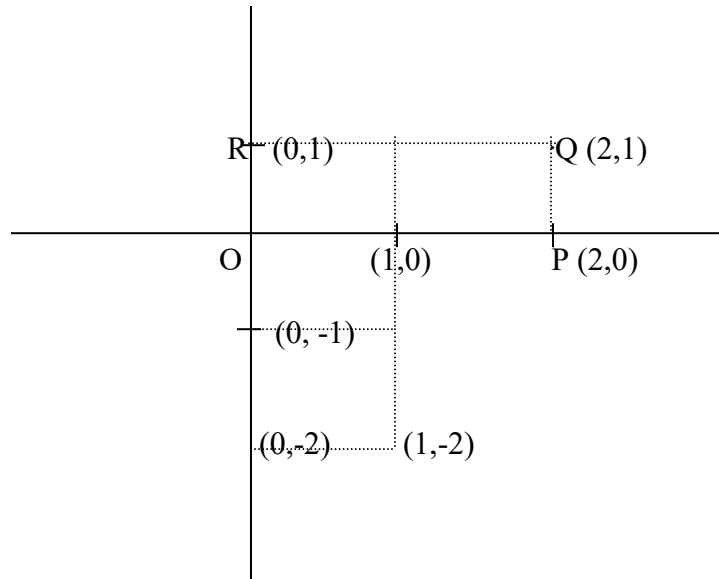


Figure no 29

The rectangle OPQR has been reflected in the x-axis. The rectangle OP'Q'R' is called the image of OPQR under the transformation described by the matrix $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ or more simply, the image under the transformation $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

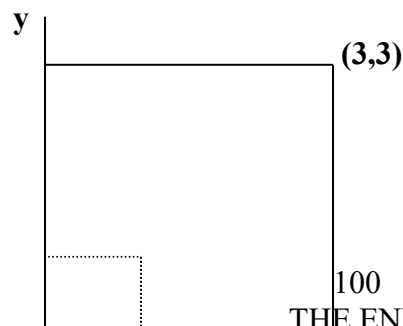
Rotation:

Similarly $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 2 & 2 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 1 & 1 \\ 0 & -2 & -2 & 0 \end{pmatrix}$ showing that OPQR has been rotated about the origin through 90° in a clockwise sense, and we see that the matrix $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ describes a rotation through 90° in a clockwise sense. The matrix $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ describes a rotation about the origin through 90° in an anticlockwise sense.

**Figure no 30**

Enlargement: If we consider the effect of the matrix $\begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix}$ on the unit square,

$$\begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 3 & 3 & 0 \\ 0 & 0 & 3 & 3 \end{pmatrix}$$



(0,1) (1,1)

(0,0)S (1,0) X

Figure no 31

We see that each side has been enlarged by a factor 3 and the area has been increased by 9, the figure remaining a square. Thus we see that matrices can describe some of the simplest geometrical operations, reflection, rotation and enlargement. We should, of course, show that these matrices would make the appropriate transformation on any figure, not just on the unit square and rectangle.

Example 34: Given that $A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, find A^2, A^3 and A^4 . Investigate the transformations described by these matrices.

$$\begin{aligned} \text{Answer: } A^2 &= \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} \\ A^3 &= A.A^2 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \\ A^4 &= A.A^3 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \end{aligned}$$

The coordinates of any point P (x,y) can be written as a matrix $\begin{pmatrix} x \\ y \end{pmatrix}$, denoted by X and

we see that:

$$\begin{aligned} A.X &= \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -y \\ x \end{pmatrix} \\ A^2.X &= \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -x \\ -y \end{pmatrix} \\ A^3.X &= \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} y \\ -x \end{pmatrix} \end{aligned}$$

and $A^4.X = X$, Since A^4 is the 2×2 unit matrix.

The matrix A has rotated OP through 90° in the anticlockwise sense, A^2 has rotated OP through 180° , A^3 has rotated OP through 270° and A^4 has rotated OP through 360° , mapping OP onto itself.

Another Example:

$$\begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 1 & 1+k & k \\ 0 & 0 & 1 & 1 \end{pmatrix}$$

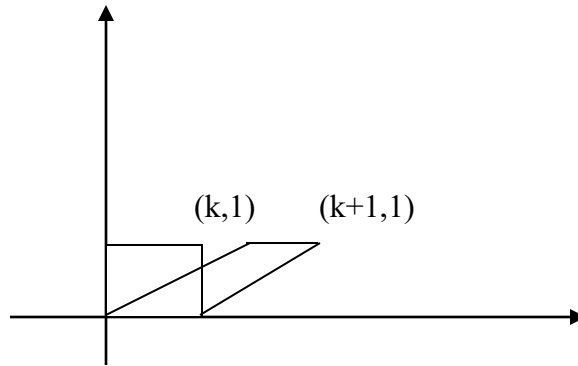


Figure no 32

The matrix $\begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix}$ has mapped the unit square into a parallelogram as the above figure.

Interpret the meaning of a determinant:

There are many ways to interpret the meaning of a determinant. One simple interpretation concerns 2D parallelograms and 3D parallelepipeds. In the 2D case,

Consider the vectors $\vec{u} = (u_1, u_2)$ and $\vec{v} = (v_1, v_2)$. Then the area of the parallelogram formed by u and v ,

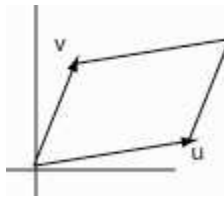


Figure no 33

$$\text{Area} = \begin{vmatrix} u_1 & v_1 \\ u_2 & v_2 \end{vmatrix} = u_1 v_2 - u_2 v_1$$

Similarly, with three 3D vectors (u_1, u_2, u_3) , (v_1, v_2, v_3) and (w_1, w_2, w_3) we can define a parallelepiped, whose volume is

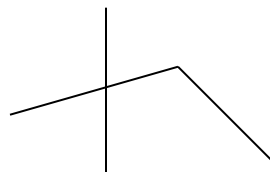
$$\text{Volume} = \begin{vmatrix} u_1 & v_1 & w_1 \\ u_2 & v_2 & w_2 \\ u_3 & v_3 & w_3 \end{vmatrix}$$

Determinants have many uses, but we will primarily use them to define and evaluate the cross product of two 3D vectors

Question#35: Find the area of the parallelogram defined by the column matrices

$$u = \begin{bmatrix} -4 \\ 4 \end{bmatrix} \text{ and } v = \begin{bmatrix} 6 \\ 2 \end{bmatrix}$$

$$\text{Answer: } \left| \text{determinant} \begin{bmatrix} -4 & 6 \\ 4 & 2 \end{bmatrix} \right| = |-32| = 32 \text{ square units.}$$



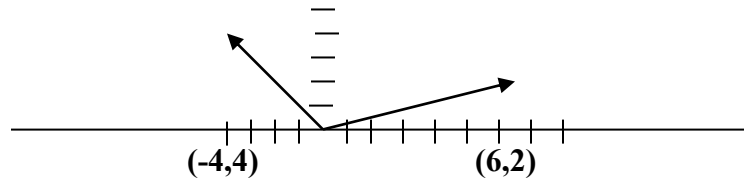


Figure no 34

Question 36: Find the area of the parallelogram defined by the column matrices

$$\mathbf{u} = \begin{bmatrix} -3 \\ -1 \end{bmatrix} \text{ and } \mathbf{v} = \begin{bmatrix} 6 \\ 2 \end{bmatrix}$$

Answer: $\left| \text{determinant} \begin{bmatrix} -3 & 6 \\ -1 & 2 \end{bmatrix} \right| = |0| = 0 \text{ square units.}$

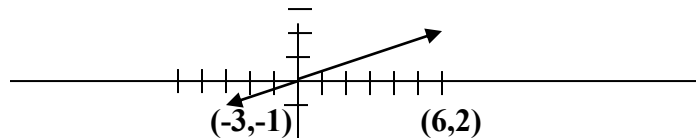


Figure no 35

Some Important Properties:

01) The determinant of an upper or lower triangular matrix is the product of the elements on the main diagonal.

Example: $\det \begin{bmatrix} 2 & 6 & -4 & 1 \\ 0 & 5 & 7 & -4 \\ 0 & 0 & -5 & 8 \\ 0 & 0 & 0 & 3 \end{bmatrix} = 2(5)(-5)(3) = -150$

02) The determinant of a diagonal matrix is the product of the elements on its main diagonal

03) If a square matrix has a zero row or a zero column, then its determinant is 0.

04) If two rows of a square matrix are identical, then its determinant is 0.

05) A square matrix has an inverse if and only if its determinant is nonzero.

06) The sum of the eigenvalues of a matrix equals the trace of the matrix

Example 37: Determine whether $\lambda_1 = 12$ and $\lambda_2 = -4$ are eigen values for

$$\mathbf{A} = \begin{bmatrix} 11 & 3 \\ -5 & -5 \end{bmatrix}$$

Solution: Here $\text{tr}(\mathbf{A}) = 11 + (-5) = 6 \neq 8 = \lambda_1 + \lambda_2$, So these numbers are not the eigen values of A.

07) The eigenvalues of an upper or lower triangular matrix are the elements on the main diagonal

Example 38: The matrix $\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & 4 & -1 \end{bmatrix}$ is lower triangular, so its eigenvalues are

$\lambda_1 = \lambda_2 = 1$ and $\lambda_3 = -1$

08) A matrix is singular if and only if it has a zero eigenvalue

09) The product of all the eigenvalues of a matrix equals to the determinant of the matrix

Example 39: The eigenvalues of $A = \begin{bmatrix} 11 & 3 \\ -5 & -5 \end{bmatrix}$ are $\lambda_1 = 10$ and $\lambda_2 = -4$

Here $\det(A) = -55 + 15 = -40 = \lambda_1 \lambda_2$

10) If λ is an eigenvalue of A , then λ also an eigenvalue of A^T

11) An $n \times n$ matrix is diagonalizable if and only if the matrix possesses n linearly independent eigenvectors

Representing Graphs Using Adjacency Matrix:

Adjacency Matrix Representation

Example 40:

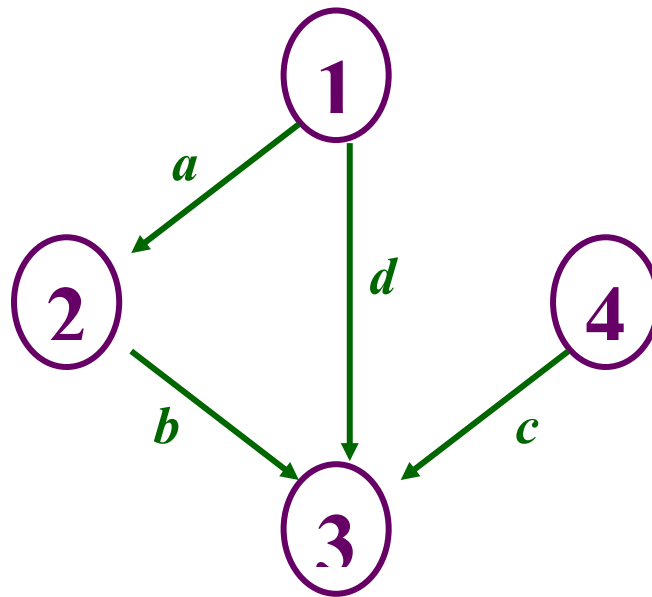


Figure no 36

<i>A</i>	<i>1</i>	<i>2</i>	<i>3</i>	<i>4</i>
<i>1</i>	<i>0</i>	<i>1</i>	<i>1</i>	<i>0</i>
<i>2</i>	<i>0</i>	<i>0</i>	<i>1</i>	<i>0</i>
<i>3</i>	<i>0</i>	<i>0</i>	<i>0</i>	<i>0</i>
<i>4</i>	<i>0</i>	<i>0</i>	<i>0</i>	<i>0</i>

An adjacency matrix represents the graph as a $n \times n$ matrix A :

$$A[i, j] = 1 \text{ if edge } (i, j) \in E \text{ (or weight of edge)}$$
$$= 0 \text{ if edge } (i, j) \notin E$$

Node:

- Nodes contain data and/or links to other nodes. Links between nodes are often implemented by pointers or references.
- Multiple computers are connected with a network in a distributed processing system (or a parallel processing system). Each computer of such a system is called a 'node'.
- Any device that is directly connected to the network, usually through Ethernet cable. Nodes include file servers and shared peripherals.
- A node is a device that is connected as part of a computer network. Nodes can be computers, personal digital assistants (PDAs), cell phones, or various other network appliances.
- The place on a stem where the leaves or branches are attached.
- A single machine on a network.
- The point on a plant stem where a leaf or leaves are attached; the point on a stem from which new leaves or stems will grow.
- A device attached to a network. A node uses the network as a means of communication and has an address on the network
- End point of a network connection. Nodes include any device attached to a network such as file servers, printers, or workstations.
- The part of the stem from which one or more leaves arise.
- The point on a stem where a leaf or bud is attached.
- This is any point of connection in a network.
- The point of intersection of a planet's orbit and the ecliptic.
- A linked list is a linear collection of data elements (called nodes), where the linear order is given by means of pointers.
- When two or more devices connected for the purpose of sharing data or resources can form a network. Devices referred to as nodes.
- A linked list is one of the fundamental data structures used in computer programming

Different of linked list: Doubly linked list, singly linked list, linked list, double linked list, circularly linked list, unrolled linked list

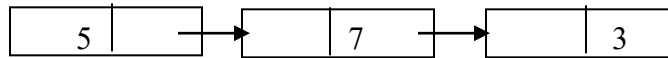
Each node is divided into 2 parts in linked list:

- i) 1st part contains the information of the element

- ii) 2nd part is called the link field or next pointer field, which contains the address of the next node in the list.

(In link list)

```
Struct node {
    int value;
    struct node *next;
}
```



Undirected Graph:

A graph $G = (V, E)$

V: vertices

E : edges, unordered pairs of vertices from $V \times V$

(u,v) is same as (v,u)

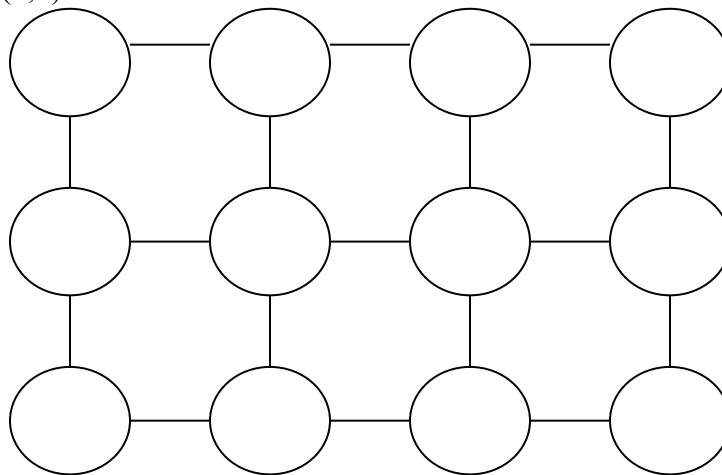


Figure no 37

Directed Graph:

A graph $G = (V, E)$

V: vertices

E: edges, ordered pairs of vertices from $V \rightarrow V$

$F(u,v)$ is different from (v,u)

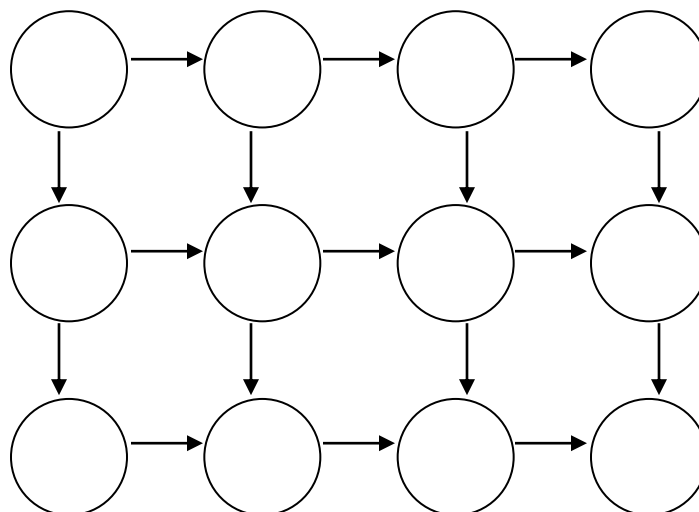


Figure no 38

Strongly Connected Components:

- Every pair of vertices are reachable from each other
- Graph G is **strongly connected** if, for every u and v in V , there is some path from u to v and some path from v to u .

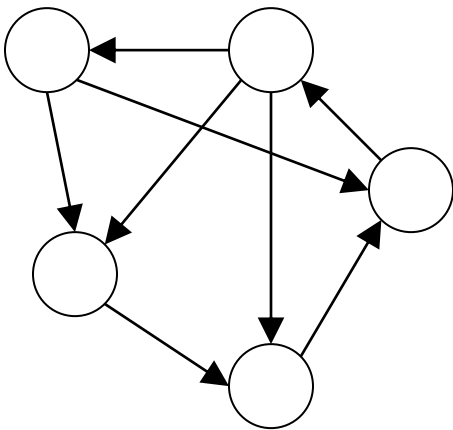


Figure no 39: Strongly Connected

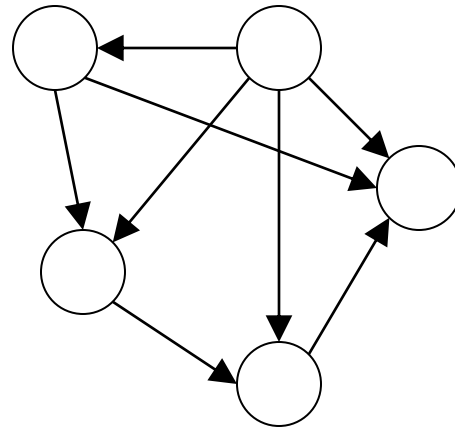


Figure no 40: Not Strongly Connected

Graphically Representation of Transpose of a Matrix:

Example: 41

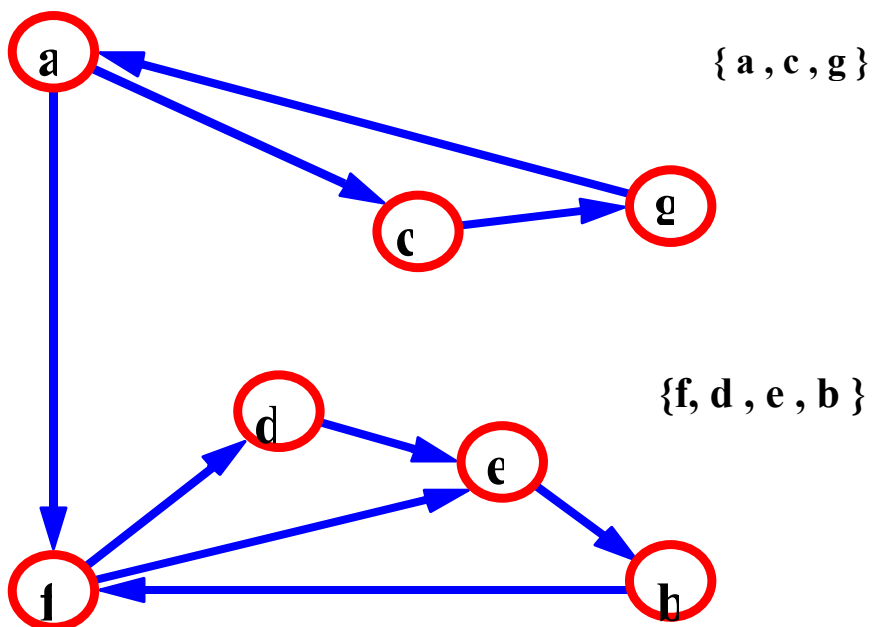
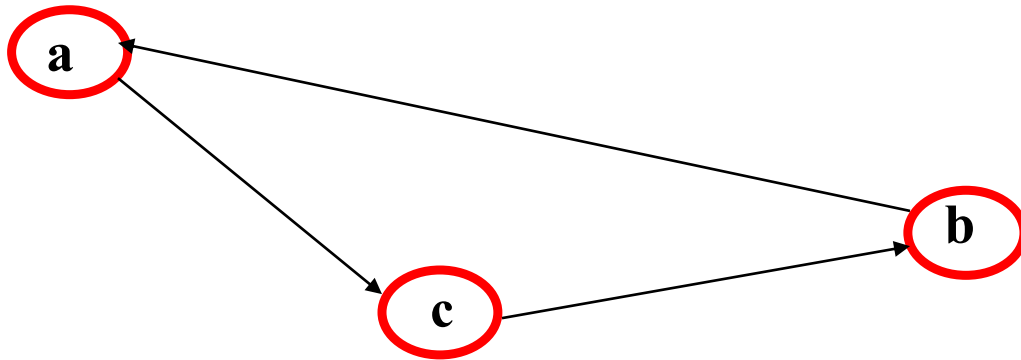


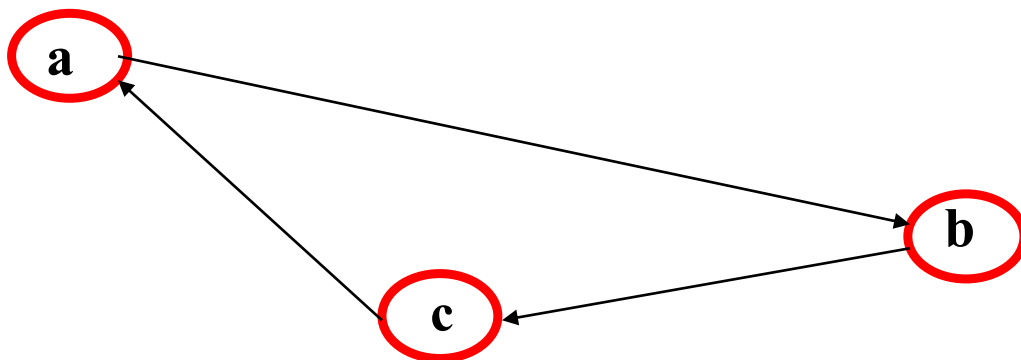
Figure no 41

The graph G^T is the transpose of G , which is visualized by reversing the arrows on the digraph.



Graph (G)

Figure no 42



Graph (G^T)

Figure no 43

	a	b	c
a	0	0	1
b	1	0	0
c	0	1	0

Matrix of Graph G

	a	b	c
a	0	1	0
b	0	0	1
c	1	0	0

Matrix of Graph G^T